5G RAN

UE Power Saving Feature Parameter Description

Issue Draft B

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HUAWEI TECHNOLOGIES CO., LTD.

Huawei Technologies Co., Ltd.

Address:	Huawei Industrial Base Bantian, Longgang Shenzhen 518129 People's Republic of China
Website:	https://www.huawei.com
Email:	support@huawei.com

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Added descriptions of the **REDCAP_RELEASE_PREFER_CTRL_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter. For details, see **7.1.1 Principles**.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added a mutually exclusive relationship between PEI and configuration-based interference avoidance enhancement. For details, see **4.6.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Removed the configuration constraints on power saving BWP with the following functions from **5.1.4.2.2 Mutually Exclusive Functions**: high speed mobility and inter-TRP SSB staggering.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between "UL and DL Decoupling" and "WUS". For details, see **4.2.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added impact relationships between "UL and DL Decoupling"and the following functions: PDCCH skipping and WUS. For details, see **4.4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between "UL and DL Decoupling" and power saving BWP. For details, see **5.1.4.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added time-domain power saving BWP. For details, see **4.7 Time-Domain Power Saving BWP (Low-Frequency TDD)**.	Added the **TDM_BWP2_SWITCH** option to the **NRDUCellUePwrSaving**.*BwpPwrSavingSw* parameter. Added parameters: **NRDUCellUePwrSaving**.*Bwp1ToTdmBwp2DlThptThld* **NRDUCellUePwrSaving**.*Bwp1ToTdmBwp2UlThptThld* **NRDUCellUePwrSaving**.*TdmBwp2SchFailRateThld*	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between SSSG switching and time-domain power saving BWP. For details, see **4.5.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between PDCCH skipping and time-domain power saving BWP. For details, see **4.4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added BWP2 PUCCH frequency-domain position adjustment. For details, see **BWP2 PUCCH Freq Position Adjustment**.	Added the **BWP2_PUCCH_FREQ_POS_ADJ** option to the **NRDUCellPucch**.***PucchAlgoExtSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Changed the name of "intra-base-station fast CA" and "fast inter-gNodeB CA " to "downlink fast intra-gNodeB CA" and "downlink fast inter-gNodeB CA", respectively. For details, see **4.4.3.2.3 Function Impacts**, **4.5.3.2.3 Function Impacts**, and **5.3.3.2.3 Function Impacts**.

  Added the cell requirements of WUS. For details, see **4.2.3.4 Cells**.

  Removed descriptions of the impact relationship between WUS and RedCap from **4.2.3.2.3 Function Impacts**. This is because WUS is not applicable to RedCap UEs.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-030210	UE Power Saving	4.1 Adaptation of the Uplink Smart Preallocation Duration
FOFD-030210	UE Power Saving	4.2 WUS
FOFD-030210	UE Power Saving	4.3 UL Skipping
FOFD-030210	UE Power Saving	4.4 PDCCH Skipping
FOFD-030210	UE Power Saving	4.5 SSSG Switching
FOFD-030210	UE Power Saving	4.6 PEI (Low-Frequency TDD)
FOFD-030210	UE Power Saving	5.1 Power Saving BWP (Low-Frequency TDD)
FOFD-030210	UE Power Saving	5.2 Reducing the Number of SCCs
FOFD-030210	UE Power Saving	5.3 Maximum-Bandwidth Reduction
FOFD-030210	UE Power Saving	6.1 Reducing the Number of MIMO Layers
FOFD-030210	UE Power Saving	6.2 BWP Spatial Adaptation (Low-Frequency TDD)
FOFD-030210	UE Power Saving	7.1 Fast RRC Connection Release
FOFD-030210	UE Power Saving	7.2 Inactive RA-SDT
FOFD-081252	UE Power Saving Based on Inactive State	7.3 UE Power Saving Based on Inactive State

3 Overview

5G networks have more frequency bands and higher bandwidths. As a result, the components within and product specifications of 5G terminals increase, and the terminals inevitably consume more power. In addition, 5G terminals are more diversified, and pose higher requirements on battery life. To save more power for 5G terminals, Huawei provides the following solutions (see **Table 3-1**) to solve the problems of continuous overheating and high power consumption of UEs, prolong UE standby time, and improve UE battery life.

Table 3-1 Overview of UE power saving technologies

Category	Power Saving Technology	Principle
Time domain	DRX (For details, see *DRX*.)	When no service data needs to be sent from the network to a UE, it is a waste of power for the UE to monitor the PDCCH. To address this issue, the discontinuous reception (DRX) function is introduced. DRX enables UEs in connected mode to periodically stop monitoring the PDCCH, reducing UE power consumption.
Time domain	**4.1 Adaptation of the Uplink Smart Preallocation Duration**	If the duration of uplink smart preallocation (for details, see *Uplink Scheduling*) is too long, UEs stay in the DRX On Duration period for a long time, wasting power. Adaptation of the uplink smart preallocation duration adaptively adjusts the preallocation duration based on the uplink response time of downlink scheduling to reduce UE power consumption.

Category	Power Saving Technology	Principle
Time domain	**4.2 WUS**	When the traffic volume of a UE is small, the scheduling probability is low. Waking up in each DRX cycle consumes excessive UE power. To address this issue, the WUS function is introduced. In this function, DCI is used to indicate whether a UE needs to wake up in the DRX On Duration period, reducing UE power consumption.
Time domain	**4.3 UL Skipping**	After uplink preallocation is enabled, a UE must respond to the base station with padding packets upon receiving UL grants even if the UE has no data to transmit, consuming excessive UE power. To address this issue, UL skipping is introduced. Sending padding packets is not required if the UE has no data to transmit after it receives UL grants from the base station. In this way, UE power consumption is reduced.
Time domain	**4.4 PDCCH Skipping**	In heartbeat or sparse packet scenarios, continuously monitoring the PDCCH consumes excessive UE power. To address this issue, PDCCH skipping is introduced. In this function, DCI is used to instruct a UE not to monitor the PDCCH in slots following the DCI, thereby saving UE power.
Time domain	**4.5 SSSG Switching**	In heartbeat or sparse packet scenarios, continuously monitoring the PDCCH consumes excessive UE power. To address this, SSSG switching is introduced. This feature enables the base station to instruct an applicable UE to switch from a dense SSSG to a sparse SSSG by using DCI. This reduces PDCCH monitoring occasions and saves power.
Time domain	**4.6 PEI (Low-Frequency TDD)**	When there is no paging message, the UE still wakes up on the PO, resulting in a waste of power. To address this issue, the PEI function is introduced. This function enables RRC_IDLE or RRC_INACTIVE UEs to wake up as required, reducing unnecessary waking-up and power consumption.

Category	Power Saving Technology	Principle
Time domain	**4.7 Time-Domain Power Saving BWP (Low-Frequency TDD)**	If a UE continuously monitors the PDCCH while its traffic volume is light, the UE power is wasted. To address this issue, the time-domain power saving BWP function is introduced. With this function, a base station can adaptively switch the UE to a BWP with a low PDCCH monitoring density based on the UE traffic volume to save UE power.
Time domain	**4.8 VoNR UE Power Saving**	In VoNR scenarios, if the uplink compensation scheduling period within silent periods is the same as that within talk spurts, some UE power is wasted during silent periods. To address this issue, VoNR UE power saving is introduced. This function allows a long uplink compensation scheduling period for VoNR within silent periods, to reduce UE power consumption.
Frequency domain	**5.1 Power Saving BWP (Low-Frequency TDD)**	When the traffic volume is small, it is a waste of power for a UE to work on a large bandwidth. To address this issue, power saving BWP is introduced. This function switches UEs from a large-bandwidth BWP to a small-bandwidth BWP, thereby saving UE power.
Frequency domain	**5.2 Reducing the Number of SCCs**	When the traffic volume is small, it is a waste of power for a UE to work on multiple carriers. In this context, the function of reducing the number of SCCs is introduced. With this function, the base station reduces the number of SCCs for UEs to achieve UE power saving and overheating protection.
Frequency domain	**5.3 Maximum-Bandwidth Reduction**	When the traffic volume is small, it is a waste of power for a UE to work on a large bandwidth. In this context, the maximum-bandwidth reduction function is introduced. With this function, the base station reduces the UE bandwidth to achieve UE power saving and overheating protection.

Category	Power Saving Technology	Principle
Frequency domain	SUL UE power saving (For details, see *Super Uplink*.)	In super uplink scenarios, when the uplink traffic volume is small, UEs working on a large bandwidth consume excessive power. In this context, SUL UE power saving is introduced. When the uplink traffic volume of a super uplink UE is lower than a specified threshold, the gNodeB can trigger SUL carrier removal to reduce UE power consumption.
Space domain	**6.1 Reducing the Number of MIMO Layers**	If the traffic volume is small, using an excessive number of antennas to transmit and receive data consumes excessive power for a UE. To address this issue, the function of reducing the number of MIMO layers is introduced. Based on the maximum number of MIMO layers in the uplink and downlink instructed by the base station, the UE shuts down some transmit and receive antennas for UE power saving or overheating protection.
Space domain	**6.2 BWP Spatial Adaptation (Low-Frequency TDD)**	If the traffic volume is small, using an excessive number of antennas to transmit and receive data consumes excessive power for a UE. To address this issue, BWP spatial adaptation is introduced. After the UE switches to the narrow-bandwidth BWP, the UE shuts down some receive antennas based on the maximum number of downlink MIMO layers configured for the narrow-bandwidth BWP to further save UE power.
Others	**7.1 Fast RRC Connection Release**	When a UE has no data to transmit, remaining in the RRC_CONNECTED state unnecessarily consumes power. To address this issue, fast RRC connection release is introduced. When a UE has no data to transmit, it instructs the base station to directly release the RRC connection without waiting for the inactivity timer to expire. This reduces the duration in which the UE stays in the RRC_CONNECTED state to save UE power.

Category	Power Saving Technology	Principle
Others	7.2 Inactive RA-SDT	When a UE is running services with a small data volume, excessive power is consumed if it stays in the RRC_CONNECTED state for a long time. To address this issue, the inactive RA-SDT function is introduced. This function allows UEs in the RRC_INACTIVE state to perform services with a small amount of data. This reduces signaling transmission and prolongs the time for UEs to stay in the RRC states other than RRC_CONNECTED, thereby achieving UE power saving.
Others	7.3 UE Power Saving Based on Inactive State	If a UE supporting the RRC_INACTIVE state has no data to transmit before the inactivity timer for the UE to enter the RRC_INACTIVE state expires, the UE enters the RRC_INACTIVE state. The UE Power Saving Based on Inactive State feature supports QCI-specific configuration of the inactivity timer for the UE to enter the RRC_INACTIVE state. When the length of this inactive timer is set to shorter value, the duration in which the UE is in the RRC_CONNECTED state is reduced, thereby saving UE power.

4 Time Domain

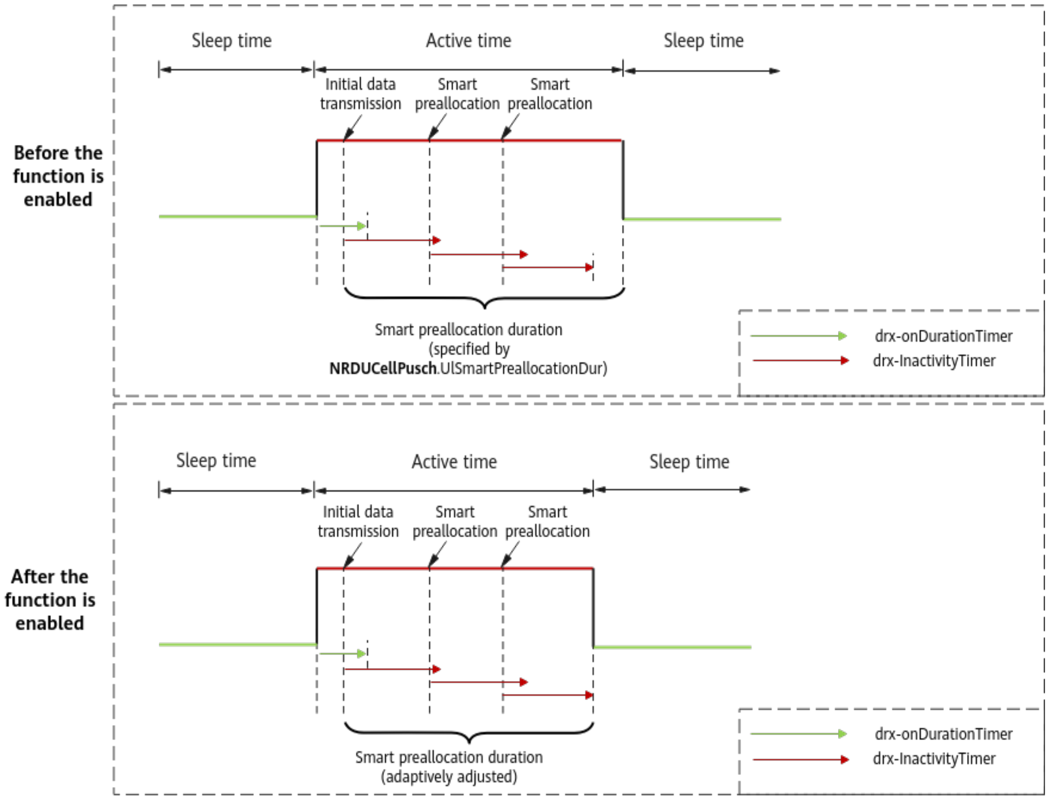
4.1 Adaptation of the Uplink Smart Preallocation Duration

4.1.1 Principles

Basic preallocation reduces the duration from when a UE sends a scheduling request (SR) to when the UE obtains the uplink scheduling grant. When basic preallocation takes effect, a UE is scheduled as long as there are available resources, regardless of whether the UE has service requests. Basic preallocation consumes an excessive amount of air interface resources and leads to a short UE standby time. To balance scheduling delay and UE power saving, smart preallocation is introduced. When smart preallocation is used, the **NRDUCellPusch**.***UlSmartPreallocationDur*** parameter can be used to specify the uplink smart preallocation duration. If the duration during which uplink smart preallocation is in effect reaches the parameter value and a UE does not have service requests, the UE enters the DRX sleep time, reducing UE power consumption and prolonging the UE's standby time. For details about preallocation and smart preallocation, see *Uplink Scheduling*.

To further reduce UE power consumption, adaptation of the uplink smart preallocation duration is introduced. This function is controlled by the **UL_SMART_PREALLOCATION_OPT_SW** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter. If this option is selected, **smart preallocation is triggered only when there is valid downlink scheduling data (excluding MAC CEs and padding data), which reduces the frequency of triggering smart preallocation. In addition, the preallocation duration is adaptively adjusted based on the uplink response time of downlink scheduling to reduce UE power consumption,** as shown in **Figure 4-1**.

Figure 4-1 DRX and uplink smart preallocation (duration adaptation)



For a fast track through the principles of smart preallocation, see **(Video) Time-Domain UE Power Saving - Smart Preallocation**.

4.1.2 Network Analysis

4.1.2.1 Benefits

After adaptation of the uplink smart preallocation duration is enabled, UEs consume less power than when only uplink smart preallocation is enabled. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

4.1.2.2 Impacts

The impacts between this function and other functions are described in **4.1.3.2.3 Function Impacts**.

Enabling adaptation of the uplink smart preallocation duration may have the following impacts:

- In light- or medium-load scenarios where the uplink PRB usage is less than 50%, the average number of PRBs used by UEs in uplink preallocation (indicated by **N.PRB.UL.Prealloc.Used.Avg**) and the uplink PRB usage decrease.
- In light- or medium-load scenarios where the uplink PRB usage is less than 50%, adaptation of the uplink smart preallocation duration reduces the preallocation duration. As a result, uplink data may be transmitted through normal scheduling instead of preallocation.
  - If most of the UEs are near the cell center, the average uplink user-perceived rate of all UEs increases.
  - If most of them are cell edge users (CEUs), the average uplink user-perceived rate of all UEs decreases.
- In light- or medium-load scenarios where the uplink PRB usage is less than 50%, uplink interference to neighboring cells decreases.
- As the number of preallocation times decreases, so does the total volume of uplink data received at the MAC layer in a cell (indicated by **N.ThpVol.UL.Cell**), but the traffic volume at the RLC layer does not decrease.

4.1.3 Requirements

4.1.3.1 Licenses

None

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Uplink smart preallocation	**UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	Adaptation of the uplink smart preallocation duration requires uplink smart preallocation to be enabled.

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink smart preallocation	**PUSCH_TIME_DOM_ADAPT_FILTER_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Uplink Scheduling*	After adaptation of the uplink smart preallocation duration is enabled, the base station adaptively adjusts the duration during which preallocation is in effect.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Others

None

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Preallocation Switch	**NRDUCellPusch.*UlPreallocationSwitch***	UL_SMART_PREALLOCATION_OPT_SW	Select this option.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to **4.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling adaptation of the uplink smart preallocation duration
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPreallocationSwitch=UL_SMART_PREALLOCATION_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling adaptation of the uplink smart preallocation duration
MOD NRDUCELLPUSCH: NrDuCellId=0, UlPreallocationSwitch=UL_SMART_PREALLOCATION_OPT_SW-0;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

If the average number of PRBs used by UEs in uplink preallocation (indicated by **N.PRB.UL.Prealloc.Used.Avg**) decreases after adaptation of the uplink smart preallocation duration is enabled, this function has taken effect.

4.1.4.3 Network Monitoring

After adaptation of the uplink smart preallocation duration is enabled, monitor the network performance based on the counters described in **4.1.2.2 Impacts**.

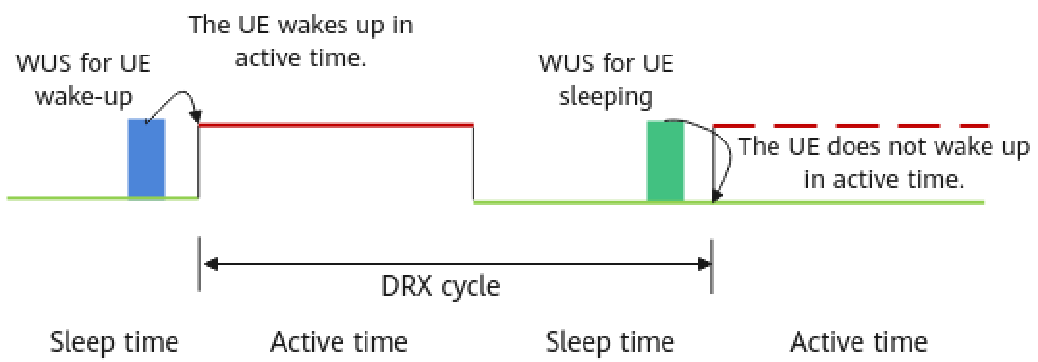
4.2 WUS

4.2.1 Principles

When DRX takes effect, UEs do not monitor the PDCCH during the DRX sleep time. DRX significantly reduces the power consumption of UEs in connected mode by reducing the number of PDCCH monitoring times. However, in each DRX active time, the UE has to wake up to monitor the PDCCH. When the traffic volume of a UE is small, the scheduling probability is low. Waking up in each cycle consumes excessive UE power.

To address this issue, the wake-up signal (WUS) technology was introduced in 3GPP Release 16. As shown in **Figure 4-2**, **a WUS DCI format 2_6 is sent to the UE before the UE enters the On Duration. This signal indicates whether the UE needs to monitor PDCCH scheduling information while the DRX On Duration Timer is running. In this way, the UE can wake up when required**. Because the length of a WUS in the time domain is shorter than the On Duration Timer, this method can save UE power. This function takes effect only on low-frequency UEs in SA networking for power saving. WUS described in this document does not apply to RedCap UEs. For details about the similar function supported by RedCap UEs, see descriptions of RedCap DRX adaptation in *RedCap*.

Figure 4-2 WUS



The WUS function is controlled by the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **DRX_ADAPTATION_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter. The settings of these options take effect only on UEs that newly access or are newly handed over to the cell.

The process of this function is as follows:

1. The UE and base station exchange capability information.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The **drx-Adaptation-r16** field in the message indicates that the UE supports WUS.

2. The base station delivers the DCI format 2_6 configuration.

   If the WUS function is enabled, the base station delivers the DCI format 2_6 configuration to UEs that support the capability indicated by **drx-Adaptation-r16**. Such configuration includes:

   – ps-Offset-r16: indicates the interval between the start position of a DCI format 2_6 and the start of the On Duration Timer corresponding to the long DRX cycle. Based on this information, the UE can calculate when to detect the DCI format 2_6.
   – sizeDCI-2-6-r16: indicates the length of a DCI format 2_6.
   – ps-PositionDCI-2-6: indicates the specific bit occupied by a DCI format 2_6 for a UE.
   – ps-WakeUp-r16: indicates that a UE needs to wake up to monitor the PDCCH during the On Duration of the next long DRX cycle if the UE does not receive a DCI format 2_6. The value is always **TRUE**.
   – ps-TransmitOtherPeriodicCSI-r16: indicates that the UE needs to report periodic CSI measurement results during the DRX sleep time specified by the On Duration timer. This field is carried only when the **SLEEP_DRXOD_CSI_RPT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected.

3. DCI format 2_6 is used to indicate whether a UE needs to wake up or can remain to sleep in the next On Duration. This applies only to the On Duration of a long DRX cycle, not that of a short DRX cycle.

   When there is no data to be sent in the RLC buffer, the base station instructs the UE to sleep through the DCI format 2_6 with a bit value of **0**. When there is data to be transmitted in the RLC buffer, the base station instructs the UE to wake up through the DCI format 2_6 with a bit value of **1**. If the UE is instructed to sleep, it does not need to wake up to monitor the PDCCH in the On Duration of the next long DRX cycle, and does not send the SRS in the On Duration period.

For a fast track through the principles of WUS, see **(Video) Time-Domain UE Power Saving**.

Blacklist and Whitelist Control Over WUS

The WUS function supports the blacklist and whitelist mechanisms.

- If the **DRX_ADAPTATION_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter is selected, the WUS function is disabled for specified UEs. That is, this function does not take effect on the specified UEs and can take effect on other UEs.
- If the **DRX_ADAPTATION_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the WUS function is enabled for specified UEs. That is, this function can take effect on the specified UEs and does not take effect on other UEs.

The **DRX_ADAPTATION_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **DRX_ADAPTATION_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

4.2.2 Network Analysis

4.2.2.1 Benefits

The WUS function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

When the WUS function is enabled and the **SLEEP_DRXOD_CSI_RPT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, the delay may decrease and the downlink user-perceived rate may increase, compared with when this option is deselected.

Assume that the WUS function is enabled. When the **DRX_ADAPTATION_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter or the **DRX_ADAPTATION_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the number of UEs with WUS in effect decreases, the BLER decreases, delay becomes shorter, the average cell and UE throughputs increase, the service drop rate decreases, and the power consumption of UEs without WUS in effect increases.

4.2.2.2 Impacts

The impacts between this function and other functions are described in **4.2.3.2.3 Function Impacts**.

After the WUS function is enabled, the following may occur:

- Decrease in average uplink and downlink cell throughput
- Significant decrease in average uplink and downlink UE throughput
- Increase in delay
- Increase or decrease in the uplink and downlink initial block error rates (IBLERs)/residual block error rates (RBLERs)
- Increase in the service drop rate

After the **SLEEP_DRXOD_CSI_RPT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, UE power consumption increases and the downlink user-perceived rate increases. However, if the CSI measurement results are inaccurate, the downlink user-perceived rate may decrease.

The involved indicators are as follows:

Counter Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Delay	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%

Counter Description	Formula
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink RBLER	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%
Service drop rate	**N.PDUSession.AbnormRel**/(**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) x 100%

4.2.3 Requirements

4.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030210	UE Power Saving	NR0S00UEPS00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00UEPS00: ● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. ● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld**.***LicenseChkSw*** parameter.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	The WUS function takes effect only when DRX is enabled.

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PEI	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter	*UE Power Saving*	The transmission of the DCI format 2_7 may affect the transmission of the WUS DCI format 2_6 and fallback DCI scheduling. As a result, the proportion of occasions where WUS takes effect decreases.
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	When High-speed Railway Superior Experience is enabled, WUS does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*CA Experience Improvement*	WUS does not take effect on a UE with inter-gNodeB CA in effect. If WUS has taken effect on a UE before inter-gNodeB CA, the UE exits WUS before inter-gNodeB CA can take effect.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	WUS does not take effect on UEs running VoNR services. Once a UE with WUS in effect is about to run a VoNR service, WUS will stop taking effect.
Low-frequency TDD	UL and DL decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** set to **ON**	*UL and DL Decoupling*	WUS does not take effect on a UE with inter-base-station SUL in effect. If WUS has taken effect on a UE before inter-base-station SUL, the UE exits WUS before inter-base-station SUL can take effect.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Cells

For a TDD cell, the **NRDUCell**.***SlotAssignment*** parameter cannot be set to **3_7_DSUUUUSUUU**.

4.2.3.5 Others

UEs must support long DRX cycles and the **drx-Adaptation-r16** capability defined in 3GPP Release 16.

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-2 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Power Saving Switch	**NRDUCellUePwrSaving**.***UePowerSavingSwitch***	UE_PWR_SAVING_ENHANCE_SW	Select this option if WUS is required.

Parameter Name	Parameter ID	Option	Setting Notes
NR DU Cell DRX Algorithm Switch	**NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch***	DRX_ADAPTATION_SW	Select this option if WUS is required.
NR DU Cell DRX Algorithm Switch	**NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch***	SLEEP_DRXOD_CSI_RPT_SW	Select this option if CSI measurement result reporting during the DRX sleep time specified by the On Duration timer is required.
Blacklist Control Switch	**gNBUeCompatSw**.***BlacklistCtrlSwitch***	DRX_ADAPTATION_SW_OFF	Select this option if a blacklist mechanism needs to be used to disable the WUS function for specified UEs.
Whitelist Control Switch	**gNBUeCompatSw**.***WhitelistCtrlExtSwitch***	DRX_ADAPTATION_SW_ON	Select this option if a whitelist mechanism needs to be used to enable the WUS function for specified UEs.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to **4.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- The **MOD GNBUECOMPATSW** command is a high-risk command. Running this command changes the feature activation states for specified UEs.
- The **DRX_ADAPTATION_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **DRX_ADAPTATION_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

Activation Command Examples

```
//Turning on the UE power saving enhancement switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-1;
```



```
//Enabling the WUS function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, NrDuCellDrxAlgoSwitch=DRX_ADAPTATION_SW-1;
//Selecting the SLEEP_DRXOD_CSI_RPT_SW option
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, NrDuCellDrxAlgoSwitch=SLEEP_DRXOD_CSI_RPT_SW-1;
//Disabling the WUS function for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=DRX_ADAPTATION_SW_OFF-1;
//Enabling the WUS function for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=DRX_ADAPTATION_SW_ON-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the WUS function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, NrDuCellDrxAlgoSwitch=DRX_ADAPTATION_SW-0;
//Deselecting the SLEEP_DRXOD_CSI_RPT_SW option
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, NrDuCellDrxAlgoSwitch=SLEEP_DRXOD_CSI_RPT_SW-0;
//Turning off the UE power saving enhancement switch. This switch also controls other functions. Before
performing this operation, check whether this switch needs to be turned off.
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-0;
//Disabling the blacklist mechanism of WUS
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=DRX_ADAPTATION_SW_OFF-0;
//Disabling the whitelist mechanism of WUS
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=DRX_ADAPTATION_SW_ON-0;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Observe the Uu interface signaling tracing result.

Step 1 Create a Uu interface tracing task on the MAE-Access. Enable a UE that supports the **drx-Adaptation-r16** capability defined in 3GPP Release 16 to initially access the network.

Step 2 Check that RRC_RECFG signaling is displayed in the Uu interface tracing result.

If the RRC_RECFG message contains **PhysicalCellGroupConfig->dcp-Config-r16**, the WUS function has taken effect. Otherwise, the WUS function does not take effect.

----End

4.2.4.3 Network Monitoring

After this function is enabled, monitor the value changes of the following, as described in **4.2.2.2 Impacts**:

- Decrease in average uplink and downlink cell throughput
- Significant decrease in average uplink and downlink UE throughput
- Increase in delay
- Possible increase or decrease in the uplink and downlink initial block error rates (IBLERs)/residual block error rates (RBLERs)
- Increase in the service drop rate

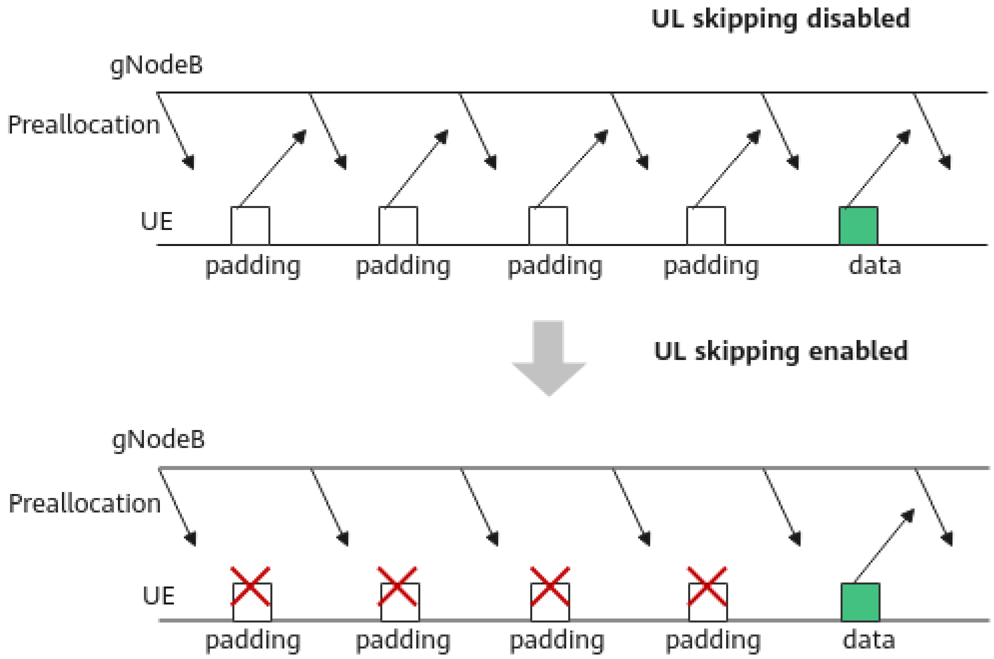
Check the values of **N.Cdrx.Sleep.Dur.Total** (total duration of RRC_CONNECTED UEs in the sleep time in a cell) and **N.Cdrx.Active.Dur.Total** (total duration of RRC_CONNECTED UEs in the active time in a cell) to evaluate the power saving effect of UEs.

4.3 UL Skipping

4.3.1 Principles

DRX saves UE power but increases service delay, affecting user experience. To reduce the impact of power saving on delay, uplink smart preallocation is enabled together with DRX. However, after uplink smart preallocation is enabled, a UE must respond to the base station with padding packets upon receiving UL grants even if the UE has no data to transmit, consuming excessive UE power. As such, 3GPP specifications proposed UL skipping for UE power saving. **For a UE supporting UL skipping, sending padding packets is not required if the UE has no data to transmit after it receives UL grants from the base station**, as shown in **Figure 4-3**. **In this way, power consumption is reduced**. This function takes effect only on low-frequency UEs in SA networking for power saving. UL skipping described in this document does not apply to RedCap UEs. For details about the similar function supported by RedCap UEs, see descriptions of RedCap UL skipping in *RedCap*.

Figure 4-3 UL skipping



The UL skipping function is controlled by the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **UPLINK_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter. The settings of these options take effect only on UEs that newly access or are newly handed over to the cell.

The process of this function is as follows:

1. The UE and base station exchange capability information.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The UE indicates the UL skipping capability by using the **SkipUplinkTxDynamic**, **enhancedSkipUplinkTxDynamic-r16**, or **enhancedSkipUplinkTxDynamic-v1660** field.
2. The base station delivers RRC configurations to UEs on which preallocation can take effect, notifying the UEs that UL skipping can take effect.
3. When the base station sends a UL grant to the UE after delivering RRC configurations to the UE, the UE does not send any padding packets if it has no data to transmit. In this way, power saving is achieved.

Blacklist and Whitelist Control Over UL Skipping

This UL skipping function supports the blacklist and whitelist mechanisms.

- If the **UPLINK_SKIP_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter is selected, the UL skipping function is disabled for specified UEs. That is, this function does not take effect on the specified UEs and can take effect on other UEs.

- If the **UPLINK_SKIP_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the UL skipping function is enabled for specified UEs. That is, this function can take effect on the specified UEs and does not take effect on other UEs.

The **UPLINK_SKIP_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UPLINK_SKIP_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

4.3.2 Network Analysis

4.3.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

Assume that the UL skipping function is enabled. When the **UPLINK_SKIP_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter or the **UPLINK_SKIP_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the number of UEs with UL skipping in effect decreases, the BLER decreases, the delay becomes shorter, the average cell and UE throughputs increase, the service drop rate decreases, and the power consumption of UEs without UL skipping in effect increases.

4.3.2.2 Impacts

The impacts between this function and other functions are described in **4.3.3.2.3 Function Impacts**.

Enabling UL skipping may have the following impacts:

- The proportion of padding packets in a cell and the data rate at the MAC layer decrease.
- For UEs enabled with UL skipping, BWP switching can be performed only through RRC connection reconfiguration. In addition, the decrease in padding packets may cause false or missing DTX detection. Therefore, UL skipping may increase the delay.
- The number of uplink padding packets sent by UEs decreases. As a result, the number of DTX detections increases, and the number of times PUSCH RSRP values fall into the range of index 0 indicated by **N.UL.PUSCH.RSRP.Index0** increases.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The service drop rate increases.

The involved indicators are as follows:

Counter Description	Formula
Delay	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%
Downlink RBLER	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) x 100%
Service drop rate	**N.PDUSession.AbnormRel**/ (**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) x 100%

4.3.3 Requirements

4.3.3.1 Licenses

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td>FOFD-030210</td><td>UE Power Saving</td><td>NR0S00UEPS00</td><td>per Cell</td></tr>
<tr><td colspan="4">For the license control item NR0S00UEPS00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
– When the license sales number pre-check function is enabled, the UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures.
– When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.
The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink preallocation	• **UL_BASIC_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter • **UL_SMART_PREALLOCATION_SWITCH** option of the **NRDUCellPusch**.***UlPreallocationSwitch*** parameter	*Uplink Scheduling*	UL skipping takes effect only after uplink preallocation takes effect.

4.3.3.2.2 Mutually Exclusive Functions

None

4.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic uplink slot aggregation	**REDCAP_UL_COVERAGE_ENH_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	UL skipping does not take effect on RedCap UEs with dynamic uplink slot aggregation in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain adaptive filtering	**PUSCH_TIME_DOM_ADAPT_FILTER_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Time-domain adaptive filtering does not take effect on UEs with UL skipping in effect.
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	When High-speed Railway Superior Experience is enabled, UL skipping does not take effect.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	UL skipping does not take effect on UEs running VoNR services. When a UE with UL skipping in effect is about to run a VoNR service, UL skipping will stop taking effect.

4.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.3.4 Others

UEs must support **SkipUplinkTxDynamic**, **enhancedSkipUplinkTxDynamic-r16**, or **enhancedSkipUplinkTxDynamic-v1660**.

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

Table 4-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Power Saving Switch	**NRDUCellUePwrSaving**.***UePowerSavingSwitch***	UE_PWR_SAVING_ENHANCE_SW	Select this option when the UL skipping function needs to be enabled.
Scheduling Power Saving Switch	**NRDUCellUePwrSaving**.***SchPwrSavingSw***	UPLINK_SKIP_SW	Select this option when the UL skipping function needs to be enabled.
Blacklist Control Switch	**gNBUeCompatSw**.***BlacklistCtrlSwitch***	UPLINK_SKIP_SW_OFF	Select this option if a blacklist mechanism needs to be used to disable the UL skipping function for specified UEs.
Whitelist Control Switch	**gNBUeCompatSw**.***WhitelistCtrlExtSwitch***	UPLINK_SKIP_SW_ON	Select this option if a whitelist mechanism needs to be used to enable the UL skipping function for specified UEs.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3.3.2 Functions** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- The **MOD GNBUECOMPATSW** command is a high-risk command. Running this command changes the feature activation states for specified UEs.
- The **UPLINK_SKIP_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UPLINK_SKIP_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

Activation Command Examples

```
//Turning on the UE power saving enhancement switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-1;
//Enabling the UL skipping function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, SchPwrSavingSw=UPLINK_SKIP_SW-1;
//Disabling the UL skipping function for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,BlacklistCtrlSwitch=UPLINK_SKIP_SW_OFF-1;
//Enabling the UL skipping function for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UPLINK_SKIP_SW_ON-1;
```

Deactivation Command Examples

```
//Disabling the UL skipping function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, SchPwrSavingSw=UPLINK_SKIP_SW-0;
//Turning off the UE power saving enhancement switch. This switch also controls other functions. Before
performing this operation, check whether this switch needs to be turned off.
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-0;
//Disabling the blacklist mechanism of UL skipping
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,BlacklistCtrlSwitch=UPLINK_SKIP_SW_OFF-0;
//Disabling the whitelist mechanism of UL skipping
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UPLINK_SKIP_SW_ON-0;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

Observe the Uu interface signaling tracing result.

Step 1 Create a Uu interface tracing task on the MAE-Access. Enable a UE with the **SkipUplinkTxDynamic** capability defined in 3GPP Release 15 or the **enhancedSkipUplinkTxDynamic-r16** or **enhancedSkipUplinkTxDynamic-v1660** capability defined in 3GPP Release 16 to initially access the network.

Step 2 Check that RRC_RECFG signaling is displayed in the Uu interface tracing result.

If the value of the **MAC-CellGroupConfig** > **SkipUplinkTxDynamic** field in the RRC_RECFG message is **True** or the RRC_RECFG message contains the **MAC-CellGroupConfig** > **enhancedSkipUplinkTxDynamic-r16** field, the UL skipping function has taken effect. Otherwise, this function has not taken effect.

----End

4.3.4.3 Network Monitoring

After this function is enabled, the value of **N.ThpVol.UL.Cell** (total volume of uplink data received at the MAC layer in a cell) decreases and that of **N.ThpVol.UL** (total volume of uplink data received at the RLC layer in a cell) remains basically unchanged.

4.4 PDCCH Skipping

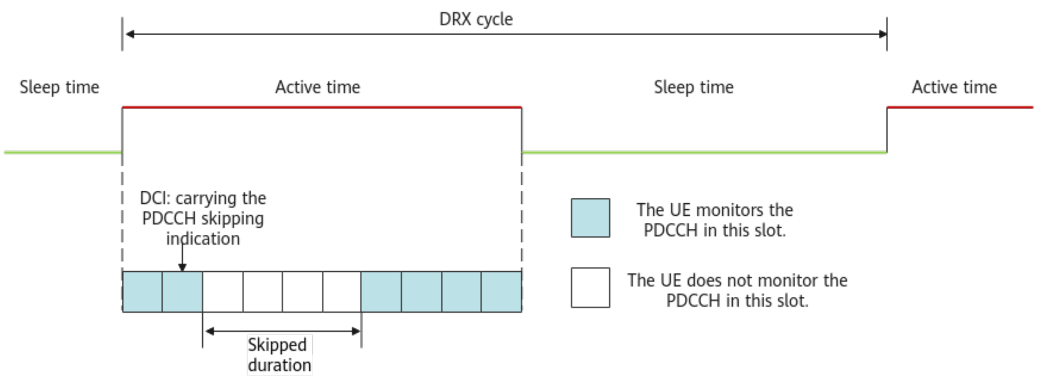
4.4.1 Principles

When DRX takes effect, UEs do not monitor the PDCCH during the DRX sleep time. DRX significantly reduces the power consumption of UEs in connected mode by reducing the number of PDCCH monitoring times. However, in the DRX active time, a UE monitors the PDCCH in each downlink slot. In heartbeat/sparse packet scenarios, if the base station does not send a PDCCH scheduling indication, monitoring the PDCCH still causes a waste of UE power.

To address this issue, the PDCCH skipping function is introduced in 3GPP Release 17. **For a UE that supports PDCCH skipping, the base station can use a DCI to instruct the UE not to monitor the PDCCH in several slots following the DCI, thereby saving power**. According to 3GPP specifications, the duration when monitoring the PDCCH is not required is called the skipped duration.

PDCCH skipping can also be used in non-DRX scenarios. However, it is recommended that DRX be enabled first because DRX provides more power saving gains. If further power saving is required, enable this function to reduce UE power consumption in DRX active time. Figure 4-4 shows the working principle of PDCCH skipping in DRX mode.

Figure 4-4 PDCCH skipping in DRX mode (using a 4-slot skipped duration as an example)



The PDCCH skipping function is controlled by the **PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter. The setting of this option takes effect only for UEs that newly access or are newly handed over to the cell.

This function is implemented as follows:

1. The base station determines whether the UE supports PDCCH skipping.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The UE indicates that it supports the PDCCH skipping capability by using the **pdcch-SkippingWithoutSSSG-r17** or **pdcch-SkippingWithSSSG-r17** field.

   NOTE

   The difference between **pdcch-SkippingWithoutSSSG-r17** and **pdcch-SkippingWithSSSG-r17** is whether the UE supports search space set group (SSSG) switching.

2. The base station indicates the skipped duration through an RRC reconfiguration message.

   The base station uses the **pdcch-SkippingDurationList-r17** field in the RRC reconfiguration message to indicate the skipped duration. The base station separately delivers skipped duration for each BWP. The skipped duration is specified by the **NRDUCellUePwrSaving**.***PdcchSkippingDuration*** parameter. If this parameter is set to **DEFAULT**, the skipped duration is 2 slots for BWP1 UEs, 4 slots for BWP2 UEs, and 4 slots for Redcap UEs. If this parameter is set to other values, the skipped duration of all UEs is the configured value. For example, if this parameter is set to **S1**, the skipped duration of all UEs is 1 slot. For details about the BWP1 and BWP2, see **5.1 Power Saving BWP (Low-Frequency TDD)**.

3. The base station sends the UE the DCI, which indicates whether the UE needs to subsequently monitor the PDCCH.

   Based on the service status, the base station indicates whether the UE subsequently needs to monitor the PDCCH by using the **PDCCH monitoring adaptation indication** field in the DCI. If the DCI indicates that the UE does not need to monitor the PDCCH, the UE does not monitor the PDCCH within the subsequent skipped duration. Otherwise, this function does not restrict the UE from monitoring the PDCCH.

4.4.2 Network Analysis

4.4.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

4.4.2.2 Impacts

The impacts between this function and other functions are described in **4.4.3.2.3 Function Impacts**.

Enabling PDCCH skipping may have the following impacts:

- The average uplink and downlink cell throughputs and the average uplink and downlink UE throughputs decrease.
- The delay of downlink data packets increases.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The service drop rate increases.
- When PDCCH skipping takes effect, a shorter skipped duration results in fewer slots in which the PDCCH does not need to be monitored, fewer power saving gains, a smaller increase in delay, and a smaller decrease in throughput; a longer skipped duration results in the opposite effects.

The involved indicators are as follows:

Counter Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Delay	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%

Counter Description	Formula
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink RBLER	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/ (**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Service drop rate	**N.PDUSession.AbnormRel**/ (**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) × 100%

4.4.3 Requirements

4.4.3.1 Licenses

There are no license requirements for basic functions.

4.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.3.2.1 Prerequisite Functions

None

4.4.3.2.2 Mutually Exclusive Functions

None

4.4.3.2.3 Function Impacts

Impact relationships with coordination-related functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	When Co-MM and PDCCH skipping are both enabled, PDCCH skipping does not take effect on Co-MM UEs.
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When intra-base-station UL CoMP and PDCCH skipping are both enabled, PDCCH skipping does not take effect on UL CoMP UEs.
Low-frequency TDD	Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When intra-base-station DL CoMP and PDCCH skipping are both enabled, PDCCH skipping does not take effect on DL CoMP UEs.
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	When both coordinated interference management and PDCCH skipping are enabled, PDCCH skipping does not take effect on CS/CBF UEs.

Impact relationships with CA-related functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both downlink intra-band CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both uplink intra-band CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both downlink intra-FR inter-band CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both uplink intra-FR inter-band CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*CA Experience Improvement*	With both inter-gNodeB CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3CC aggregation	**NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL3CC**	*CA Experience Improvement*	With both 3CC aggregation and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink fast intra-gNodeB CA	**CA_ULTRA_LOW_LATENCY_SPLIT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*CA Experience Improvement*	With both downlink fast intra-gNodeB CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink fast inter-gNodeB CA	**INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*CA Experience Improvement*	With both downlink fast inter-gNodeB CA and PDCCH skipping enabled: ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. ● If the **CA_UE_PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

Impact relationships with URLLC-related functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When the low latency and high reliability function and PDCCH skipping are both enabled, PDCCH skipping does not take effect on UEs performing low latency and high reliability services.

Impact relationships with other functions are as follows.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SUL CoMP	**INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter	*UL and DL Decoupling*	When both SUL CoMP and PDCCH skipping are enabled, PDCCH skipping does not take effect on SUL UEs.
TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	PDCCH skipping does not take effect on UEs running VoNR services. When a UE with PDCCH skipping in effect is about to run a VoNR service, PDCCH skipping will stop taking effect.
Low-frequency TDD	Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	**4.7 Time-Domain Power Saving BWP (Low-Frequency TDD)**	Time-domain power saving BWP does not take effect on UEs with PDCCH skipping in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG Switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	**4.5 SSSG Switching**	Assume that both PDCCH skipping and SSSG switching are enabled. If a UE reports both **pdcch-SkippingWithoutSSSG-r17** and **SSSG-Switching-1BitInd-r17** but not **pdcch-SkippingWithSSSG-r17**, only SSSG switching takes effect on the UE.
Low-frequency TDD	RedCap SSSG switching	**REDCAP_SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter **REDCAP_SSSG_SWITCH_EXT_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*RedCap*	Assume that both PDCCH skipping and RedCap SSSG switching are enabled. If a UE reports both **pdcch-SkippingWithoutSSSG-r17** and **SSSG-Switching-1BitInd-r17** but not **pdcch-SkippingWithSSSG-r17**, only RedCap SSSG switching takes effect on the UE.

4.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.4.3.4 Others

UEs must support **pdcch-SkippingWithoutSSSG-r17** or **pdcch-SkippingWithSSSG-r17**.

4.4.4 Operation and Maintenance

4.4.4.1 Data Configuration

4.4.4.1.1 Data Preparation

Table 4-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Scheduling Power Saving Switch	**NRDUCellUePwrSaving**.***SchPwrSavingSw***	PDCCH_SKIP_SW	Select this option when the PDCCH skipping function needs to be enabled.
PDCCH Skipping Duration	**NRDUCellUePwrSaving**.***PdcchSkippingDuration***	None	Retain the default value.

4.4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling the PDCCH skipping function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, SchPwrSavingSw=PDCCH_SKIP_SW-1;
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, PdcchSkippingDuration=DEFAULT;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the PDCCH skipping function
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, SchPwrSavingSw=PDCCH_SKIP_SW-0;
```

4.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4.2 Activation Verification

Observe the signaling tracing result.

Step 1 Enable a UE that supports **pdcch-SkippingWithoutSSSG-r17** or **pdcch-SkippingWithSSSG-r17** to initially access the network or initiate a handover procedure.

Step 2 Observe the RRC connection reconfiguration message in the signaling tracing result on the MAE-Access.

If the RRC connection reconfiguration message contains "PDCCH-Config > pdcch-SkippingDurationList-r17", PDCCH skipping has taken effect. Otherwise, PDCCH skipping has not taken effect.

----End

4.4.4.3 Network Monitoring

After this function is enabled, observe the **N.UePowerSaving.PdcchSkip.Dur** counter to obtain the total duration of PDCCH skipping for all UEs in a cell.

4.5 SSSG Switching

4.5.1 Principles

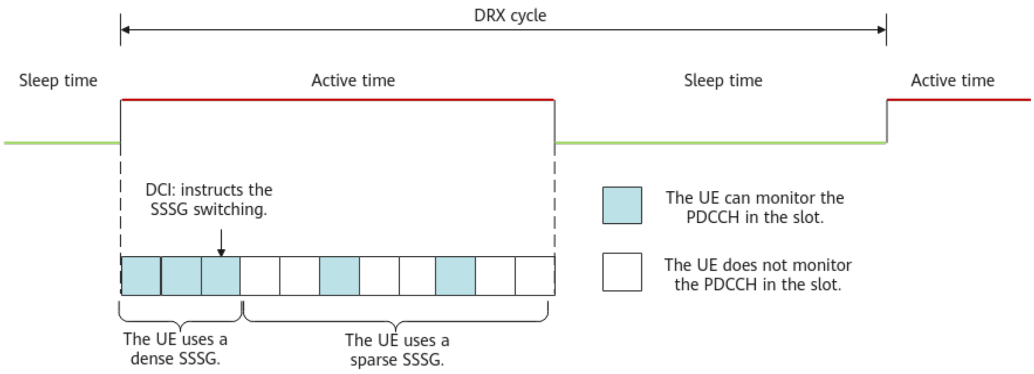
When DRX takes effect, UEs do not monitor the PDCCH during the DRX sleep time. This mechanism significantly reduces the power consumption of UEs in the

RRC_CONNECTED state by minimizing the frequency of PDCCH monitoring. However, in the DRX active time, a UE monitors the PDCCH in each downlink slot. In heartbeat/sparse packet scenarios, if the base station does not send a PDCCH scheduling indication, monitoring the PDCCH still causes a waste of UE power.

To further reduce UE power, 3GPP Release 17 proposed the search space set group (SSSG) switching function. A base station supports multiple search space sets to form different SSSGs. PDCCH monitoring densities of the search space may vary with the SSSG. An SSSG with a high PDCCH monitoring density is referred to as a dense SSSG, and an SSSG with a low PDCCH monitoring density is referred to as a sparse SSSG. **For a UE that supports SSSG switching, the base station can instruct the UE to switch from a dense SSSG to a sparse SSSG by using DCI. This reduces PDCCH monitoring occasions and saves power.** SSSG switching described in this document does not apply to RedCap UEs. For details about the similar function supported by RedCap UEs, see descriptions of RedCap SSSG switching in *RedCap*.

SSSG switching can also be used in non-DRX scenarios. However, it is recommended that DRX be enabled preferentially because DRX provides more power saving gains. If further power saving is required, enable this function to reduce UE power consumption in DRX active time. **Figure 4-5** shows the principle for SSSG switching during DRX operation.

Figure 4-5 SSSG switching in DRX mode



SSSG switching is controlled by both the **UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter. The level involved in SSSG switching can be configured using the **NRDUCellUePwrSaving**.***SssgSwitchingLevel*** parameter. Compared with **LEVEL0**, **LEVEL1** indicates that the SSSG is more sparse and the energy saving effect is better.

This function is implemented as follows:

1. The base station determines whether the UE supports SSSG switching.

   After the base station sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the base station. The UE may indicate support for SSSG switching by using a **pdcch-SkippingWithSSSG-r17**, **SSSG-Switching-1BitInd-r17**, or **SSSG-Switching-2BitInd-r17** field.
2. The base station indicates an SSSG set by using an RRC reconfiguration message.

For UEs that support SSSG switching, the **searchSpaceGroupIDList-r17** field is delivered through the RRC reconfiguration message to indicate the SSSG set that can be used by the UE.

3. The base station instructs, by using the DCI, the UE to switch between SSSGs.

   Based on the service status, the base station indicates to the UE which SSSG in the aforementioned set should be activated through the **PDCCH monitoring adaptation indication** field in the DCI, completing the SSSG switching. For details about the switching conditions and methods, see **Switching from a Dense SSSG to a Sparse SSSG** and **Switching from a Sparse SSSG to a Dense SSSG**.

Switching from a Dense SSSG to a Sparse SSSG

The base station determines whether the UE meets all of the following conditions every 1s. If so, it instructs the UE to switch from a dense SSSG to a sparse SSSG.

- The downlink RLC throughput of a UE is lower than threshold A, and the downlink RLC buffer data volume is less than threshold B.
- The uplink RLC throughput of the UE is less than threshold C, and the uplink BSR data volume is less than threshold D.

For details about the methods for calculating thresholds for BWP1 UEs and BWP2 UEs, see **Table 4-5**. In addition, BWP2 UEs exist only when the power saving BWP function is enabled (by selecting the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter).

Table 4-5 Thresholds for switching from a dense SSSG to a sparse SSSG

Threshold	Meaning	Calculation Method for BWP1 UEs	Calculation Method for BWP2 UEs
Threshold A	Threshold of downlink throughput for switching from a dense SSSG to a sparse SSSG	min{Downlink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***DnsToSparseSssgDlThptThld***}	min{Downlink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld*** x **NRDUCellUePwrSaving**.***Bwp2DnsToSparseSssgThldFac***}

Threshold	Meaning	Calculation Method for BWP1 UEs	Calculation Method for BWP2 UEs
Threshold B	Threshold of downlink traffic volume for switching from a dense SSSG to a sparse SSSG	min{**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.3, Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** x 0.1}	min{Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum*** x **NRDUCellUePwrSaving**.***Bwp2DnsToSparseSssgThldFac***, **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.5 x (BWP2 bandwidth/BWP1 bandwidth) x **NRDUCellUePwrSaving**.***Bwp2DnsToSparseSssgThldFac***}
Threshold C	Threshold of uplink throughput for switching from a dense SSSG to a sparse SSSG	min{Uplink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***DnsToSparseSssgUlThptThld***}	min{Uplink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld*** x **NRDUCellUePwrSaving**.***Bwp2DnsToSparseSssgThldFac***}
Threshold D	Threshold of uplink traffic volume for switching from a dense SSSG to a sparse SSSG	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum*** x 0.1	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum*** x **NRDUCellUePwrSaving**.***Bwp2DnsToSparseSssgThldFac***

Switching from a Sparse SSSG to a Dense SSSG

The base station determines whether the UE meets either of the following conditions at a granularity of ms. If so, it instructs the UE to switch from a sparse SSSG to a dense SSSG.

- The downlink RLC buffer data volume of a UE is greater than or equal to threshold E.
- The uplink BSR data volume of a UE is greater than or equal to threshold F.

For details about the methods for calculating thresholds for BWP1 UEs and BWP2 UEs, see **Table 4-6**. In addition, BWP2 UEs exist only when the power saving BWP function is enabled (by selecting the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.**BwpPwrSavingSw** parameter).

Table 4-6 Thresholds for switching from a sparse SSSG to a dense SSSG

Threshold	Meaning	Calculation Method for BWP1 UEs	Calculation Method for BWP2 UEs
Threshold E	Threshold of downlink traffic volume for switching from a sparse SSSG to a dense SSSG	min{**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.4, Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** x 0.2}	min{Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** x **NRDUCellUePwrSaving**.***Bwp2SparseToDnsSssgThldFac***, **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x (BWP2 bandwidth/BWP1 bandwidth) x **NRDUCellUePwrSaving**.***Bwp2SparseToDnsSssgThldFac***}
Threshold F	Threshold of uplink traffic volume for switching from a sparse SSSG to a dense SSSG	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum*** x 0.15	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum*** x **NRDUCellUePwrSaving**.***Bwp2SparseToDnsSssgThldFac***

4.5.2 Network Analysis

4.5.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

4.5.2.2 Impacts

The impacts between this function and other functions are described in **4.5.3.2.3 Function Impacts**.

Enabling SSSG switching may have the following impacts:

- The average uplink and downlink cell throughputs and the average uplink and downlink UE throughputs decrease.

- Uplink and downlink cell edge throughputs decrease.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The delay increases.
- The CCE usage decreases.
- The uplink and downlink CCE allocation success rates decrease.
- The QoS flow drop rate of all services increases.
- The values of the counters related to the uplink and downlink RB usage may increase or decrease.
- The RRC connection setup success rate fluctuates.

The involved indicators are as follows:

Indicator Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Uplink cell edge throughput	(**N.ThpVol.UL.CellOverlap** – **N.ThpVol.UL.SmallPkt.CellOverlap**)/ **N.ThpTime.UL.RmvSmallPkt.CellOverlap**
Downlink cell edge throughput	(**N.ThpVol.DL.CellOverlap** – **N.ThpVol.DL.LastSlot.CellOverlap**)/ **N.ThpTime.DL.RmvLastSlot.CellOverlap**
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%

Indicator Description	Formula
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink RBLER	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Average transmission delay of the first packet	**N.Traffic.DL.RlcFirstPktDelay.Time**/**N.Qos.DL.RlcFirstPktDelay.Num**
Average air interface processing delay of downlink data packets	**N.QoS.DL.PktDelayAirInterface.Time**/**N.QoS.DL.PktDelayAirInterface.Num**
Average RLC processing delay of downlink data packets	**N.QoS.DL.PktDelaygNBDU.Time**/**N.QoS.DL.PktDelaygNBDU.Num**
Average gNodeB DU processing delay of downlink data packets	**N.QoS.DL.PktDelayAirInterface.Time**/**N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/**N.QoS.DL.PktDelaygNBDU.Num**
CCE usage	**N.CCE.Used.Avg**/**N.CCE.Avail.Avg**
Uplink CCE allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num** × 100%

Indicator Description	Formula
Downlink CCE allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/ **N.CCE.DL.AllocReq.Num** × 100%
Service drop rate	**Service Call Drop Rate (CU)**
Downlink PRB usage in a cell	**Downlink Resource Block Utilizing Rate (DU)**
Uplink PRB usage in a cell	**Uplink Resource Block Utilizing Rate (DU)**
Downlink PDSCH DRB PRB usage	**N.PRB.DL.DrbUsed.Avg**/ **N.PRB.DL.Avail.Avg**
Uplink PUSCH DRB PRB usage	**N.PRB.UL.DrbUsed.Avg**/ **N.PRB.UL.PUSCH.Avail.Avg**
Average PUSCH PRB usage	**N.PRB.PUSCH.Used.Avg**/ **N.PRB.UL.PUSCH.Avail.Avg**
RRC connection setup success rate	**RRC Setup Success Rate (CU)**

4.5.3 Requirements

4.5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030210	UE Power Saving	NR0S00UEPS00	per Cell

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td colspan="4">For the license control item NR0S00UEPS00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
– When the license sales number pre-check function is enabled, the UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures.
– When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.
The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Enhanced UE Power Saving Switch	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	The **SSSG_SWITCHING_SW** option can be selected only when the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter is selected.

4.5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	IAB deployment	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	If **NRDUCell**.***DeployPosition*** is set to **DEPLOY_IAB**, SSSG switching must be disabled.

4.5.3.2.3 Function Impacts

Impact relationships with coordination-related functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	UEs for which Co-MM has taken effect will not switch from a dense search space set group (SSSG) to a sparse one. For Co-MM to take effect for a UE for which a sparse SSSG is in effect, the UE must switch to a dense SSSG.
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	UEs for which intra-base-station UL CoMP has taken effect will not switch from a dense SSSG to a sparse one. For intra-base-station UL CoMP to take effect for a UE for which a sparse SSSG is in effect, the UE must switch to a dense SSSG.
Low-frequency TDD	Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	UEs for which intra-base-station DL CoMP has taken effect will not switch from a dense SSSG to a sparse one. For intra-base-station DL CoMP to take effect for a UE for which a sparse SSSG is in effect, the UE must switch to a dense SSSG.

Impact relationships with CA-related functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both downlink intra-band CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both uplink intra-band CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both downlink intra-FR inter-band CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	With both uplink intra-FR inter-band CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*CA Experience Improvement*	With both inter-gNodeB CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.
Low-frequency TDD	3CC aggregation	**NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL3CC**	*CA Experience Improvement*	With both 3CC aggregation and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink fast intra-gNodeB CA	**CA_ULTRA_LOW_LATENCY_SPLIT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*CA Experience Improvement*	With both downlink fast intra-gNodeB CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.
Low-frequency TDD	Downlink fast inter-gNodeB CA	**INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*CA Experience Improvement*	With both downlink fast inter-gNodeB CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	In inter-FR CA, a sparse SSSG can be configured for PCCs but not for SCCs, and the SSSG switching function cannot be enabled for high-frequency SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.
Low-frequency TDD	Massive CA	The **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter and **NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL4CC** or **DL5CC**	*CA Experience Improvement*	With both massive CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.

Impact relationships with URLLC-related functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	UEs with low latency and high reliability in effect do not switch to a sparse SSSG.

Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH rate matching	• **PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter • **PDCCH_NO_PATTERN_RATE_MATCH_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Downlink Scheduling*	UEs for which PDCCH rate matching is in effect do not switch to a sparse SSSG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Super uplink	**FLEX_SUPER_UPLINK_SW** and **INTER_GNB_SUPER_UPLINK_SW** options of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	UEs with super uplink in effect do not switch to a sparse SSSG.
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	UEs running VoNR services do not switch to a sparse SSSG.
Low-frequency TDD	FWA	**gNBQciBearer**.***ServiceType*** set to **FWA_INTERNET**, **FWA_VIDEO**, or **FWA_PRIVATE_LINE**	*FWA*	UEs running FWA services do not switch to a sparse SSSG.
Low-frequency TDD	Live streaming UE identification	**gNBQciBearer**.***ServiceType*** set to **EMBB_LIVE_VIDEO**	*Ultimate Video Experience*	Live streaming UEs do not switch to a sparse SSSG.
Low-frequency TDD	Ultimate Cloud X experience	**gNBQciBearer**.***ServiceType*** set to **EMBB_CLOUD_VR**, **EMBB_CLOUD_GAMING**, or **EMBB_CLOUD_PHONE**	*Ultimate Cloud X Experience*	UEs with ultimate Cloud X experience in effect do not switch to a sparse SSSG.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-level threshold control for BWP switching	**QCI_CA_BWP_SWITCH_THLD_CTRL_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*UE Power Saving*	If the **QCI_CA_BWP_SWITCH_THLD_CTRL_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected, the QCI-specific switching thresholds are configured for BWP2 UEs, and some cell-level parameters in the threshold calculation methods (**Table 4-5** and **Table 4-6**) for BWP2 UEs in SSSG switching are replaced with QCI-specific parameters. The parameter replacement principles are as follows: • **NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld*** is replaced with **NRDUCellQciBearer**.***QciBwp1ToBwp2DlThptThld***. • **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** is replaced with **NRDUCellQciBearer**.***QciDlStTransitBufVolThld***.

RAT	Function Name	Function Switch	Reference	Description
				• **NRDUCellUePwrSaving**.*Bwp1ToBwp2UlThptThld* is replaced with **NRDUCellQciBearer**.*QciBwp1ToBwp2UlThptThld*.
Low-frequency TDD	Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.*BwpPwrSavingSw* parameter	**4.7 Time-Domain Power Saving BWP (Low-Frequency TDD)**	If the **SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.*UePowerSavingSwitch* is selected for an SSSG-switching-capable UE, only SSSG takes effect for the UE. The time-domain power saving BWP function takes effect for UEs only when the **SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.*UePowerSavingSwitch* is deselected and the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.*UePowerSavingSwitch* parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	4.4 PDCCH Skipping	Assume that both PDCCH skipping and SSSG switching are enabled. If a UE reports both **pdcch-SkippingWithoutSSSG-r17** and **SSSG-Switching-1BitInd-r17** but not **pdcch-SkippingWithSSSG-r17**, only SSSG switching takes effect on the UE.

4.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.5.3.4 Cells

For a TDD cell, the **NRDUCell**.***SlotAssignment*** parameter must be set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.

4.5.3.5 Others

UEs must support the capabilities indicated by **pdcch-SkippingWithSSSG-r17**, **SSSG-Switching-1BitInd-r17**, or **SSSG-Switching-2BitInd-r17**.

4.5.4 Operation and Maintenance

4.5.4.1 Data Configuration

4.5.4.1.1 Data Preparation

Table 4-7 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Power Saving Switch	**NRDUCellUePwrSaving.*UePowerSavingSwitch***	UE_PWR_SAVING_ENHANCE_SW	Select this option if SSSG switching is required.
UE Power Saving Switch	**NRDUCellUePwrSaving.*UePowerSavingSwitch***	SSSG_SWITCHING_SW	Select this option if SSSG switching is required.
SSSG Switching Level	**NRDUCellUePwrSaving.*SssgSwitchingLevel***	None	Use the recommended value.
Dense-to-Sparse SSSG SW DL Throughput Thld	**NRDUCellUePwrSaving.*DnsToSparseSssgDlThptThld***	None	Use the recommended value.
Dense-to-Sparse SSSG SW UL Throughput Thld	**NRDUCellUePwrSaving.*DnsToSparseSssgUlThptThld***	None	Use the recommended value.
BWP2 Dense-to-Sparse SSSG SW Thld Factor	**NRDUCellUePwrSaving.*Bwp2DnsToSparseSssgThldFac***	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
BWP2 Sparse-to-Dense SSSG SW Thld Factor	**NRDUCellUePwrSaving**.***Bwp2SparseToDnsSssgThldFac***	None	Use the recommended value.
BWP1 to BWP2 DL Throughput Threshold	**NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld***	None	Use the recommended value.
BWP1 to BWP2 UL Throughput Threshold	**NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld***	None	Use the recommended value.
CA Deactivate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum***	None	Use the recommended value.
CA Deactivate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum***	None	Use the recommended value.
CA Activate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum***	None	Use the recommended value.
CA Activate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum***	None	Use the recommended value.
DL State Transition Buffer Volume Threshold	**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld***	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Switch	**NRDUCellPdcch**.***PdcchAlgoSwitch***	PDCCH_TYPE3_CSS_CONFIG_SW	It is recommended that this option be selected to increase the proportion of occasions where SSSG switching takes effect when a sparse SSSG can take effect only in Type-3 CSS for UEs.

4.5.4.1.2 Using MML Commands

Before using MML commands, refer to **4.5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the UE power saving enhancement switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-1;
//Turning on the SSSG switching switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=SSSG_SWITCHING_SW-1;
//Configuring the SSSG switching level
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, SssgSwitchingLevel=LEVEL0;
//Configuring the thresholds related to SSSG switching for non-RedCap UEs
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, DnsToSparseSssgDlThptThld=5000,
DnsToSparseSssgUlThptThld=1000, Bwp2DnsToSparseSssgThldFac=30, Bwp2SparseToDnsSssgThldFac=50,
Bwp1ToBwp2DlThptThld=1000, Bwp1ToBwp2UlThptThld=250;
//Setting the parameters based on the recommended values for low-frequency NR TDD scenarios
MOD NRDUCELLCARRMGMT: NrDuCellId=1, CaDlScellDeactThldSlotNum=4, CaUlScellDeactThldSlotNum=1,
CaDlScellActThldSlotNum=16, CaUlScellActThldSlotNum=4, DlStateTransitBufVolThld=20;
//(Recommended) Turning on PDCCH_TYPE3_CSS_CONFIG_SW when a sparse SSSG can take effect only in
Type-3 CSS for UEs
MOD NRDUCELLPDCCH: NrDuCellId=1,PdcchAlgoSwitch=PDCCH_TYPE3_CSS_CONFIG_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the SSSG switching switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=SSSG_SWITCHING_SW-0;
//Turning off the UE power saving enhancement switch. This switch also controls other functions. Before
performing this operation, check whether this switch needs to be turned off.
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-0;
```

4.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.4.2 Activation Verification

Observe the signaling tracing result.

Step 1 Enable a UE that supports **pdcch-SkippingWithSSSG-r17**, **SSSG-Switching-1BitInd-r17**, or **SSSG-Switching-2BitInd-r17** to initially access the network or initiate a handover.

Step 2 Observe the RRC connection reconfiguration message in the signaling tracing result on the MAE-Access.

If the RRC connection reconfiguration message contains "SearchSpace> searchSpaceGroupIDList-r17", an SSSG has been configured for the UE. Otherwise, no SSSG has been configured for the UE, and SSSG switching cannot take effect for the UE.

----End

The following table lists the related counters.

If the value of **N.User.SSSG.Sparse.Num.Avg** is not 0, some UEs have switched to a sparse SSSG.

Counter Name	Indicator Description
N.User.SSSG.Sparse.Num.Avg	Average number of UEs for all sparse SSSGs in a cell

4.5.4.3 Network Monitoring

The performance indicators described in **4.5.2.2 Impacts** can be used for network monitoring of this function.

4.6 PEI (Low-Frequency TDD)

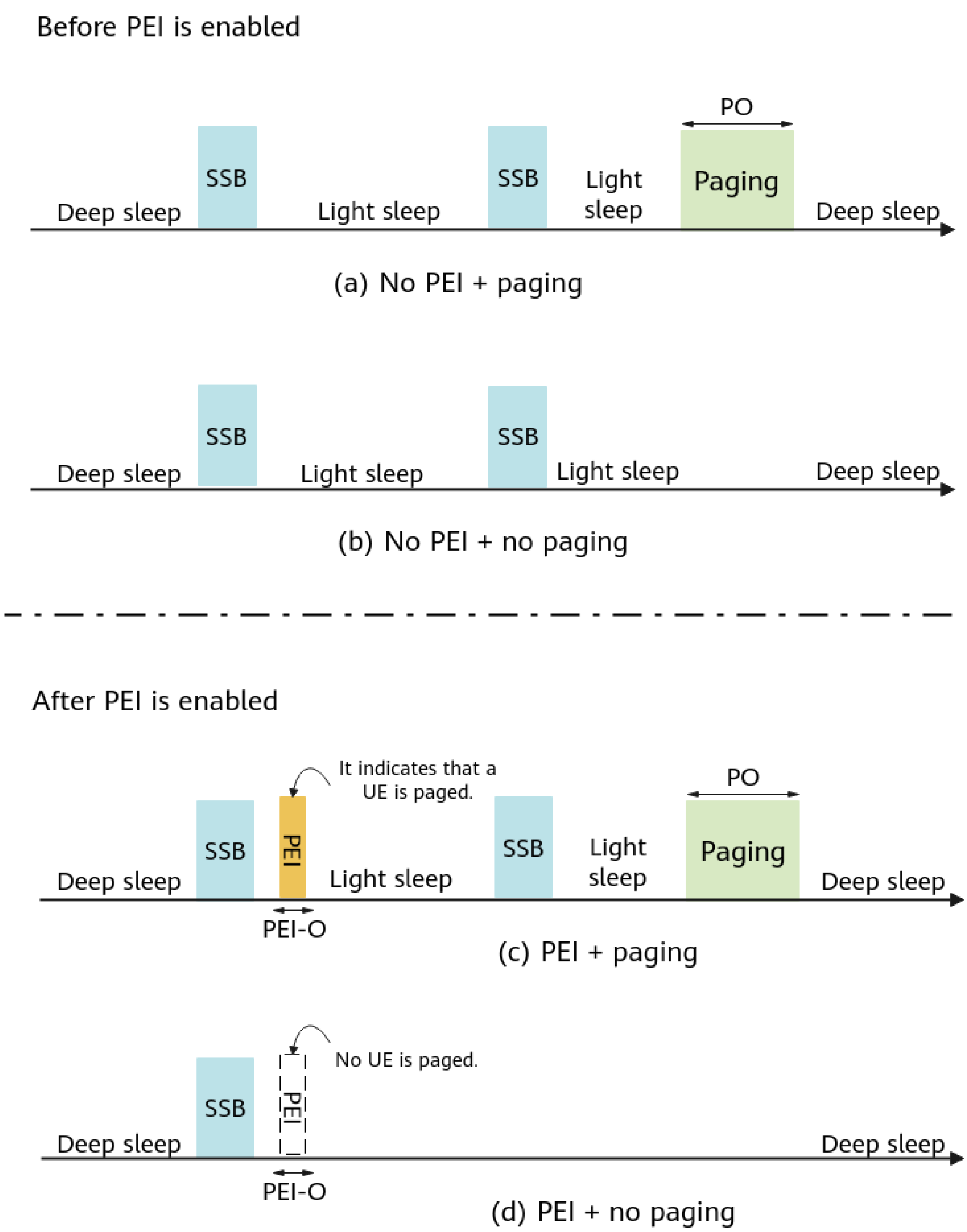
4.6.1 Principles

In each paging cycle, a UE needs to wake up before a paging occasion (PO) to receive a certain number of synchronization signal and PBCH blocks (SSBs) for downlink time and frequency synchronization, no matter whether there is a paging message for the UE, as shown by a and b in **Figure 4-6**. If there are no paging messages for the UE to receive in the paging cycle, UE waking-up wastes power.

To address this issue, 3GPP Release 17 introduces the paging early indication (PEI) technology. **If a PEI-capable UE needs to be paged on a PO, the gNodeB sends a PEI WUS DCI format 2_7 on the paging early indication occasion (PEI-O)**

corresponding to the PO, to indicate whether the UE needs to wake up on the PO to receive the paging message, as shown by c and d in **Figure 4-6**. The PEI function enables RRC_IDLE or RRC_INACTIVE UEs to wake up as required, reducing unnecessary waking-up and power consumption.

Figure 4-6 Schematic diagram of PEI

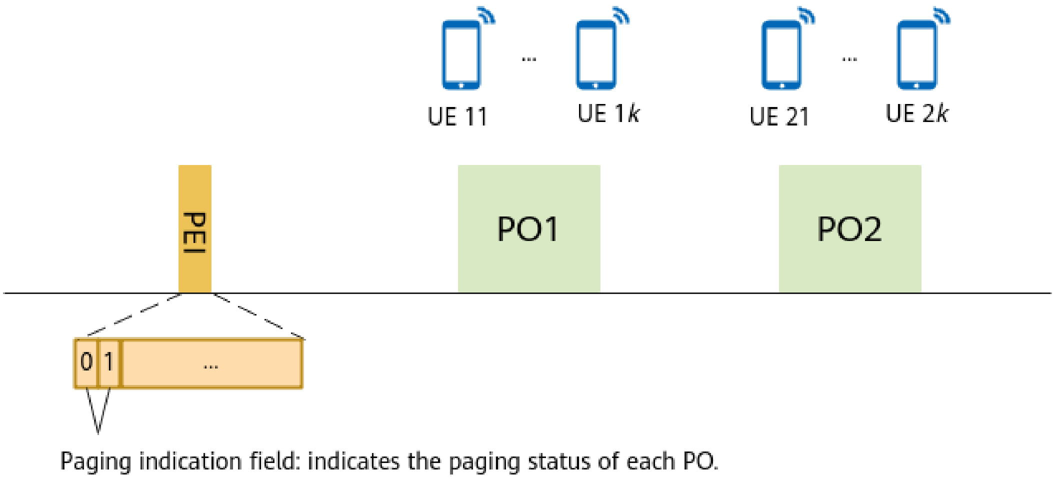


The PEI function is controlled by the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter. The settings of these options take effect only on UEs that newly access or are newly handed over to the cell.

PEI works as follows:

1. The gNodeB broadcasts PEI information in SIB1. The major IE is *pei-Config-r17*. For details about the configuration information in *pei-Config-r17*, see 3GPP TS 38.331 Release 17.
2. The gNodeB obtains information about whether a UE supports PEI in the following way: After the gNodeB sends a UECapabilityEnquiry message to the UE, the UE returns a UECapabilityInformation message to the gNodeB. Specifically, the UE reports its PEI capability through the **UE-RadioPagingInfo-r17** field.
3. If a PEI-capable UE needs to be paged on a PO, the gNodeB sends a PEI WUS DCI format 2_7 on the PEI-O corresponding to the PO. If no PEI-capable UE needs to be paged on a PO, the gNodeB does not send the PEI WUS. Specifically, the first two bits of the Paging indication field in the DCI format 2_7 correspond to two POs. If a UE needs to be paged on a PO, the corresponding bit is set to **1**. If no UE needs to be paged on a PO, the corresponding bit is set to **0**.

Figure 4-7 PEI WUS



4. Subsequently, when the UE returns to the RRC_IDLE or RRC_INACTIVE state, if the UE detects a PEI WUS DCI format 2_7 on the PEI-O and the bit corresponding to a PO in the Paging indication field is **0**, each UE to be paged on the PO does not need to wake up on the PO to receive the paging message. If the UE does not detect DCI format 2_7 or the bit corresponding to a PO in the Paging indication field of the DCI format 2_7 is **1**, each UE to be paged on the PO needs to wake up on the PO to receive the paging message.

Paging Subgroup

When the PEI paging load is high, that is, at least one PEI-capable UE is paged on each PO, the PEI associated with each PO carries a WUS, a UE that does not need to be paged on a PO also wakes up on the PO to receive a paging message, and the power of the UE is still wasted to some extent. To address this issue, the paging subgroup function is introduced.

The paging subgroup function is controlled by the **PAGING_SUBGROUP_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter. After this function is enabled, as shown in **Figure 4-8**, the gNodeB divides PEI-capable UEs on the same PO into eight subgroups based on UE_IDs. **Each bit in the Paging indication field of the PEI WUS DCI format 2_7 is used to indicate whether UEs in individual subgroups in the PO associated with the PEI need to wake up to receive paging messages**.

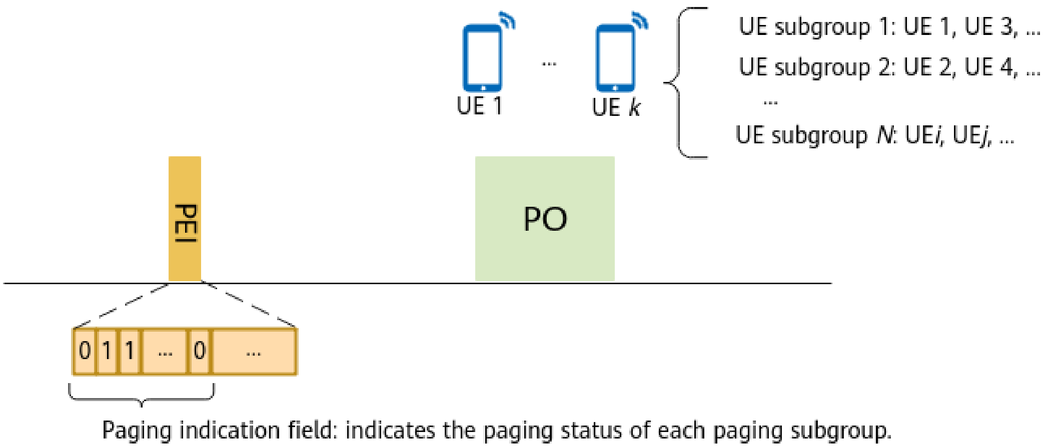
The implementation on the gNodeB side is as follows:

On a PO, if a to-be-paged UE supports PEI and belongs to subgroup *N*, the gNodeB sets the bit corresponding to subgroup *N* in the Paging indication field of the DCI format 2_7 to **1**. For other subgroups that are not paged, the gNodeB sets the bits corresponding to the subgroups to **0**. If none of the UEs to be paged on a PO supports PEI, the gNodeB does not send the PEI WUS DCI format 2_7 on PEI-O corresponding to the PO.

The implementation on the UE side is as follows:

When a UE receives a DCI format 2_7 and the bit corresponding to the subgroup to which the UE belongs in the Paging indication field is **1**, the UE wakes up on a corresponding PO to receive a paging message, and all UEs in the subgroup wake up on the PO. Otherwise, none of the UEs in the subgroup wakes up on the PO.

Figure 4-8 Schematic diagram of paging subgroup



4.6.2 Network Analysis

4.6.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

4.6.2.2 Impacts

The impacts between this function and other functions are described in **4.6.3.2.3 Function Impacts**.

- After this function is enabled, the paging delay of certain PEI-capable UEs paged in the interval between a PEI and PO may increase by one paging cycle, prolonging the average access delay.
- If PEI is enabled when the random access load is high, there is a higher probability that PEIs conflict with random access responses (RARs). If RAR scheduling is delayed due to the conflicts, but the service load of a neighboring cell is high or common signaling such as the remaining minimum system information (RMSI) of the neighboring cell is scheduled in slots where the delayed RAR scheduling is performed, the neighboring cell will cause interference to RARs of the local cell, affecting the probability of successful RAR demodulation. As a result, the contention-based and non-contention-based random access success rates decrease and the access delay increases. The more UEs that access the network, the greater the impact.
- The WUS DCI format 2_7 consumes CORESET 0 resources. As a result, the average number of PDCCH CCEs used for common DCI (indicated by **N.CCE.CommDCI.Used.Avg**) increases, and the average aggregation level may increase. It is recommended that PEI be enabled when the CCE usage of a cell (indicated by **N.CCE.Used.Avg**/**N.CCE.Avail.Avg**) is lower than 30%.
- The DCI format 2_7 increases the downlink CCE resource failure rate, increases the proportion of downlink CCEs, decreases the total number of uplink CCEs, and decreases the uplink CCE allocation success rate (adaptive uplink-to-downlink CCE ratio adjustment). In symbol adaptation scenarios, the probability of symbol expansion increases.
- When a DCI format 2_7 is sent in power aggregation mode, the number of available CCEs may decrease.

After this function is enabled, the values of the following indicators may increase:

Counter Description	Formula
PDCCH CCE usage	(**N.CCE.Used.Avg**/**N.CCE.Avail.Avg**) x 100%
Average number of PDCCH CCEs used for common DCI	**N.CCE.CommDCI.Used.Avg**
Average number of PDCCH CCEs used for downlink DCI	**N.CCE.DLDCI.Used.Avg**
Number of times of successful PDCCH downlink DCI allocations with aggregation level 8	**N.CCE.DL.AggLvl8Num**
Average number of used PDCCH CCEs	**N.CCE.Used.Avg**

When the PEI paging load on the network is heavy (The PEI WUS DCI format 2_7 is sent on the PEI-O corresponding to each PO), the values of the following indicators may decrease:

Counter Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Total number of scheduled UEs in the uplink in a cell	**N.User.Schedule.Ul.Sum**
Total number of scheduled UEs in the downlink in a cell	**N.User.Schedule.Dl.Sum**
Average number of used uplink PRBs	**N.PRB.UL.Used.Avg**
Average number of used downlink PRBs	**N.PRB.DL.Used.Avg**
Average number of used PUSCH PRBs	**N.PRB.PUSCH.Used.Avg**
Average number of PRBs used by PUSCH DRBs	**N.PRB.DL.DrbUsed.Avg**

The values of the following counters fluctuate, depending on the traffic model:

- Counters related to average number of times a specific MCS index is selected in the uplink, average uplink rank, and uplink block error rate:
  – **N.PUSCH.TbUl.Rank1**
  – **N.PUSCH.TbUl.Rank2**
  – **N.ChMeas.PUSCH.MCS.0** to **N.ChMeas.PUSCH.MCS.28**
  – **N.UL.SCH.QPSK.ErrTB.Rbler**
  – **N.UL.SCH.16QAM.ErrTB.Rbler**
  – **N.UL.SCH.64QAM.ErrTB.Rbler**
  – **N.UL.SCH.256QAM.ErrTB.Rbler**
  – **N.UL.SCH.QPSK.ErrTB.Ibler**
  – **N.UL.SCH.16QAM.ErrTB.Ibler**
  – **N.UL.SCH.64QAM.ErrTB.Ibler**
  – **N.UL.SCH.256QAM.ErrTB.Ibler**
- Counters related to average number of times a specific MCS index is selected in the downlink, average downlink rank, and downlink block error rate:
  – **N.PDSCH.NbrTbDl.Rank1**
  – **N.PDSCH.NbrTbDl.Rank2**
  – **N.PDSCH.NbrTbDl.Rank3**
  – **N.PDSCH.NbrTbDl.Rank4**

- **N.ChMeas.PDSCH.MCS.0** to **N.ChMeas.PDSCH.MCS.31**
- **N.DL.SCH.QPSK.ErrTB.Rbler**
- **N.DL.SCH.16QAM.ErrTB.Rbler**
- **N.DL.SCH.64QAM.ErrTB.Rbler**
- **N.DL.SCH.256QAM.ErrTB.Rbler**
- **N.DL.SCH.QPSK.ErrTB.Ibler**
- **N.DL.SCH.16QAM.ErrTB.Ibler**
- **N.DL.SCH.64QAM.ErrTB.Ibler**
- **N.DL.SCH.256QAM.ErrTB.Ibler**

PEI-related paging messages may fail to be delivered when the PEI function status is changing once the function switch setting is changed. As a result, the value of the **N.paging.Dis.PchCong** counter increases.

4.6.3 Requirements

4.6.3.1 Licenses

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td>FOFD-030210</td><td>UE Power Saving</td><td>NR0S00UEPS00</td><td>per Cell</td></tr>
<tr><td colspan="4">For the license control item NR0S00UEPS00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
– When the license sales number pre-check function is enabled, the UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures.
– When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.
The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.6.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
PEI	**PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter	*UE Power Saving*	The **PAGING_SUBGROUP_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter can be selected only when the **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter is selected.

4.6.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Configuring-based interference avoidance enhancement	**CFG_INTRF_AVOID_ENH_SW** option of the **NRDUCellIntrfIdent**.***DlIntrfAvoidSw*** parameter	None	If the **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter is selected, the **CFG_INTRF_AVOID_ENH_SW** option of the **NRDUCellIntrfIdent**.***DlIntrfAvoidSw*** parameter must be deselected.

Function Name	Function Switch	Reference	Description
Paging capacity mode	**NRDUCellPagingConfig**.***PagingCapacityMode***	*5G Networking and Signaling*	The PEI function cannot be enabled when the **NRDUCellPagingConfig**.***PagingCapacityMode*** parameter is set to **ENHANCE_MODE_1**.
Flexible Smart Dormancy	**FLEX_SMART_DORMANCY_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	None	The **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter cannot be selected when the **FLEX_SMART_DORMANCY_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter is selected.
Maximum Common DCI Power Offset	**NRDUCellChnPwr**.***MaxCommonDciPwrOffset***	*Power Control*	The **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter cannot be selected when the **NRDUCellChnPwr**.***MaxCommonDciPwrOffset*** parameter is set to value greater than **0**.

Function Name	Function Switch	Reference	Description
LTE TDD and NR spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter **gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** **gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex***	None	The **PEI_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter cannot be selected if the following conditions are met: (1) The **LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter is selected. (2) The **gNBDULteNrSpctShrCg**.***NrReservedRbEndIndex*** parameter value minus the **gNBDULteNrSpctShrCg**.***NrReservedRbStartIndex*** parameter value is less than 95.

4.6.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
RedCap DRX adaptation	**REDCAP_DRX_ADAPTATION_SW** or **REDCAP_DRX_ADAPTATION_EXT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*RedCap*	The transmission of the DCI format 2_7 may affect the transmission of the WUS DCI format 2_6 and fallback DCI scheduling. As a result, the proportion of occasions where RedCap DRX adaptation takes effect decreases.

Function Name	Function Switch	Reference	Description
eDRX in idle mode	**IDLE_MODE_EDRX_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter	*RedCap*	When the setting of the **PEI_SW** or **PAGING_SUBGROUP_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter changes, UEs with eDRX in idle mode in effect may not obtain the latest paging configuration information in a timely manner. As a result, they cannot receive paging messages for a maximum of three hours.
WUS	**DRX_ADAPTATION_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*UE Power Saving*	The transmission of the DCI format 2_7 may affect the transmission of the WUS DCI format 2_6 and fallback DCI scheduling. As a result, the proportion of occasions where WUS takes effect decreases.
Dynamic adjustment of the number of PFs	**PAGING_FRAME_NUM_DYN_ADJ_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When the function of dynamic adjustment of the number of PFs is enabled, the PEIs may be inaccurate during the adjustment of the number of paging frames. As a result, the PEI paging fails.

Function Name	Function Switch	Reference	Description
eDRX in idle mode	**IDLE_MODE_EDRX_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter	None	When the setting of the **PEI_SW** or **PAGING_SUBGROUP_SW** option of the **NRDUCellUePwrSaving**.***IdleUePwrSavingSwitch*** parameter changes, UEs with eDRX in idle mode in effect may not obtain the latest paging configuration information in a timely manner. As a result, they cannot receive paging messages for a maximum of three hours.

4.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.6.3.4 Cells

- The **NRDUCell**.***SlotAssignment*** parameter must be set to **4_1_DDDSU**, **8_2_DDDDDDDSUU**, or **8_2_DDDSUUDDDD**.

- The cell bandwidth is greater than or equal to 40 MHz.

4.6.3.5 Others

Requirements for UEs: UEs must support the **UE-RadioPagingInfo-r17** capability.

Requirements for the core network: The *Core Network Assistance Information for RRC INACTIVE* IE carries the **Extended UE Identity Index Value** field in the following messages over the NG interface when the UE is in RRC_INACTIVE mode: INITIAL CONTEXT SETUP REQUEST, UE CONTEXT MODIFICATION REQUEST, HANDOVER REQUEST, or PATH SWITCH REQUEST ACKNOWLEDGE.

4.6.4 Operation and Maintenance

4.6.4.1 Data Configuration

4.6.4.1.1 Data Preparation

Table 4-8 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Power Saving Switch	**NRDUCellUePwrSaving.*UePowerSavingSwitch***	UE_PWR_SAVING_ENHANCE_SW	Select this option if PEI is required.
Idle UE Power Saving Switch	**NRDUCellUePwrSaving.*IdleUePwrSavingSwitch***	PEI_SW	Select this option if PEI is required.
Idle UE Power Saving Switch	**NRDUCellUePwrSaving.*IdleUePwrSavingSwitch***	PAGING_SUBGROUP_SW	Select this option if paging subgroup is required.

4.6.4.1.2 Using MML Commands

Before using MML commands, refer to **4.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling PEI
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-1, IdleUePwrSavingSwitch=PEI_SW-1;
//Enabling paging subgroup
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=PAGING_SUBGROUP_SW-1;
```

Deactivation Command Examples

```
//Disabling paging subgroup
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=PAGING_SUBGROUP_SW-0;
//Disabling PEI
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, IdleUePwrSavingSwitch=PEI_SW-0;
//Turning off the UE power saving enhancement switch. This switch also controls other functions. Before performing this operation, check whether this switch needs to be turned off.
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-0;
```

4.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.6.4.2 Activation Verification

Tracing Signaling

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 In the navigation tree on the left of the displayed **Signaling Trace Management** page, choose **Uu Interface Trace**.

Step 3 On the displayed **Uu Interface Trace** page, check the RMSI message. If the RMSI message contains **downlinkConfigCommon->pei-Config-r17** and **pdcch-ConfigCommon->commonSearchSpaceListExt2-r17**, it indicates that PEI has taken effect. Otherwise, it indicates that PEI has not taken effect.

----End

Counter Observation

Observe the following counters to check whether the PEI function has taken effect. If the value of the counter listed in **Table 4-9** is not zero, the PEI function has taken effect.

Table 4-9 Performance counters for PEI

Counter ID	Counter Name
1911840399	**N.Paging.PEI.UU.Ue.Num**

4.6.4.3 Network Monitoring

The indicators listed in **4.6.2 Network Analysis** can be used to monitor the benefits and impacts of this function.

4.7 Time-Domain Power Saving BWP (Low-Frequency TDD)

4.7.1 Overview

PDCCH monitoring affects UE power consumption. The longer the PDCCH monitoring duration, the higher the UE power consumption. If a UE continuously monitors the PDCCH while its traffic volume is light, the UE power is wasted. To address this issue, the time-domain power saving BWP function is introduced. With this function, **a base station can adaptively switch the UE to a BWP with a low PDCCH monitoring density based on the UE traffic volume to save UE power**. The time-domain power saving BWP function is controlled by the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.

NOTE

A BWP is a part of continuous bandwidth resources configured by the network (base station) for a UE, allowing for flexible transmission bandwidth configurations. Based on the access status of a UE, BWPs can be categorized as initial BWP and dedicated BWP.

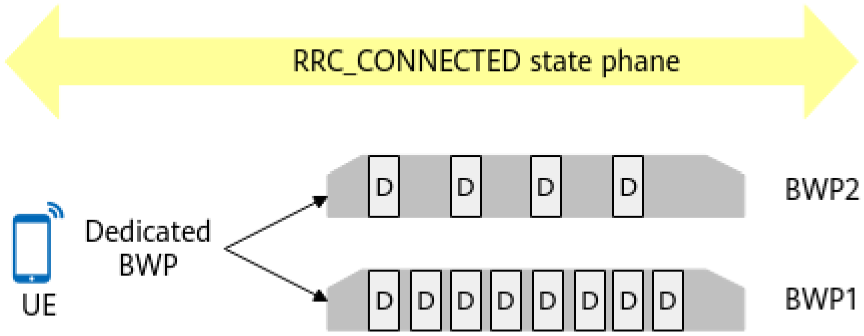
- An initial BWP is used for data transmission of the UE during initial access.
- A dedicated BWP is used for data transmission if the UE is in the RRC_CONNECTED state.

For details about BWP, see 3GPP TS 38.331, 3GPP TS 38.321, and 3GPP TS 38.213.

4.7.2 Principles

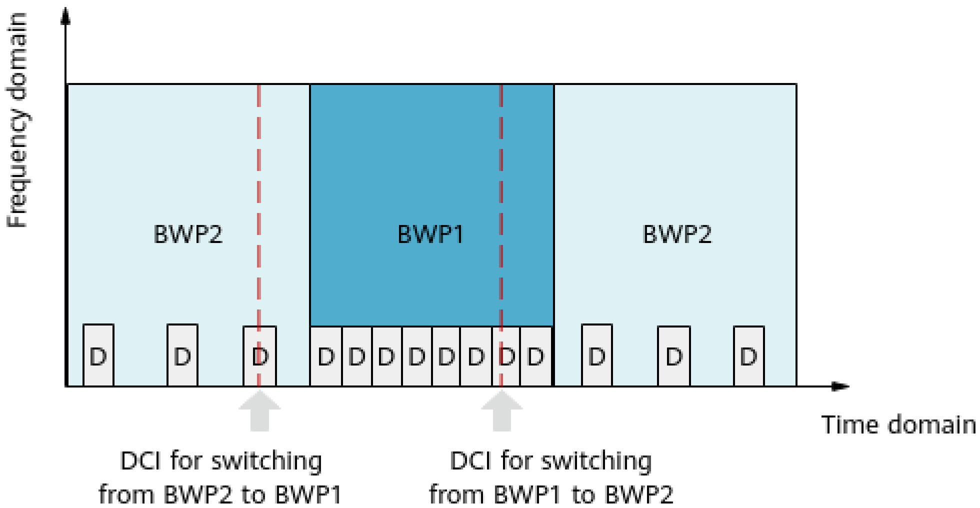
The time-domain power saving BWP function can configure two dedicated uplink BWPs and two dedicated downlink BWPs for a UE. Among the two dedicated BWPs, the PDCCH monitoring density of BWP1 is high, and that of BWP2 is low, as shown in **Figure 4-9**. In addition, the bandwidths of BWP1 and BWP2 are equal to the cell bandwidth, and the two BWPs completely overlap each other in the frequency domain. Dedicated BWPs must be configured separately in the uplink and downlink through RRC signaling. At a time, a UE must use the same kind of BWP, either BWP1 or BWP2, in both the uplink and downlink.

Figure 4-9 BWPs for a UE in the RRC_CONNECTED state



After the time-domain power saving BWP function takes effect, the base station and UE can switch between the BWP1 and BWP2 to select an appropriate BWP based on the UE traffic volume. As the PDCCH monitoring density of BWP2 is low, when the base station and UE switch to BWP2, the UE power consumption reduces. Currently, downlink control information (DCI) is used to instruct a UE to switch between two BWPs, as shown in **Figure 4-10**. DCI 0_1 (for uplink) or DCI 1_1 (for downlink) includes the field **Bandwidth part indicator**, which indicates the target BWP for switching.

Figure 4-10 DCI-based BWP switching



Switching from BWP1 to BWP2

The base station checks whether conditions for BWP switching are met in a period specified by **NRDUCellUePwrSaving**.***Bwp1ToBwp2EvaluatePeriod***. When all the following conditions are met, the base station and UE perform switching from BWP1 to BWP2.

- The downlink RLC throughput of the UE is less than threshold A, and the downlink RLC buffer data volume is less than threshold B.
- The uplink RLC throughput of the UE is less than threshold C, and the uplink BSR-indicated data volume is less than threshold D.

Table 4-10 Thresholds related to switching from BWP1 to BWP2

Threshold	Meaning	Calculation Method
Threshold A	Downlink UE throughput threshold for BWP1-to-BWP2 switching	min{Downlink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***Bwp1ToTdmBwp2DlThptThld***}
Threshold B	Downlink buffered traffic volume threshold for BWP1-to-BWP2 switching	min{Downlink throughput estimated based on radio channel quality, **NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum*** x 0.1, **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.3}
Threshold C	Uplink UE throughput threshold for BWP1-to-BWP2 switching	min{Uplink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***Bwp1ToTdmBwp2UlThptThld***}
Threshold D	Uplink traffic volume threshold for BWP1-to-BWP2 switching	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum*** x 0.1

Switching from BWP2 to BWP1

The base station checks whether conditions for BWP switching are met in a ms-level period (the period is set by the system and cannot be changed). When any one of the following conditions is met, the base station and the UE perform switching from BWP2 to BWP1.

- The downlink RLC buffer data volume of the UE is greater than or equal to threshold E.
- The uplink BSR-indicated data volume of the UE is greater than or equal to threshold F.

Table 4-11 Thresholds related to switching from BWP2 to BWP1

Threshold	Meaning	Calculation Method
Threshold E	Downlink buffered traffic volume threshold for BWP2-to-BWP1 switching	min{Downlink throughput estimated based on radio channel quality, **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** x 0.2,**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.4}

Threshold	Meaning	Calculation Method
Threshold F	Uplink traffic volume threshold for BWP2-to-BWP1 switching	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum*** x 0.15

4.7.2.1 Time-Domain Power Saving BWP2 Scheduling Failure Rate Threshold

In the time-domain power saving BWP function, the **NRDUCellUePwrSaving**.***TdmBwp2SchFailRateThld*** parameter is used to specify the threshold expressed as a probability of UE scheduling failures on BWP2. When there are a large number of UEs in a cell, the probability of UE scheduling failures on BWP2 is larger because of a lower PDCCH monitoring density of BWP2 and fewer scheduling occasions. Adjusting the value of **NRDUCellUePwrSaving**.***TdmBwp2SchFailRateThld*** can change the probability of switching between BWP2 and BWP1, to keep a balance between network performance and UE power consumption.

A smaller value of this parameter results in a higher probability that the probability of UE scheduling failures on BWP2 in a cell is higher than the value of this parameter. In this case, the base station dynamically adjusts the time-domain power saving BWP algorithm to increase the difficulty of switching UEs from BWP1 to BWP2 and decrease the difficulty of switching UEs from BWP2 to BWP1. In this way, more UEs can camp on BWP1, the network performance is better, but the power saving effect is worse. If this parameter is set to a larger value, UEs are more likely to be switched from BWP1 to BWP2, resulting in poorer network performance and better power saving effect.

4.7.2.2 Exit of Weak-Coverage UEs from BWP2

After the time-domain power saving BWP function is enabled, the probability that the BWP on the UE side is inconsistent with that on the base station side in weak coverage scenarios is higher. For example, the base station has switched to BWP2, but the UE still uses BWP1. In this case, the service drop rate increases. To prevent this, exit of weak-coverage UEs from BWP2 is introduced. This function is controlled by the **WEAK_COV_BWP2_EXIT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.

When this option is selected, the processing mechanism on the base station is as follows:

- If a UE initially accesses a cell and meets any of the following conditions, the UE is identified as a weak-coverage UE that initially accesses the network in UE power saving scenarios. In this case, BWP2 resource information will not be included in the RRC reconfiguration message for the UE.
  - The Msg3-based DMRS RSRP is less than the sum of values of **NRDUCellDmrs**.***InitDmrsRsrpWeakCovThld*** and **NRDUCellUePwrSaving**.***WeakCoverageThldOffset***.
  - The Msg3-based DMRS SINR is less than the sum of values of **NRDUCellDmrs**.***InitDmrsSinrWeakCovThld*** and **NRDUCellUePwrSaving**.***WeakCoverageThldOffset***.

- If BWP2 resource information is configured for a UE through RRC signaling and any of the following conditions is met, the UE will be identified as a weak-coverage UE in UE power saving scenarios and it will switch from BWP2 to BWP1.
  - The CQI measurement result periodically reported by the UE is less than the value of **NRDUCellUePwrSaving**.***WeakCoverageCqiThld***.
  - The uplink SINR is less than the value of **NRDUCellUePwrSaving**.***WeakCoverageUlSinrThld***.

4.7.2.3 QCI-Level Configuration for Time-Domain Power Saving BWP

The switch for the time-domain power saving BWP function is a cell-level switch. Therefore, this function cannot be disabled separately for specific services on the wide area network (WAN). **To avoid delay issues of UEs running low-latency services, QCI-level configuration for time-domain power saving BWP is introduced. With this function, time-domain power saving BWP can be disabled separately for QCI-specific services.**

This function is controlled by the **NRDUCellQciBearer**.***QciBwp2Ind*** parameter.

- If this parameter is set to **NOT_CONFIG** (default value), BWP2 is not configured for services with the specific QCI. This QCI-specific parameter does not take effect.
- If this parameter is set to **NOT_SUPPORT**, services with the specific QCI cannot use BWP2.
- If this parameter is set to **SUPPORT**, services with the specific QCI can use BWP2.

A UE may have multiple radio bearers and each bearer corresponds to a QCI. Different QCIs have different priorities (specified by the **gNBQciBearer**.***PriorityLevel*** parameter). When QCI-level configuration for time-domain power saving BWP takes effect on a UE, the value of the **NRDUCellQciBearer**.***QciBwp2Ind*** parameter corresponding to the highest-priority QCI takes effect.

4.7.2.4 Type of UEs (in Terms of the Networking Mode) on Which Time-Domain Power Saving BWP Can Take Effect

When the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the **NRDUCellUePwrSaving**.***Bwp2NetworkingMode*** parameter specifies the type of UEs (in terms of the networking mode) on which time-domain power saving BWP can take effect. If the **NRDUCellUePwrSaving**.***Bwp2NetworkingMode*** parameter is set to **SA_AND_NSA**, time-domain power saving BWP can take effect on both SA UEs and NSA UEs. If this parameter is set to **SA_ONLY**, time-domain power saving BWP can take effect only on SA UEs.

4.7.3 Network Analysis

4.7.3.1 Benefits

This function reduces the power consumption of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains. This function is most effective when UEs are performing services with a small amount of data. In other cases, the UE power saving effect is not as pronounced.

4.7.3.2 Impacts

The impacts between this function and other functions are described in **4.7.4.2.3 Function Impacts**.

This function may have the following impacts:

- The average uplink and downlink cell throughputs and the average uplink and downlink UE throughputs decrease.
- Uplink and downlink cell edge throughputs increase or decrease.
- The uplink and downlink IBLERs and RBLERs increase or decrease.
- The delay increases.
- The CCE usage decreases.
- The uplink and downlink CCE allocation success rates decrease.
- The QoS flow drop rate of all services increases.
- The values of the counters related to the uplink and downlink RB usage increase or decrease.
- The RRC connection setup success rate fluctuates.
- The average uplink and downlink ranks may increase or decrease.
- The total number of CQI reporting times increases, and the measured CQI value increases or decreases.

The involved indicators are as follows:

Indicator Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Uplink cell edge throughput	(**N.ThpVol.UL.CellOverlap** – **N.ThpVol.UL.SmallPkt.CellOverlap**)/ **N.ThpTime.UL.RmvSmallPkt.CellOverlap**
Downlink cell edge throughput	(**N.ThpVol.DL.CellOverlap** – **N.ThpVol.DL.LastSlot.CellOverlap**)/ **N.ThpTime.DL.RmvLastSlot.CellOverlap**

Indicator Description	Formula
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Uplink RBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink RBLER	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Average transmission delay of the first packet	**N.Traffic.DL.RlcFirstPktDelay.Time**/ **N.Qos.DL.RlcFirstPktDelay.Num**
Average air interface processing delay of downlink data packets	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num**
Average RLC processing delay of downlink data packets	**N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
Average gNodeB DU processing delay of downlink data packets	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
CCE usage	**N.CCE.Used.Avg**/**N.CCE.Avail.Avg**

Indicator Description	Formula
Uplink DCI allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num** × 100%
Downlink DCI allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num** × 100%
Service drop rate	**Service Call Drop Rate (CU)**
Downlink PRB usage in a cell	**Downlink Resource Block Utilizing Rate (DU)**
Uplink PRB usage in a cell	**Uplink Resource Block Utilizing Rate (DU)**
Downlink PDSCH DRB PRB usage	**N.PRB.DL.DrbUsed.Avg**/**N.PRB.DL.Avail.Avg**
Uplink PUSCH DRB PRB usage	**N.PRB.UL.DrbUsed.Avg**/**N.PRB.UL.PUSCH.Avail.Avg**
Average PUSCH PRB usage	**N.PRB.PUSCH.Used.Avg**/**N.PRB.UL.PUSCH.Avail.Avg**
RRC connection setup success rate	**RRC Setup Success Rate (CU)**
Average downlink rank	(1 × **N.PDSCH.InitTbDl.Rank1** + 2 × **N.PDSCH.InitTbDl.Rank2** + 3 × **N.PDSCH.InitTbDl.Rank3** + ... + 8×**N.PDSCH.InitTbDl.Rank8**)/(**N.PDSCH.InitTbDl.Rank1** + ... + **N.PDSCH.InitTbDl.Rank8**)
Average uplink rank	(1 × **N.PUSCH.InitTbUl.Rank1** + 2 × **N.PUSCH.InitTbUl.Rank2**)/(**N.PUSCH.InitTbUl.Rank1** + **N.PUSCH.InitTbUl.Rank2**)
Number of CQI reporting times	**N.ChMeas.CQI.SingleCW.0** to **N.ChMeas.CQI.SingleCW.15**
Average CQI	(0 × **N.ChMeas.CQI.SingleCW.0** + ... + 15 × **N.ChMeas.CQI.SingleCW.15**)/(**N.ChMeas.CQI.SingleCW.0** + ... + **N.ChMeas.CQI.SingleCW.15**)

4.7.4 Requirements

4.7.4.1 Licenses

There are no license requirements for basic functions.

4.7.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.7.4.2.1 Prerequisite Functions

None

4.7.4.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Power saving BWP	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	When the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the **BWP2_SWITCH** option must be deselected.
Uplink DCI extension-based scheduling	**DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	If the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the **DCI_EXTEND_UL_CONGEST_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter must be deselected.

4.7.4.2.3 Function Impacts

Impact relationships with functions related to scheduling:

Function Name	Function Switch	Reference	Description
Rate-matching-pattern-configuration-based PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	When both rate-matching-pattern-configuration-based PDCCH rate matching and time-domain power saving BWP are enabled, newly admitted UEs are not allowed to enter BWP2 as long as rate-matching-pattern-configuration-based PDCCH rate matching is in effect. The air interface resources of the UEs that have already switched to BWP2 are deleted through RRC reconfiguration messages, and such UEs are switched back to BWP1 from BWP2.

Impact relationships with functions related to carrier management

Function Name	Function Switch	Reference	Description
Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*	When both time-domain power saving BWP and single DCI are enabled on the PCC, whether DCI format 1_3 can be configured on BWP2 of the PCC depends on whether the "single DCI configuration on BWP2" function (controlled by the **BWP2_CONFIG_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaCompatibilitySwitch*** parameter) is enabled. ● If single DCI configuration on BWP2 is enabled, DCI format 1_3 can be configured on the BWP2 of the PCC. ● If single DCI configuration on BWP2 is disabled, DCI format 1_3 cannot be configured on the BWP2 of the PCC. If UEs have compatibility issues when both single DCI and time-domain power saving BWP take effect, it is recommended that "single DCI configuration on

Function Name	Function Switch	Reference	Description
			BWP2" be enabled to reduce the downlink IBLER and RBLER of these UEs.

Function Name	Function Switch	Reference	Description
Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	BWP2 is not configured for CA UEs in their SCells. When CA and time-domain power saving BWP are both enabled, there are the following impacts: ● A larger value of **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP2 to BWP1, a lower probability of such switching, and lower UE power consumption. ● A larger value of **NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP1 to BWP2, a higher probability of such switching, and lower UE power consumption. ● A larger value of **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** results in a lower probability of UE switching from
Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	

Function Name	Function Switch	Reference	Description
			BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. ● A smaller value of **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more frequently. For cell-edge UEs under weak coverage, the service drop rate may increase. ● SCells are more likely to be deactivated if the **NRDUCellCarrMgmt**.***RlcDlThroughputThld***

Function Name	Function Switch	Reference	Description
			parameter value is less than the **NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld*** parameter value. If **NRDUCellCarrMgmt**.***RlcDlThroughputThld*** is set to **0**, SCells will not be deactivated and UEs will not switch to BWP2.

Function Name	Function Switch	Reference	Description
Uplink intra-band CA (TDD)	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	BWP2 is not configured for CA UEs in their SCells. When CA and time-domain power saving BWP are both enabled, there are the following impacts: ● A larger value of **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP2 to BWP1, a lower probability of such switching, and lower UE power consumption. ● A larger value of **NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP1 to BWP2, a higher probability of such switching, and lower UE power consumption. ● SCells are more likely to be deactivated if the **NRDUCellCarrMgmt**.***RlcUlThroughputThld***
Uplink intra-FR inter-band CA (low-frequency TDD)	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	

Function Name	Function Switch	Reference	Description
			parameter value is less than the **NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld*** parameter value. If **NRDUCellCarrMgmt**.***RlcUlThroughputThld*** is set to **0**, SCells will not be deactivated and UEs will not switch to BWP2.
Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Time-domain power saving BWP does not take effect on inter-FR CA UEs.

Impact relationships with MIMO-related functions

Function Name	Function Switch	Reference	Description
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	When there is a large proportion of BWP2 UEs, the gains on the downlink user-perceived rate brought by Fusion Cell are affected.

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Downlink MU-MIMO is not performed on UEs that work on BWP2 for power saving. If time-domain power saving BWP is enabled after MIMO, the following may occur:(1) ● The number of MIMO layers, BLER, and average number of downlink RBs for MU-MIMO UE pairing in the downlink fluctuate. ● The number of MIMO layers, BLER, and average number of RBs for MU-MIMO UE pairing in the uplink fluctuate. ● The numbers of uplink and downlink scheduling times decrease. ● The average cell throughput and average UE throughput fluctuate.
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	
Downlink SU-MIMO	**DL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	
Uplink SU-MIMO	**UL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	

Function Name	Function Switch	Reference	Description
(1) For details about the formulas for calculating the uplink and downlink BLERs, average cell throughput, and average UE throughput, see **4.7.3.2 Impacts**. The following are indicators related to the number of layers, number of scheduling times, and average number of RBs for MU pairing: Average number of downlink MU-MIMO layers per PRB in a cell = **N.ChMeas.MIMO.DL.Pair.Layer**/**N.ChMeas.MIMO.DL.Pair.PRB** Average number of downlink layers per PRB in a cell = **N.ChMeas.MIMO.DL.Transmission.Layer**/[(**N.PRB.DL.DrbUsed.Avg** + **N.PRB.DL.SrbUsed.Avg**) x **N.DL.PDSCH.Tti.Num**] Maximum number of downlink layers per PRB in a cell = **N.ChMeas.MIMO.DL.Transmission.Layer.Max** Average number of RBs for downlink MU pairing = (1 x **N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + ...+16 x **N.ChMeas.MIMO.DL.MuPairing.16Layers.RB**)/(**N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + ... + **N.ChMeas.MIMO.DL.MuPairing.16Layers.RB**) Average number of uplink MU-MIMO layers per PRB in a cell = **N.ChMeas.MIMO.UL.Pair.Layer**/**N.ChMeas.MIMO.UL.Pair.PRB** Average number of uplink layers per PRB in a cell = **N.ChMeas.MIMO.UL.Trans.Layer**/(**N.PRB.PUSCH.Used.Avg** x **N.UL.PUSCH.Tti.Num**) Average number of RBs for uplink MU pairing = (1 x **N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + ... + 8 x **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**)/(**N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + ... + **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**) Number of uplink scheduling times = **N.User.Schedule.Ul.Sum** Number of downlink scheduling times = **N.User.Schedule.Dl.Sum**			

Impact relationships with UCN-related functions

Function Name	Function Switch	Reference	Description
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**, and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	With time-domain power saving BWP enabled, UEs are more likely to use BWP2 after the value of the **NRDUCellMultiTrp**.***DataTransMode*** parameter is changed from **BASIC_MODE** to **DPS_MODE** for Cell Combination. Related performance indicators on the base station side deteriorate. For details, see **4.7.3.2 Impacts**.
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	If the **COOR_UE_PWR_SAVE_PRI_ADJ_SW** option of the **NRDUCellCollabServ**.***CommAlgoSwitch*** parameter is selected, the downlink throughput of Co-MM UEs increases. However, the proportion of time in which UEs work on BWP2 may decrease, the value of the **N.Bwp2.Dur.Total** counter decreases, and the UE power saving effect may deteriorate. The SRS SINR measurement value may increase (indicated by a decrease in the proportion of SINR measurement values that fall into small-index ranges).

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	UEs with CoMP in effect will not be switched to BWP2. When a UE is working on BWP2 and the conditions for CoMP to take effect are met, the UE is first switched to BWP1 and then CoMP takes effect.
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	UEs with CoMP in effect will not be switched to BWP2. When a UE is working on BWP2 and the conditions for CoMP to take effect are met, the UE is first switched to BWP1 and then CoMP takes effect.

Impact relationships with functions related to radio services

Function Name	Function Switch	Reference	Description
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	● Time-domain power saving BWP does not take effect on RedCap UEs. ● When time-domain power saving BWP takes effect, the scheduling of RedCap UEs on BWP1 is affected, and RedCap user experience fluctuates.

Function Name	Function Switch	Reference	Description
Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	After the downlink resource isolation function is enabled, if the time-domain power saving BWP function takes effect on FWA UEs, the downlink experience of FWA UEs deteriorates more severely. In this case, it is good practice to select the **FWA_BWP2_DISABLE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter so that BWP2 will not be configured for FWA UEs.
Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When "low latency and high reliability" and time-domain power saving BWP are both enabled, the UEs using the "low latency and high reliability" function will not be switched to BWP2.
3GPP R15-compliant uplink slot aggregation	**VONR_UL_SLOT_AGGREGATION_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter	*VoNR*	VoNR UEs with uplink slot aggregation in effect will not be switched to BWP2.

Function Name	Function Switch	Reference	Description
Optimization of collaboration between VoNR and UE power saving	**VONR_UE_PWR_SAVING_COOP_OPT_SW** option of the **NRDUCellAlgoSwitch.*ServiceDiffSwitch*** parameter	*VoNR*	When "optimization of collaboration between VoNR and UE power saving" is enabled, the duration in which BWP2 is used by UEs reduces.
SRS field strength-based positioning	**SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	Time-domain power saving BWP can take effect for UEs with SRS field strength-based positioning in effect if the UEs can report ***srs-PosResources-r15***.
Live streaming UE identification	**gNBQciBearer**.***ServiceType*** set to **EMBB_LIVE_VIDEO**	*Ultimate Video Experience*	Live streaming UEs will not be switched to BWP2.
Cloud X UE identification	**gNBQciBearer**.***ServiceType*** set to **EMBB_CLOUD_VR** or **EMBB_CLOUD_GAMING**	*Ultimate Cloud X Experience*	UEs with Cloud X services in effect will not be switched to BWP2.

Impact relationships with functions related to power saving

Function Name	Function Switch	Reference	Description
SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	If the **SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** is selected for an SSSG-switching-capable UE, only SSSG takes effect for the UE. The time-domain power saving BWP function takes effect for UEs only when the **SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** is deselected and the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter is selected.

Function Name	Function Switch	Reference	Description
DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	When both time-domain power saving BWP and DRX are enabled: ● If the duration configured using the **gNBDUDrxParamGroup**.***DrxInactivityTimer*** parameter is less than 20 ms, UEs may remain working on BWP2 and cannot be switched to BWP1. ● If the **BWP2_APER_CSI_UE_DRX_ENH_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, UEs in BWP2 do not aperiodically report CSI-RS for CM measurement results even though configured to do so when there is no uplink data transmission in non-On Duration periods. This way, these UEs can enter the DRX sleep time

Function Name	Function Switch	Reference	Description
			and the number of CSI-RS for CM measurement reports decreases, affecting network performance. For details about the impact analysis, see *DRX*. ● If the base station triggers a BWP2-to-BWP1 switching during the DRX sleep time, the corresponding DCI may fail to be delivered. As a result, the base station stops downlink scheduling for the UE during the BWP switching, leading to downlink scheduling failures. After the **DRX_BWP2_OPT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the base station can trigger UE switching from BWP2 to BWP1 only when the

Function Name	Function Switch	Reference	Description
			DRX inactivity timer is running.
PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	Time-domain power saving BWP does not take effect on UEs with PDCCH skipping in effect.
Data convergence in slots in power saving scheduling	**POWER_SAVING_SCHEDULE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	A larger value of the **NRDUCellPowerSaving**.***RlcFirstPktDelayThld*** parameter results in a higher degree of data convergence with a longer time. As a result, the number of scheduling failures of UEs on BWP2 in a cell increases. When the scheduling failure rate of BWP2 UEs in a cell exceeds the value of the **NRDUCellUePwrSaving**.***Bwp2SchFailRateThld*** parameter, UEs are more likely to switch from BWP2 back to BWP1.

Other impacts:

Function Name	Function Switch	Reference	Description
Super uplink	**SUPER_UL_TDM_SCH_SW**, **FLEX_SUPER_UPLINK_SW**, or **INTER_GNB_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	Only UEs on the NUL carrier can be switched to BWP2. UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay for UEs on the NUL carrier increases when the SUL carrier is activated.
UL and DL decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling*	Only UEs on the NUL carrier can be switched to BWP2. UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay for UEs on the NUL carrier increases when the SUL carrier is activated.

4.7.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.7.4.4 Cells

None

4.7.4.5 Others

Requirements on UEs

UEs must support multiple (two or more) dedicated BWPs. As defined in 3GPP TS 38.331, a UE supports multiple dedicated BWPs if its capability field contains **bwp-SameNumerology**.

4.7.5 Operation and Maintenance

4.7.5.1 Data Configuration

4.7.5.1.1 Data Preparation

The following tables describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-12 Parameters used for activation (time-domain power saving BWP)

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	TDM_BWP2_SWITCH	Select this option.
BWP1 to TDM BWP2 DL Throughput Threshold	**NRDUCellUePwrSaving**.***Bwp1ToTdmBwp2DlThptThld***	None	Use the recommended value.
BWP1 to TDM BWP2 UL Throughput Threshold	**NRDUCellUePwrSaving**.***Bwp1ToTdmBwp2UlThptThld***	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
CA Deactivate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum***	None	Use the recommended value.
CA Deactivate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum***	None	Use the recommended value.
CA Activate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum***	None	Use the recommended value.
CA Activate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum***	None	Use the recommended value.
DL State Transition Buffer Volume Threshold	**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld***	None	The value **20** is recommended.
BWP1 to BWP2 Evaluation Period	**NRDUCellUePwrSaving**.***Bwp1ToBwp2EvaluatePeriod***	None	Use the recommended value.

Table 4-13 Parameters used for activation (BWP2 scheduling failure rate threshold for time-domain power saving BWP)

Parameter Name	Parameter ID	Option	Setting Notes
TDM BWP2 Scheduling Failure Rate Threshold	**NRDUCellUePwrSaving**.***TdmBwp2SchFailRateThld***	None	Use the recommended value.

Table 4-14 Parameters used for activation (exit of weak-coverage UEs from BWP2)

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	WEAK_COV_BWP2_EXIT_SW	Select this option if you want to enable exit of weak-coverage UEs from BWP2.
Weak Coverage Threshold Offset	**NRDUCellUePwrSaving**.***WeakCoverageThldOffset***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Initial DMRS RSRP Weak Coverage Threshold	**NRDUCellDmrs**.***InitDmrsRsrpWeakCovThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Initial DMRS SINR Weak Coverage Threshold	**NRDUCellDmrs**.***InitDmrsSinrWeakCovThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Weak Coverage CQI Threshold	**NRDUCellUePwrSaving**.***WeakCoverageCqiThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Weak Coverage UL SINR Threshold	**NRDUCellUePwrSaving**.***WeakCoverageUlSinrThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.

Table 4-15 Parameters used for activation (QCI-level configuration for time-domain power saving BWP)

Parameter Name	Parameter ID	Option	Setting Notes
QCI BWP2 Indicator	**NRDUCellQciBearer**.***QciBwp2Ind***	None	If time-domain power saving BWP needs to be disabled separately for ToB services, set this parameter to **NOT_SUPPORT** for corresponding QCIs.

Table 4-16 Parameters used for activation (Type of UEs [in terms of the networking mode] on which time-domain power saving BWP can take effect)

Parameter Name	Parameter ID	Option	Setting Notes
BWP2 Networking Mode	**NRDUCellUePwrSaving**.***Bwp2NetworkingMode***	None	Use the recommended value.

4.7.5.1.2 Using MML Commands

Before using MML commands, refer to **4.7.4 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Time-domain power saving BWP

```
//Turning on TDM_BWP2_SWITCH
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=TDM_BWP2_SWITCH-1;
//Configuring the downlink and uplink throughput thresholds for BWP1-to-BWP2 switching
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp1ToTdmBwp2DlThptThld=5000,
Bwp1ToTdmBwp2UlThptThld=1000;

//Setting the number of slots used for calculating the downlink and uplink traffic volume thresholds for CA
SCell deactivation and BWP switching
MOD NRDUCELLCARRMGMT: NrDuCellId=1, CaDlScellDeactThldSlotNum=4, CaUlScellDeactThldSlotNum=1;
//Setting the number of slots used for calculating the downlink and uplink traffic volume thresholds for CA
SCell activation and BWP switching
MOD NRDUCELLCARRMGMT: NrDuCellId=1, CaDlScellActThldSlotNum=16, CaUlScellActThldSlotNum=4;
//Configuring the threshold of buffered traffic volume for downlink state transitions
MOD NRDUCELLCARRMGMT: NrDuCellId=1, DlStateTransitBufVolThld=20;
//Configuring the period for determining whether to switch from BWP1 to BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp1ToBwp2EvaluatePeriod=10;
```

- Time-domain power saving BWP2 scheduling failure rate threshold

```
//Configuring the threshold expressed as a probability of UE scheduling failures on BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, TdmBwp2SchFailRateThld=1;
```

- Exit of weak-coverage UEs from BWP2

```
//Turning on the switch for exit of weak-coverage UEs from BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=WEAK_COV_BWP2_EXIT_SW-1;
//Configuring the initial DMRS RSRP weak coverage threshold and initial DMRS SINR weak coverage
threshold
MOD NRDUCELLDMRS: NrDuCellId=1, InitDmrsRsrpWeakCovThld=-130, InitDmrsSinrWeakCovThld=0;
//Configuring the weak coverage threshold offset, weak coverage CQI threshold, and weak coverage uplink
SINR threshold
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, WeakCoverageThldOffset=5, WeakCoverageCqiThld=5,
WeakCoverageUlSinrThld=5;
```

- QCI-level configuration for time-domain power saving BWP


```
//Setting the NRDUCellQciBearer.QciBwp2Ind parameter to NOT_SUPPORT for a QCI (QCI 82 is used as an example) so that time-domain power saving BWP can be disabled separately for ToB services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, QciBwp2Ind=NOT_SUPPORT;
```

- Type of UEs (in terms of the networking mode) on which time-domain power saving BWP can take effect

```
//Configuring the type of UEs (in terms of the networking mode) on which time-domain power saving BWP can take effect
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp2NetworkingMode=SA_AND_NSA;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Exit of weak-coverage UEs from BWP2

```
//Turning off the switch for exit of weak-coverage UEs from BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=WEAK_COV_BWP2_EXIT_SW-0;
```

- Time-domain power saving BWP

```
//Turning off TDM_BWP2_SWITCH
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=TDM_BWP2_SWITCH-0;
```

4.7.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.7.5.2 Activation Verification

Tracing Signaling

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 In the navigation tree on the left, choose **Uu Interface Trace**.

Step 3 On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If both **BWP-Downlink** and **BWP-Uplink** contain two BWP IDs, the UE is configured with two BWPs.

----End

Counter Observation

Observe the following counters to check whether the time-domain power saving BWP function has taken effect. If the value of the **N.Bwp2.Dur.Total** counter is not 0, time-domain power saving BWP has taken effect.

4.7.5.3 Network Monitoring

The following indicators can be used for network monitoring of this function.

Counter ID	Counter Name	Monitoring Purpose
1911825117	**N.Bwp2.Dur.Total**	The counter measures the total duration of all UEs working in BWP2 in the cell.
1911825121	**N.BWP.ThpVol.DL**	These counters measure the average throughputs of UEs working in BWP0 and BWP1 in a cell. UEs working in BWP0 are those working in the initial BWP.
1911825125	**N.BWP.ThpVol.DL.LastSlot**	
1911825128	**N.BWP.ThpTime.DL.RmvLastSlot**	
1911825129	**N.BWP.ThpVol.UL**	
1911825123	**N.BWP.ThpVol.UE.UL.SmallPkt**	
1911825127	**N.BWP.ThpTime.UE.UL.RmvSmallPkt**	

Average downlink throughput of UEs working in BWP0 and BWP1 = (**N.BWP.ThpVol.DL – N.BWP.ThpVol.DL.LastSlot**) x 1000/ **N.BWP.ThpTime.DL.RmvLastSlot**

Average uplink throughput of UEs working in BWP0 and BWP1 = (**N.BWP.ThpVol.UL – N.BWP.ThpVol.UE.UL.SmallPkt**) x 1000/ **N.BWP.ThpTime.UE.UL.RmvSmallPkt**

NOTE

- The gain of UE power saving is closely related to the BWP capability of UEs in a cell and the traffic model. Therefore, the gain of UE power saving needs to be observed and evaluated on the UE side, not by using the **N.Bwp2.Dur.Total** counter.
- When the switch for time-domain power saving BWP is off, the average UE throughputs (indicated by **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**), targeted for measurement of all UEs in a cell, differ from the average UE throughputs in BWP0 and BWP1 in terms of the measurement dimensions. Therefore, the final measurement results differ slightly.
- The average throughputs of UEs working in BWP0 and BWP1 is related to the traffic model and the measured values may be 0.

4.8 VoNR UE Power Saving

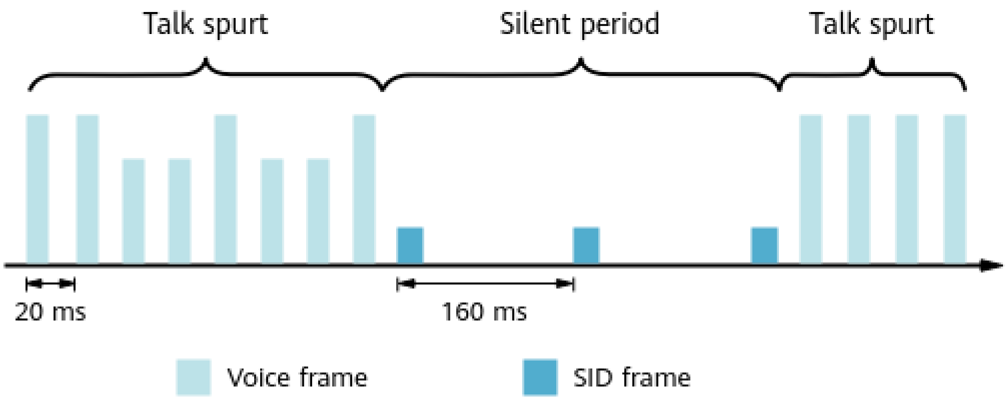
4.8.1 Principles

Voice services have two states, as shown in **Figure 4-11**.

Talk spurts: UEs transmit voice frames in the uplink or receive voice frames in the downlink at 20 ms intervals.

Silent periods: UEs transmit silence insertion descriptor (SID) frames in the uplink or receive SID frames in the downlink at 160 ms intervals.

Figure 4-11 VoNR service states



When VoNR takes effect, the base station performs uplink compensation scheduling for VoNR UEs to ensure sufficient uplink scheduling opportunities for VoNR UEs. For details, see *VoNR*. The uplink packet transmission interval of the UE during silent periods is long. If the uplink compensation scheduling period within silent periods is the same as that within talk spurts, power of the UE is unnecessarily consumed during silent periods. To address this issue, the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter is introduced to specify the uplink compensation scheduling period for VoNR within silent periods.

- If this parameter is set to **OFF**, the uplink compensation scheduling period within silent periods, which is specified by the **NRDUCellPusch**.***UlVonrCompensationSchPrd*** parameter, is the same as that within talk spurts.
- If this parameter is not set to **OFF**, the uplink compensation scheduling period within silent periods is specified by the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter, and the uplink compensation scheduling period within talk spurts is specified by the **NRDUCellPusch**.***UlVonrCompensationSchPrd*** parameter. If the uplink compensation scheduling period within silent periods is longer than that within talk spurts, UEs can save power.

For details about how the **NRDUCellPusch**.***UlVonrCompensationSchPrd*** parameter controls the uplink compensation scheduling period, see *VoNR*.

4.8.2 Network Analysis

4.8.2.1 Benefits

If the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter is set to a value other than **OFF** and the uplink compensation scheduling period within silent periods is longer than that within talk spurts, the power consumption of communication modules of UEs decreases. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

4.8.2.2 Impacts

The impacts between this function and other functions are described in **4.8.3.2.3 Function Impacts**.

If the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter is set to a value other than **OFF** and the uplink compensation scheduling period within silent periods is longer than that within talk spurts, the following are true:

- The first several voice packets during the transition from the silent period to the talk spurts suffer extra jitter. As a result, voice experience deteriorates, for example, voice frame freezing occurs.
- The uplink PRB usage in the cell (indicated by **Uplink Resource Block Utilizing Rate (DU)**) may decrease, and the average uplink cell throughput (indicated by **Cell Uplink Average Throughput (DU)**) may increase.

4.8.3 Requirements

4.8.3.1 Licenses

This function takes effect only in VoNR scenarios and does not have additional license requirements. For details about the license requirements for the VoNR feature, see *VoNR*.

4.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.8.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	VoNR UE power saving takes effect only in VoNR scenarios.

4.8.3.2.2 Mutually Exclusive Functions

None

4.8.3.2.3 Function Impacts

None

4.8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.8.3.4 Others

This function takes effect only in VoNR scenarios. This function has the same requirements for UEs and the core network as the VoNR feature. For details, see *VoNR*.

4.8.4 Operation and Maintenance

4.8.4.1 Data Configuration

4.8.4.1.1 Data Preparation

Table 4-17 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-17 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
UL VoNR Silence Compen Scheduling Period	**NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd***	If UE power saving is required, the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter cannot be set to **OFF**.

4.8.4.1.2 Using MML Commands

Before using MML commands, refer to **4.8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,UlVonrSilenceCompenSchPrd=PERIOD_40MS;
```

Deactivation Command Examples

```
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0,UlVonrSilenceCompenSchPrd=OFF;
```

4.8.4.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.8.4.3 Activation Verification

If the value of **N.PRB.UL.DrbUsed.Avg** or **N.CCE.ULDCI.Used.VoNR** decreases after the modification of **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd***, VoNR UE power saving has taken effect.

4.8.4.4 Network Monitoring

The indicators listed in **4.8.2.2 Impacts** can be used to monitor the benefits and impacts of this function.

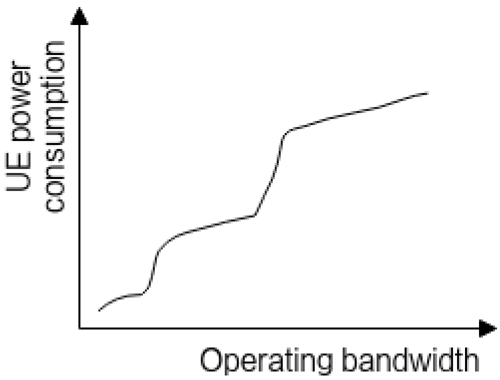
5 Frequency Domain

5.1 Power Saving BWP (Low-Frequency TDD)

5.1.1 Overview

The amount of power a UE consumes depends on its operating bandwidth. In theory, UEs with a larger operating bandwidth consume more power. **Figure 5-1** shows the relationship between UE power consumption and operating bandwidth.

Figure 5-1 Relationship between UE power consumption and operating bandwidth



When a UE is running services with a small data volume, excessive power is consumed if the UE still uses a large operating bandwidth. To address this issue, the power saving BWP function is introduced. **With this function, a base station can adaptively adjust the BWP for a UE based on the UE traffic volume to save UE power**. The power saving BWP function is controlled by the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.

NOTE

A BWP is a part of continuous bandwidth resources configured by the network (base station) for a UE, allowing for flexible transmission bandwidth configurations. Based on the access status of a UE, BWPs can be categorized as initial BWPs and dedicated BWPs.

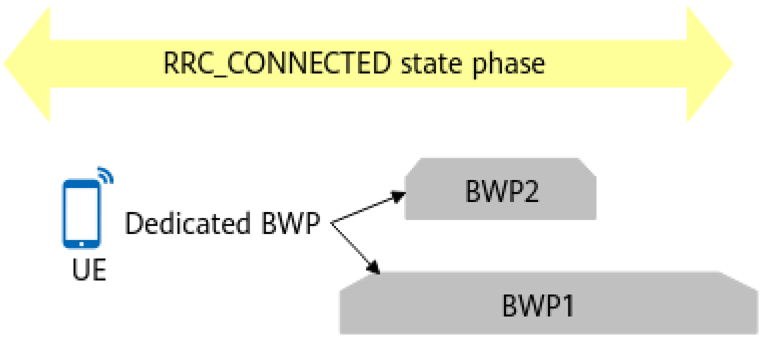
- An initial BWP is used for data transmission of the UE during initial access.
- A dedicated BWP is used for data transmission if the UE is in the RRC_CONNECTED state.

For details about the BWP, see 3GPP TS 38.331, 3GPP TS 38.321, and 3GPP TS 38.213.

5.1.2 Principles

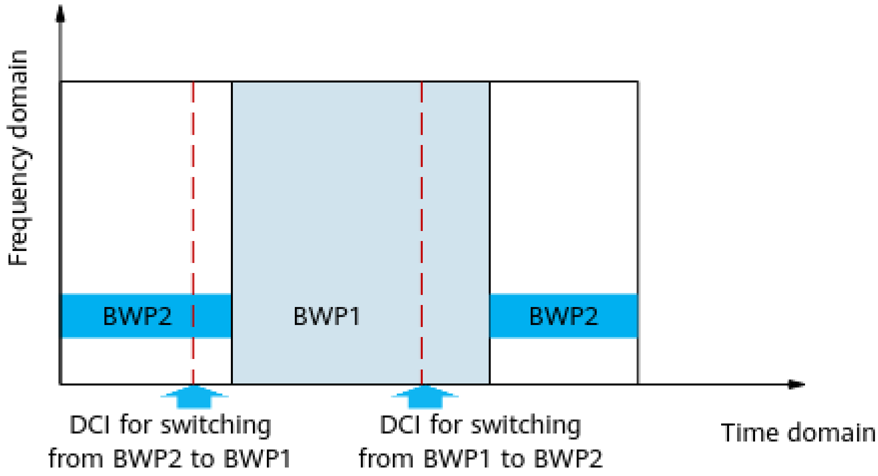
The power saving BWP function can configure two dedicated uplink BWPs and two dedicated downlink BWPs for a UE. Of the two dedicated BWPs, one (named BWP1) is a full bandwidth, whereas the other (named BWP2 or the power saving BWP) is a narrow bandwidth, as shown in **Figure 5-2**. Dedicated BWPs must be configured separately in the uplink and downlink through RRC signaling. At a time, a UE must use the same kind of BWP, either BWP1 or BWP2, in both the uplink and downlink. The size of the full-bandwidth BWP1 is the cell bandwidth, and the size of the narrow-bandwidth BWP2 is adaptively allocated by the base station based on the UE capability. The **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** parameter can be used to specify the maximum number of RBs supported by the BWP2.

Figure 5-2 BWPs for a UE in the RRC_CONNECTED state



After the power saving BWP function takes effect, the base station and UE can switch between the BWP1 and BWP2 to select an appropriate BWP based on the UE traffic volume. Currently, downlink control information (DCI) is used to instruct a UE to switch between two dedicated BWPs, as shown in **Figure 5-3**. DCI 0_1 (for uplink) or DCI 1_1 (for downlink) includes the field **Bandwidth part indicator**, which indicates the target BWP for switching.

Figure 5-3 DCI-based BWP switching



- Switching from BWP1 to BWP2:

  The base station checks whether conditions for BWP switching are met in a period specified by **NRDUCellUePwrSaving**.***Bwp1ToBwp2EvaluatePeriod***. When all the following conditions are met, the base station and UE perform switching from BWP1 to BWP2.

  – The downlink RLC throughput of the UE is less than threshold A, and the downlink RLC buffer data volume is less than threshold B.

  – The uplink RLC throughput of the UE is less than threshold C, and the uplink BSR-indicated data volume is less than threshold D.

Table 5-1 Thresholds related to switching from BWP1 to BWP2

Threshold	Meaning	Calculation Method
Threshold A	Downlink UE throughput threshold for BWP1-to-BWP2 switching	min{Downlink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld***}
Threshold B	Downlink buffered traffic volume threshold for BWP1-to-BWP2 switching	min{Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum***, **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x 0.5 x (BWP2 bandwidth/BWP1 bandwidth)}
Threshold C	Uplink UE throughput threshold for BWP1-to-BWP2 switching	min{Uplink throughput estimated based on radio channel quality, **NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld***}

Threshold	Meaning	Calculation Method
Threshold D	Uplink traffic volume threshold for BWP1-to-BWP2 switching	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum***

- Switching from BWP2 to BWP1:

  The base station checks whether conditions for BWP switching are met in a ms-level period (the period is set by the system and cannot be changed). When any one of the following conditions is met, the base station and the UE perform switching from BWP2 to BWP1.

  – The downlink RLC buffer data volume of the UE is greater than or equal to threshold E.

  – The uplink BSR-indicated data volume of the UE is greater than or equal to threshold F.

Table 5-2 Thresholds related to switching from BWP2 to BWP1

Threshold	Meaning	Calculation Method
Threshold E	Downlink buffered traffic volume threshold for BWP2-to-BWP1 switching	min{Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum***, **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** x (BWP2 bandwidth/BWP1 bandwidth)}
Threshold F	Uplink traffic volume threshold for BWP2-to-BWP1 switching	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum***

For a fast track through the principles of the power saving BWP, see **(Video) Frequency-Domain UE Power Saving**.

BWP Switching Reliability Optimization

After the power saving BWP function is enabled, the BWP on the UE side may be inconsistent with that on the base station side. For example, the base station has switched to the BWP2, but the UE still uses the BWP1. In this case, the service drop rate increases. In this context, BWP switching reliability optimization is introduced to reduce the probability that BWP is switched only on the base station side but not on the UE side, thereby reducing the service drop rate. This function is controlled by the **BWP_SWITCHING_OPT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter and is enabled by default.

BWP2 Scheduling Failure Rate Threshold

In the power saving BWP function, the **NRDUCellUePwrSaving**.***Bwp2SchFailRateThld*** parameter is used to specify the threshold expressed as a probability of UE scheduling failures on BWP2s. When there are a large number of UEs in a cell, the probability of UE scheduling failures on BWP2s is larger because BWP2s have smaller bandwidths. Adjusting the value of **NRDUCellUePwrSaving**.***Bwp2SchFailRateThld*** can change the probability of switching between BWP2 and BWP1, to keep a balance between network performance and UE power consumption.

A smaller value of this parameter results in a higher probability that the probability of failure to schedule UEs on BWP2s in a cell is higher than the value of this parameter. In this case, the base station dynamically adjusts the power saving BWP algorithm to increase the difficulty of switching UEs from BWP1 to BWP2 and decrease the difficulty of switching UEs from BWP2 to BWP1. In this way, more UEs can camp on BWP1s, the network performance is better, but the power saving effect is worse. If this parameter is set to a larger value, UEs are more likely to be switched from BWP1 to BWP2, resulting in poorer network performance and better power saving effect.

Power Saving BWP CCE Adjustment Policy

When the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDuCellPdcch**.***PdcchAlgoSwitch*** parameter is selected, the **NRDuCellPdcch**.***PwrSaveBwpCceAdjustPolicy*** parameter can be used to specify the CCE adjustment policy in power saving BWP.

- If this parameter set to **PWR_SAVING_BWP_NOT_CONSIDERED**:

  CCE ratio adjustment is not performed for UEs involved in power saving BWP. In this way, the probability that UEs switch to BWP2s increases, the average uplink throughput of UEs involved in power saving BWP may decrease, the average downlink throughput of UEs on BWP1s may increase, and the overall network performance may deteriorate.
- If this parameter set to **PWR_SAVING_BWP_CONSIDERED**:

  CCE ratio adjustment is performed for UEs involved in power saving BWP. In this way, the probability that UEs switch to BWP2s decreases, the average downlink throughput of UEs on BWP1s may decrease, and the overall network performance improves.

If power saving is preferred, the value **PWR_SAVING_BWP_NOT_CONSIDERED** is recommended. If performance is preferred, the value **PWR_SAVING_BWP_CONSIDERED** is recommended.

Exit of Weak-Coverage UEs from BWP2

After the power saving BWP function is enabled, the probability that the BWP on the UE side is inconsistent with that on the base station side in weak coverage scenarios is higher. For example, the base station has switched to the BWP2, but the UE still uses the BWP1. In this case, the service drop rate increases. To prevent this, exit of weak-coverage UEs from BWP2 is introduced. This function is controlled by the **WEAK_COV_BWP2_EXIT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter.

When this option is selected, the processing mechanism on the base station is as follows:

- If a UE initially accesses a cell and meets any of the following conditions, the UE is identified as a weak-coverage UE that initially accesses the network in UE power saving scenarios. In this case, BWP2 resource information will not be included in the RRC reconfiguration message for the UE.
  - The Msg3-based DMRS RSRP is less than the sum of values of **NRDUCellDmrs**.***InitDmrsRsrpWeakCovThld*** and **NRDUCellUePwrSaving**.***WeakCoverageThldOffset***.
  - The Msg3-based DMRS SINR is less than the sum of values of **NRDUCellDmrs**.***InitDmrsSinrWeakCovThld*** and **NRDUCellUePwrSaving**.***WeakCoverageThldOffset***.
- If BWP2 resource information is configured for a UE through RRC signaling and any of the following conditions is met, the UE will be identified as a weak-coverage UE in UE power saving scenarios and it will switch from BWP2 to BWP1.
  - The CQI measurement result periodically reported by the UE is less than the value of **NRDUCellUePwrSaving**.***WeakCoverageCqiThld***.
  - The uplink SINR is less than the value of **NRDUCellUePwrSaving**.***WeakCoverageUlSinrThld***.

NOTE

The **NRDUCellUePwrSaving**.***WeakCoverageCqiThld*** parameter is designed in line with the 64QAM-based CQI and modulation scheme mapping table. If 256QAM is used, the gNodeB converts the parameter value based on the same spectral efficiency and then compares the converted value with the CQI measurement result reported by the UE.

DCI Scheduling Optimization for Switching from BWP2 to BWP1

CCE resources on BWP2 are limited. To reduce the probability of failed DCI scheduling for switching from BWP2 to BWP1 and reduce the service drop rate, DCI scheduling optimization for switching from BWP2 to BWP1 is introduced. This function is controlled by the **BWP2_SWITCHING_DCI_SCH_OPT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter and is disabled by default.

QCI-Level Configuration for Power Saving BWP

The BWP2 switch for UE power saving is a cell-level parameter. Therefore, the power saving BWP function cannot be disabled separately for specific services on the wide area network (WAN). **To avoid delay issues of UEs running low-latency services, QCI-level configuration for power saving BWP is introduced.**

With this function, power saving BWP can be disabled separately for QCI-specific services.

This function is controlled by the **NRDUCellQciBearer**.***QciBwp2Ind*** parameter.

- If this parameter is set to **NOT_CONFIG** (default value), power saving BWP is not configured for services with a specific QCI.
- If this parameter is set to **NOT_SUPPORT**, power saving BWP is disabled for services with a specific QCI.
- If this parameter is set to **SUPPORT**, power saving BWP can take effect for services with a specific QCI.

A UE may have multiple radio bearers and each bearer corresponds to a QCI. Different QCIs have different priorities (specified by the **gNBQciBearer**.***PriorityLevel*** parameter). When QCI-level configuration for power saving BWP takes effect on a UE, the value of the **NRDUCellQciBearer**.***QciBwp2Ind*** parameter corresponding to the highest-priority QCI takes effect.

QCI-level Threshold Control for BWP Switching

Currently, the thresholds for BWP switching in the power saving BWP function is of cell-level configuration and cannot be separately set for services with different QCIs. Therefore, QCI-level threshold control for BWP switching is introduced to separately control the probability of BWP switching for UEs running different services.

This function is controlled by the **QCI_CA_BWP_SWITCH_THLD_CTRL_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. **After this function is enabled, if a UE is configured with a power saving BWP, the BWP can be switched based on the QCI-level switching thresholds.** If the UE is running services with multiple QCIs, the switching thresholds for the QCI with the highest priority take effect.

After this function takes effect, BWP switching is performed for UEs running services with a specific QCI as follows:

- Switching from BWP1 to BWP2:

  The base station checks whether conditions for BWP switching are met in a period specified by **NRDUCellUePwrSaving**.***Bwp1ToBwp2EvaluatePeriod***. When all the following conditions are met, the base station and UE perform switching from BWP1 to BWP2. For details about how to calculate related thresholds, see **Table 5-3**.

  – The downlink RLC throughput of a UE is lower than threshold A, and the downlink RLC buffer data volume is less than threshold B.
  – The uplink RLC throughput of the UE is less than threshold C, and the uplink BSR data volume is less than threshold D.

Table 5-3 Thresholds related to switching from BWP1 to BWP2

Threshold	Meaning	Calculation Method
Threshold A	Downlink UE throughput threshold for switching from BWP1 to BWP2	min{Downlink throughput estimated based on radio channel quality, **NRDUCellQciBearer**.***QciBwp1ToBwp2DlThptThld***}
Threshold B	Downlink buffered traffic volume threshold for switching from BWP1 to BWP2	min{Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum***, **NRDUCellQciBearer**.***QciDlStTransitBufVolThld*** x 0.5 x (BWP2 bandwidth/ BWP1 bandwidth)}
Threshold C	Uplink UE throughput threshold for switching from BWP1 to BWP2	min{Uplink throughput estimated based on radio channel quality, **NRDUCellQciBearer**.***QciBwp1ToBwp2UlThptThld***}
Threshold D	Uplink traffic volume threshold for switching from BWP1 to BWP2	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum***

- Switching from BWP2 to BWP1:

  The base station checks whether conditions for BWP switching are met in a ms-level period (the period is set by the system and cannot be changed). When any one of the following conditions is met, the base station and the UE perform switching from BWP2 to BWP1. For details about how to calculate related thresholds, see **Table 5-4**.

  – The downlink RLC buffer data volume of a UE is greater than or equal to threshold E.

  – The uplink BSR data volume of a UE is greater than or equal to threshold F.

Table 5-4 Thresholds related to switching from BWP2 to BWP1

Threshold	Meaning	Calculation Method
Threshold E	Downlink buffered traffic volume threshold for switching from BWP2 to BWP1	min{Downlink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum***, **NRDUCellQciBearer**.***QciDlStTransitBufVolThld*** x (BWP2 bandwidth/BWP1 bandwidth)}
Threshold F	Uplink traffic volume threshold for switching from BWP2 to BWP1	Uplink throughput estimated based on radio channel quality x **NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum***

Downlink Enhancement for Power Saving

This function is controlled by the **PWR_SAVING_DOWNLINK_ENH_SW** option of the **NRDUCellPdsch**.***DlLinkAdaptEnhancementSw*** parameter. Before this function is enabled, the base station triggers power saving BWP switching based on the traffic volume. After this function is enabled, the base station can trigger power saving BWP switching based on SRS weight validity and traffic volume. This function applies only to low-frequency TDD cells served by the MetaAAU.

Periodic CSI Resource Allocation for BWP2 UEs

When PUCCH RB adaptation is enabled (by selecting the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter) and the number of RRC_CONNECTED UEs is small, it is difficult for BWP2 UEs to enter the DRX sleep time when aperiodic CSI scheduling is used, increasing UE power consumption. To address this issue, periodic CSI resource allocation for BWP2 UEs is introduced.

This function is controlled by the **BWP2_PERIOD_CSI_SW** option of the **NRDuCellUePwrSaving**.***BwpPwrSavingSw*** parameter. After this function is enabled, **the base station initially allocates periodic CSI resources to BWP2 UEs by default. The number of BWP2 UEs that can be allocated periodic CSI resources is determined by the number of CSI RBs. This reduces aperiodic CSI scheduling for BWP2 UEs, prolongs the DRX sleep time, and reduces UE power consumption**. This function applies only to low-frequency TDD cells.

Type of UEs (in Terms of the Networking Mode) on Which Power Saving BWP Can Take Effect

When the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the **NRDUCellUePwrSaving**.***Bwp2NetworkingMode*** parameter specifies the type of UEs (in terms of the networking mode) on which power saving BWP can take effect. If the **NRDUCellUePwrSaving**.***Bwp2NetworkingMode*** parameter is set to **SA_AND_NSA**, power saving BWP can take effect on both SA UEs and NSA UEs. If this parameter is set to **SA_ONLY**, power saving BWP can take effect only on SA UEs.

SRS Resource Reservation for BWP2 UEs in RedCap Scenarios

When the RedCap load is heavy, RedCap UEs occupy SRS resources of BWP2 UEs. As a result, the number of BWP2 UEs decreases. To address this, SRS resource reservation for BWP2 UEs in RedCap scenarios is introduced.

This function is controlled by the **BWP2_SRS_RSV_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter. **This feature reserves certain SRS resources for BWP2 UEs, allowing preferential allocation of SRS resources to RedCap UEs in other time-frequency domains. This ensures consistent SRS resource allocation for BWP2 UEs, increases their number, and enhances the power saving effect.**

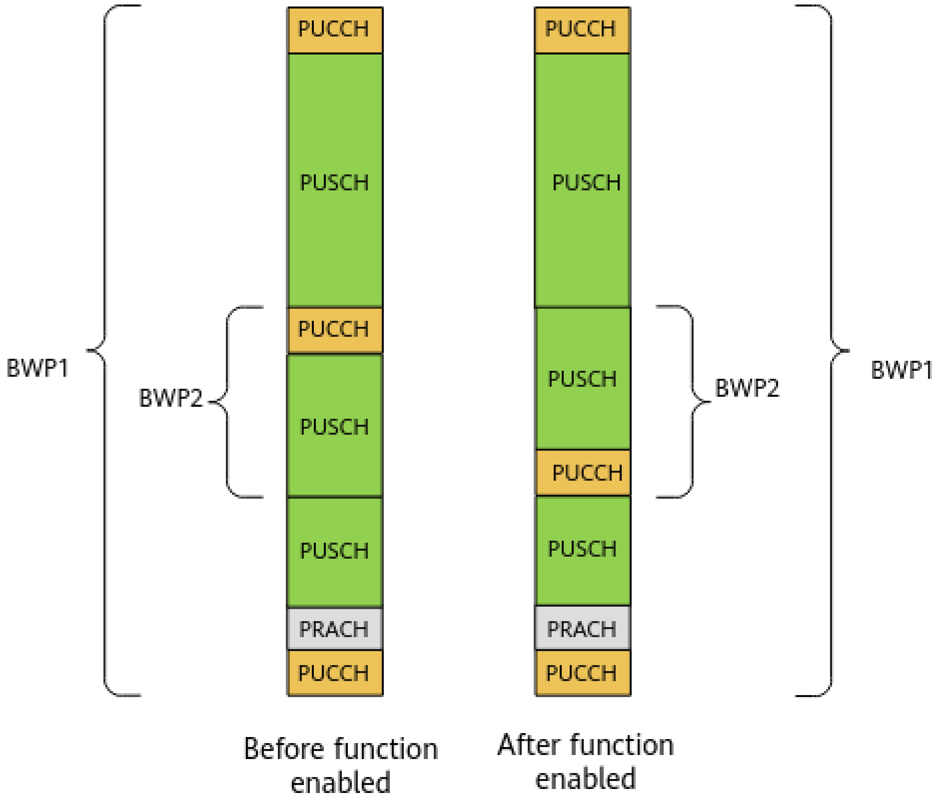
This function takes effect only when both the **REDCAP_SW** option of the **NRDUCellFeatureSW**.***UeCapbBasedFunSw*** parameter and the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter are selected.

BWP2 PUCCH Freq Position Adjustment

After power saving BWP takes effect, the PUCCH of UEs using BWP2 truncates the PUSCH resources, as shown in the content before this function is enabled in **Figure 5-4**. As a result, more contiguous PUSCH RBs cannot be allocated to UEs using BWP1, affecting uplink UE performance. To address this issue, BWP2 PUCCH frequency-domain position adjustment is introduced to increase the number of available RBs for the PUSCH.

This function is controlled by the **BWP2_PUCCH_FREQ_POS_ADJ** option of the **NRDUCellPucch**.***PucchAlgoExtSwitch*** parameter. After this function is enabled, the BWP2 PUCCH frequency-domain position is adjusted as much as possible towards the cell frequency-domain edge, so that the PUSCH resources are contiguous in the frequency-domain. As shown in **Figure 5-4**, resources are allocated according to the new PUCCH and PUSCH positions.

Figure 5-4 BWP2 PUCCH frequency-domain position adjustment (using one scenario as an example)



5.1.3 Network Analysis

5.1.3.1 Benefits

This function reduces the power consumption of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains. This function is most effective when UEs are performing services with a small amount of data. In other cases, the UE power saving effect is not as pronounced.

Assume that power saving BWP is enabled. The average uplink UE throughput increases if BWP2 PUCCH frequency-domain position adjustment is also enabled.

5.1.3.2 Impacts

The impacts between this function and other functions are described in **5.1.4.2.3 Function Impacts**.

On the UE side: After this function is enabled, the throughput of UEs supporting BWP2 may decrease and the service delay of such UEs may increase.

On the base station side:

- Changing the setting for the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter will cause the cell to automatically reset. As a result, the cell will not provide services during the reset period, affecting UE services.

- After the power saving BWP function is enabled, the following may occur:
  - The average uplink and downlink cell throughputs decrease, and the average uplink and downlink UE throughputs decrease significantly.
  - The RRC connection setup success rate, QoS flow setup success rate, and contention-based/non-contention-based random access success rate decrease.
  - The uplink and downlink CCE allocation success rates in the cell decrease.
  - The SRS SINR measurement value increases (indicated by an increase in the proportion of SINR measurement values that fall into large-index ranges).
  - The interference and noise received on each uplink PRB increases, the delay prolongs, and the service drop rate increases.
  - The total number of CQI reporting times increases, and the measured CQI value increases or decreases. As a result, the CCE aggregation level decreases or increases.
  - The proportions of different ranks, uplink and downlink HARQ retransmission rates, uplink and downlink IBLERs and RBLERs, average MCS indexes for uplink and downlink scheduling, CCE usage, and average uplink and downlink RB usages in a cell increase or decrease.
  - When the proportion of CEUs is high, the increase in the service drop rate may decrease the number of CEUs, the uplink and downlink traffic volumes, delay, and BLER, and increase the MCS index, and average uplink and downlink UE throughputs.
- Compared with only enabling the power saving BWP function, enabling both power saving BWP and downlink enhancement for power saving may increase the average uplink and downlink UE throughput, but decreases the proportion of the duration in which a UE stays in the power saving state (namely, the UE works on BWP2). In addition, the value of **N.Bwp2.Dur.Total** decreases, and the UE power consumption increases. Downlink enhancement for power saving also has the following impacts:
  - Counters such as CQIs measured on different BWPs vary. When the duration of UEs working on BWP2 decreases, the following impacts may occur: The CQI measurement value, proportions of different ranks, uplink and downlink MCS indexes, and the CCE aggregation level increases or decreases. The uplink and downlink CCE allocation success rates of a cell, the total number of PRBs paired for downlink MIMO (**N.ChMeas.MIMO.DL.Pair.PRB**), and the total number of layers paired for downlink MIMO (**N.ChMeas.MIMO.DL.Pair.Layer**) increase.
  - Aperiodic CSI scheduling is triggered when a UE works on BWP2. When the duration of a UE on the BWP2 decreases, the number of uplink aperiodic CSI scheduling times decreases. As a result, the number of UEs scheduled per TTI and the average uplink RB usage in the cell may decrease, the value of **N.ThpVol.UL.Cell** (total volume of uplink data received at the MAC layer in a cell) decreases, but the value of **N.ThpVol.UL** (total volume of uplink data received at the RLC layer in a cell) remains unchanged.

  - This function prevents UEs performing downlink large-packet services from switching from BWP1 to BWP2. Therefore, this function reduces BWP switching and may decrease the proportion of CQI 0.
- When power saving BWP is enabled, enabling the periodic CSI resource allocation for BWP2 UEs function has the following impacts compared with disabling this function:
  - The DRX sleep time for UEs in a cell increases, and the average number of PDCCH CCEs used for uplink DCI decreases.
  - The average number of available uplink PRBs for the PUSCH decreases, which may decrease the uplink PRB usage, the number of RBs paired for uplink MU-MIMO, and the average number of uplink MIMO layers per TTI.
  - In a cell, the proportion of wideband CQI reports with a value of 0 in single-codeword transmission increases, the total number of CQI reports decreases, and the measured CQI value may increase or decrease.
  - The service delay may increase, and the uplink and downlink user-perceived rates may decrease.
  - The number of UEs scheduled in the uplink decreases, uplink IBLERs/RBLERs may increase or decrease, and the average MCS index for uplink scheduling may increase or decrease.
- If SRS resource reservation for BWP2 UEs in RedCap scenarios is enabled together with power saving BWP, the number BWP2 UEs may increase under heavy RedCap load, further enhancing the benefits yielded by power saving BWP.
- When power saving BWP is enabled, enabling BWP2 PUCCH frequency-domain position adjustment has the following impacts compared with disabling this function:
  - The number of available PUSCH RBs may increase, and the interference may change. As a result, the values of the following indicators may fluctuate:
    - Average uplink UE throughput
    - Average uplink cell throughput
    - Average uplink RB usage in a cell
    - Total number of PRBs that are paired for uplink MIMO
    - Uplink and downlink IBLERs
    - Uplink and downlink RBLERs
    - Uplink and downlink CCE aggregation levels
    - Uplink and downlink CCE allocation success rates
  - Due to RB adjustment, the PUCCH interference differs, and therefore the PUCCH SINR fluctuates. The values of the following indicators may fluctuate:
    - Average MCS indexes for uplink and downlink scheduling

- Uplink and downlink IBLERs
- Uplink and downlink RBLERs
- Average downlink UE throughput
- Average downlink cell throughput
- Uplink and downlink CCE aggregation levels
- Uplink and downlink CCE allocation success rates

– Due to RB adjustment, the PRACH frequency-domain position changes, and therefore the contention-based and non-contention-based random access success rates may fluctuate.

The following table lists the preceding indicators and their calculation formulas.

Indicator Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Delay	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
Number of CQI reporting times	**N.ChMeas.CQI.SingleCW.0** to **N.ChMeas.CQI.SingleCW.15**
Uplink HARQ retransmission rate	(**N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB** + **N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**) × 100%

Indicator Description	Formula
Downlink HARQ retransmission rate	(**N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB** + **N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**) × 100%
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Uplink RBLER at the MAC layer	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink RBLER at the MAC layer	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%

Indicator Description	Formula
Proportions of TBs transmitted with different ranks	• Rank1 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank1**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank2 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank2**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank3 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank3**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank4 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank4**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank5 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank5**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank6 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank6**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank7 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank7**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank8 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank8**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100%
Service drop rate	**N.PDUSession.AbnormRel**/(**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) × 100%
Interference and noise received per PRB in the uplink	**N.UL.NI.Avg**, **N.UL.NI.Max**, and **N.UL.NI.Min**
Distribution of SRS SINR measurement values	**N.UL.SRS.PreSINR.Index0** to **N.UL.SRS.PreSINR.Index5**
CCE usage	**N.CCE.Used.Avg**/**N.CCE.Avail.Avg** × 100%
Average MCS index for uplink scheduling	(0 × **N.ChMeas.PUSCH.MCS.0** + 1 × **N.ChMeas.PUSCH.MCS.1** + 2 × **N.ChMeas.PUSCH.MCS.2** + 3 × **N.ChMeas.PUSCH.MCS.3** + ... + 26 × **N.ChMeas.PUSCH.MCS.26** + 27 × **N.ChMeas.PUSCH.MCS.27** + 28 × **N.ChMeas.PUSCH.MCS.28**)/(**N.ChMeas.PUSCH.MCS.0** + **N.ChMeas.PUSCH.MCS.1** + **N.ChMeas.PUSCH.MCS.2** + **N.ChMeas.PUSCH.MCS.3** + ... + **N.ChMeas.PUSCH.MCS.26** + **N.ChMeas.PUSCH.MCS.27** + **N.ChMeas.PUSCH.MCS.28**)

Indicator Description	Formula
Average MCS index for downlink scheduling	(0 × **N.ChMeas.PDSCH.MCS.0** + 1 × **N.ChMeas.PDSCH.MCS.1** + 2 × **N.ChMeas.PDSCH.MCS.2** + 3 × **N.ChMeas.PDSCH.MCS.3** + ... + 26 × **N.ChMeas.PDSCH.MCS.26** + 27 × **N.ChMeas.PDSCH.MCS.27** + 28 × **N.ChMeas.PDSCH.MCS.28**)/(**N.ChMeas.PDSCH.MCS.0** + **N.ChMeas.PDSCH.MCS.1** + **N.ChMeas.PDSCH.MCS.2** + **N.ChMeas.PDSCH.MCS.3** + ... + **N.ChMeas.PDSCH.MCS.26** + **N.ChMeas.PDSCH.MCS.27** + **N.ChMeas.PDSCH.MCS.28**)
Uplink DCI allocation success rate	(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num** × 100%
Downlink DCI allocation success rate	(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/**N.CCE.DL.AllocReq.Num** × 100%
Total number of PRBs that are paired for uplink MIMO	**N.ChMeas.MIMO.UL.Pair.PRB**
Average uplink RB usage in a cell	**Uplink Resource Block Utilizing Rate (DU)**
Average downlink RB usage in a cell	**Downlink Resource Block Utilizing Rate (DU)**
RRC connection setup success rate	**RRC Setup Success Rate (CU)**
QoS flow setup success rate	**QoS Flow Setup Success Rate (CU)**
Contention-based random access success rate	**N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att** × 100%
Non-contention-based random access success rate	**N.RA.Dedicated.Msg3**/**N.RA.Dedicated.Att** × 100%
Uplink CCE aggregation level	(1 * **N.CCE.UL.AggLvl1Num** + 2 * **N.CCE.UL.AggLvl2Num** + 4 * **N.CCE.UL.AggLvl4Num** + 8 * **N.CCE.UL.AggLvl8Num** + 16 * **N.CCE.UL.AggLvl16Num**)/(**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)

Indicator Description	Formula
Downlink CCE aggregation level	(1 * **N.CCE.DL.AggLvl1Num** + 2 * **N.CCE.DL.AggLvl2Num** + 4 * **N.CCE.DL.AggLvl4Num** + 8 * **N.CCE.DL.AggLvl8Num** + 16 * **N.CCE.DL.AggLvl16Num**)/(**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)

5.1.4 Requirements

5.1.4.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-030210	UE Power Saving	NR0S00UEPS00	per Cell
For the license control item NR0S00UEPS00: ● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. ● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld**.***LicenseChkSw*** parameter.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.4.2.1 Prerequisite Functions

Power saving BWP

<table>
<tr><th>Function Name</th><th>Function Switch</th><th>Reference</th><th>Description</th></tr>
<tr><td>Enhanced UE Power Saving Switch</td><td>UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter</td><td>UE Power Saving</td><td>The BWP2_SWITCH option can be selected only if the UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter is selected.</td></tr>
<tr><td>SRS period adaptation</td><td>SRS_PERIOD_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter</td><td>Channel Management</td><td rowspan="2">To enable power saving BWP (BWP2_SWITCH), the SRS_PERIOD_ADAPT_SW or USER_CHARACTER_SRS_ADAPT_SW option must be selected.</td></tr>
<tr><td>UE-characteristic-based SRS period adaptation</td><td>USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter</td><td>Channel Management</td></tr>
<tr><td>Uplink Bandwidth</td><td>NRDUCell.UlBandwidth</td><td>Scalable Bandwidth</td><td>The BWP2_SWITCH option can be selected only if the NRDUCell.UlBandwidth parameter is set to CELL_BW_30M or a larger bandwidth.</td></tr>
</table>

Function Name	Function Switch	Reference	Description
Downlink Bandwidth	**NRDUCell**.***DlBandwidth***	*Scalable Bandwidth*	The **BWP2_SWITCH** option can be selected only if the **NRDUCell**.***DlBandwidth*** parameter is set to **CELL_BW_30M** or a larger bandwidth.

Function Name	Function Switch	Reference	Description
Full-bandwidth initial BWP	**INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter	*5G Networking and Signaling*	When **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** is set to **RB51**, the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter must be selected. If the **PHASE_JUMP_MODULE_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***SpecRfModuleNrAdaptSwitch*** parameter and the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter are selected, the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter must be selected.

Function Name	Function Switch	Reference	Description
UE bandwidth adaptation	**UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter	*Scalable Bandwidth*	The **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter can only be deselected if all of the following conditions are met: (1) The **NRDUCell**.***FrequencyBand*** parameter is set to **N38**, **N40**, **N41**, **N77**, **N78**, or **N79**. (2) The **NRDUCell**.***DlBandwidth*** and **NRDUCell**.***UlBandwidth*** parameters are set to **CELL_BW_30M**. (3) The **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is deselected.

Periodic CSI resource allocation for BWP2 UEs

Function Name	Function Switch	Reference	Description
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	The **BWP2_PERIOD_CSI_SW** option of the **NRDuCellUePwrSaving**.***BwpPwrSavingSw*** parameter can be selected only when the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected.
PUCCH RB Adaptation Switch	**PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter	*Channel Management*	The **BWP2_PERIOD_CSI_SW** option of the **NRDuCellUePwrSaving**.***BwpPwrSavingSw*** parameter can be selected only when the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** is selected.

5.1.4.2.2 Mutually Exclusive Functions

If **NRDUCell**.***NrDuCellNetworkingMode*** is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL** (for details, see *Hyper Cell* and *Cell Combination*), the functions mutually exclusive with power saving BWP are as follows.

Function Name	Function Switch	Reference	Description
Narrowband TRS rate matching	**NARROW_BAND_TRS_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	None	If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected and the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **NARROW_BAND_TRS_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter must be deselected.

Function Name	Function Switch	Reference	Description
Percentage-based dynamic RB sharing among operators	**RB_DYNAMIC_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected and the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **RB_DYNAMIC_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter must be deselected.

Function Name	Function Switch	Reference	Description
Operator-specific limitation on the maximum number of available RBs	**MAX_RB_LIMIT_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*	If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected and the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL_COMBINE_MODE** or **HYPER_CELL**, the **MAX_RB_LIMIT_SW** option of the **NRDUCellAlgoSwitch**.***RanSharingAlgoSwitch*** parameter must be deselected.

In other scenarios, the functions mutually exclusive with power saving BWP are as follows.

Function Name	Function Switch	Reference	Description
Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	4.7 Time-Domain Power Saving BWP (Low-Frequency TDD)	When the **TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the **BWP2_SWITCH** option must be deselected.
Configuration of multiple PDCCH search spaces	**MULTI_SEARCH_SPACE_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Channel Management*	When configuration of multiple PDCCH search spaces and power saving BWP are both enabled, the **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** parameter can only be set to **RB51**.
DC Component Rate Matching	**DC_COMPONENT_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	The DC component rate matching function is mutually exclusive with power saving BWP.

Function Name	Function Switch	Reference	Description
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	When both RedCap Introduction and the power saving BWP function are enabled, the **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** parameter cannot be set to a value other than **RB51** for an NR TDD cell.
PLLC primary cell	**PLLC_PRIMARY_CELL_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	None	The PLLC Primary Cell Switch and power saving BWP are mutually exclusive.
Staggered PRACH frequency-domain resource allocation	**PRACH_FREQ_POS_STAGGER_SW** option of the **NRDUCellPrach**.***PrachAlgoExtSwitch*** parameter	*Random Access Control*	Power saving BWP is mutually exclusive with staggered PRACH frequency-domain resource allocation.

Function Name	Function Switch	Reference	Description
PRACH frequency-domain start position	**NRDUCell**.***PrachFreqStartPosition***	*Random Access Control*	If the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter and the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter are selected, the **NRDUCell**.***PrachFreqStartPosition*** parameter must be set to **65535**.
Uplink blocking interference avoidance mode	**NRDUCellIntrfIdent**.***UlBlockingIntrfAvoidMode***	None	If the **BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the **NRDUCellIntrfIdent**.***UlBlockingIntrfAvoidMode*** parameter cannot be set to **ASYMMETRIC_BWP_MODE**.

5.1.4.2.3 Function Impacts

Impact relationships with functions related to channel management and scheduling

Function Name	Function Switch	Reference	Description
PUCCH format-1 RB resource control	**FORMAT1_RB_RES_CTRL_SW** option of the **gNodeBAlgo**.***PucchSwitch*** parameter	*Channel Management*	When both PUCCH format-1 RB resource control and power saving BWP are enabled, more BWP2-capable UEs can access the network. In this case, however, the average uplink and downlink UE throughputs may decrease.

Function Name	Function Switch	Reference	Description
PUCCH channel management	Structure type: **NRDUCellPucch.*StructureType*** Number of format-1 RBs: **NRDUCellPucch.*Format1RbNum*** Number of format-3 RBs: **NRDUCellPucch.*Format3RbNum***	*Channel Management*	● The impact relationship between power saving BWP and the structure type is as follows: When the power saving BWP function takes effect, cells support only long PUCCH (with the **NRDUCellPucch.*StructureType*** parameter set to **LONG_STRUCTURE**). ● The impact relationship between power saving BWP and the number of format-1 RBs is as follows: When PUCCH RB adaptation (controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.*PucchAlgoSwitch*** parameter) is disabled: – In NSA networking, if **NRDUCellPucch.*Format1RbNum*** is

Function Name	Function Switch	Reference	Description
			set to **RB2**, BWP2 cannot be configured for UEs. – In SA networking, when the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch*** parameter is selected and the **NRDUCellPucch**.***Format1RbNum*** parameter is set to **RB2**, BWP2 cannot be configured for UEs. ● The impact relationship between power saving BWP and the number of format-3 RBs is as follows: When PUCCH RB adaptation (controlled by the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch**.***PucchAlgoSwitch*** parameter) is

Function Name	Function Switch	Reference	Description
			disabled and the **NRDUCellPucch**.***Format3RbNum*** parameter is set to **RB2**, BWP2 (narrow bandwidth) cannot be allocated to UEs.
PDCCH uplink-to-downlink CCE ratio adaptation	**UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Channel Management*	When DCI scheduling optimization for switching from BWP2 to BWP1 is enabled, CCE resource allocation for the DCI that indicates switching from BWP2 to BWP1 is not restricted by the uplink CCE proportion configured by PDCCH uplink-to-downlink CCE ratio adaptation.

Function Name	Function Switch	Reference	Description
Rate-matching-pattern-configuration-based PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	When both the rate-matching-pattern-configuration-based PDCCH rate matching and power saving BWP functions are enabled, newly admitted UEs are not allowed to enter BWP2 as long as rate-matching-pattern-configuration-based PDCCH rate matching is in effect. The air interface resources of the UEs that have already switched to BWP2 are released through RRC reconfiguration messages, and such UEs switch back to BWP1 from BWP2.

Impact relationships with functions related to carrier management

Function Name	Function Switch	Reference	Description
Single DCI	**DL_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Multi-Frequency Smart Aggregation*	When both power saving BWP and single DCI are enabled on the PCC, whether DCI format 1_3 can be configured on BWP2 of the PCC depends on whether single DCI configuration on BWP2 (controlled by the **BWP2_CONFIG_SINGLE_DCI_SW** option of the **NRDUCellCarrMgmt**.***CaCompatibilitySwitch*** parameter) is enabled. ● If single DCI configuration on BWP2 is enabled, DCI format 1_3 can be configured on the BWP2 of the PCC. ● If single DCI configuration on BWP2 is disabled, DCI format 1_3 cannot be configured on the BWP2 of the PCC. If UEs have compatibility issues when both single DCI and power saving BWP are enabled, it is recommended that single DCI configuration on BWP2 be enabled to reduce the

Function Name	Function Switch	Reference	Description
			downlink IBLER and downlink RBLER of these UEs.

Function Name	Function Switch	Reference	Description
Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	BWP2, which provides a narrow bandwidth, is not configured for CA UEs in their SCells. When power saving BWP or "QCI-level threshold control for BWP switching" is enabled together with CA, there are the following impacts: ● A larger value of **NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP2 to BWP1, a lower probability of such switching, and lower UE power consumption. ● A larger value of **NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP1 to BWP2, a higher probability of such switching, and lower UE power consumption. ● A larger value of the threshold of buffered traffic
Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	

Function Name	Function Switch	Reference	Description
			volume for downlink state transitions (specified by the **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** or **NRDUCellQciBearer**.***QciDlStTransitBufVolThld*** parameter) results in a lower probability of UE switching from BWP2 to BWP1 and a higher probability of UE switching from BWP1 to BWP2. As a result, UEs stay in BWP2 for a longer time and their power consumption decreases. ● A smaller value of the threshold of buffered traffic volume for downlink state transitions (specified by the **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** or **NRDUCellQciBearer**.***QciDlStTransitBufVolThld*** parameter) results in a higher probability of UE switching from BWP2 to BWP1 and a lower probability of UE switching from

Function Name	Function Switch	Reference	Description
			BWP1 to BWP2. As a result, UEs stay in BWP2 for a shorter time and their power consumption increases. BWP switching and RRC reconfiguration may occur more frequently. For cell-edge UEs under weak coverage, the service drop rate may increase. ● SCells are more likely to be deactivated if the **NRDUCellCarrMgmt**.***RlcDlThroughputThld*** parameter value is less than the downlink throughput threshold for BWP1-to-BWP2 switching (specified by the **NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld*** or **NRDUCellQciBearer**.***QciBwp1ToBwp2DlThptThld*** parameter). If **NRDUCellCarrMgmt**.***RlcDlThroughputThld*** is set to **0**, SCells will not be deactivated and UEs will not switch to BWP2.

Function Name	Function Switch	Reference	Description
Uplink intra-band CA (TDD)	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	BWP2, which provides a narrow bandwidth, is not configured for CA UEs in their SCells. When power saving BWP or "QCI-level threshold control for BWP switching" is enabled together with CA, there are the following impacts: ● A larger value of **NRDUCellCarrMgmt.*CaUlScellActThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP2 to BWP1, a lower probability of such switching, and lower UE power consumption. ● A larger value of **NRDUCellCarrMgmt.*CaUlScellDeactThldSlotNum*** results in a higher traffic volume threshold for UE switching from BWP1 to BWP2, a higher probability of such switching, and lower UE power consumption. ● SCells are more likely to be deactivated if
Uplink intra-FR inter-band CA (low-frequency TDD)	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	

Function Name	Function Switch	Reference	Description
			the **NRDUCellCarrMgmt**.***RlcUlThroughputThld*** parameter value is less than the uplink throughput threshold for BWP1-to-BWP2 switching (specified by the **NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld*** or **NRDUCellQciBearer**.***QciBwp1ToBwp2UlThptThld*** parameter). If **NRDUCellCarrMgmt**.***RlcUlThroughputThld*** is set to **0**, SCells will not be deactivated and UEs will not switch to BWP2.
Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Power saving BWP does not take effect on inter-FR CA UEs.

Impact relationships with MIMO-related functions

Function Name	Function Switch	Reference	Description
Smart PDCCH symbol allocation	**PDCCH_SYM_SMART_ALLOC_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	● Enabling smart PDCCH symbol allocation decreases the uplink CCE allocation success rate in a cell when all the following conditions are met in the cell: (1) Power saving BWP takes effect. (2) The duration in which UEs working on BWP2 accounts for a large proportion. (3) The **NRDUCellUePwrSaving**.***Bwp2MaxRbNum*** parameter is set to **RB64** for BWP2. ● Smart PDCCH symbol allocation increases the CCE allocation success rate for UEs. When power saving BWP is enabled, the CCE allocation success rate of BWP2 UEs also increases. As a result, more UEs may work on BWP2s, and the average uplink UE throughput does not

Function Name	Function Switch	Reference	Description
			increase or even decreases. To resolve this issue, you can disable power saving BWP or set the **NRDUCellUePwrSaving**.***Bwp2SchFailRateThld*** parameter to a smaller value (for example, **1** when the downlink CCE allocation success rate is lower than 80%).

Function Name	Function Switch	Reference	Description
Packet-length-based scheduling optimization	**PKT_LEN_BASED_SCH_OPT_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	A heavy load in the uplink is likely to cause PDCCH resource insufficiency and a decreased CCE allocation success rate. With packet-length-based scheduling optimization enabled, it is recommended that the **NRDuCellPdcch**.***PwrSaveBwpCceAdjustPolicy*** parameter be set to **PWR_SAVING_BWP_NOT_CONSIDERED** if the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter is selected. This prevents UEs involved in power saving BWP from participating in CCE ratio adjustment.
Tight SRS multiplexing	**SRS_TIGHT_MULTIPLEXING_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	After tight SRS multiplexing is enabled, the average uplink throughput of UEs working on the power saving BWP may decrease.

Function Name	Function Switch	Reference	Description
Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	When there is a large proportion of UEs in power saving BWP, the downlink user-perceived rate gain brought by Fusion Cell is affected.

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Downlink MU-MIMO is not performed on UEs that work on BWP2 for power saving. If power saving BWP is enabled after MIMO, the following may occur:(1) ● The number of MIMO layers, BLER, and average number of RBs for MU-MIMO UE pairing in the downlink increase. ● The number of MIMO layers, BLER, and average number of RBs for MU-MIMO UE pairing in the uplink decrease. ● The numbers of uplink and downlink scheduling times increases. ● The average cell throughput and average UE throughput decrease.
PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	
Downlink SU-MIMO	**DL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	
Uplink SU-MIMO	**UL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	

Function Name	Function Switch	Reference	Description
(1) The following are indicators related to the number of layers, number of scheduling times, average number of RBs for MU pairing, and BLER: Average number of downlink MU-MIMO layers per PRB in a cell = **N.ChMeas.MIMO.DL.Pair.Layer**/**N.ChMeas.MIMO.DL.Pair.PRB** Average number of downlink layers per PRB in a cell = **N.ChMeas.MIMO.DL.Transmission.Layer**/[(**N.PRB.DL.DrbUsed.Avg** + **N.PRB.DL.SrbUsed.Avg**) x **N.DL.PDSCH.Tti.Num**] Maximum number of downlink layers per PRB in a cell = **N.ChMeas.MIMO.DL.Transmission.Layer.Max** Average number of RBs for downlink MU pairing = (1 x **N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + …+16 x **N.ChMeas.MIMO.DL.MuPairing.16Layers.RB**)/(**N.ChMeas.MIMO.DL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.DL.MuPairing.2Layers.RB** + … + **N.ChMeas.MIMO.DL.MuPairing.16Layers.RB**) Average number of uplink MU-MIMO layers per PRB in a cell = **N.ChMeas.MIMO.UL.Pair.Layer**/**N.ChMeas.MIMO.UL.Pair.PRB** Average number of uplink layers per PRB in a cell = **N.ChMeas.MIMO.UL.Trans.Layer**/(**N.PRB.PUSCH.Used.Avg** x **N.UL.PUSCH.Tti.Num**) Average number of RBs for uplink MU pairing = (1 x **N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + 2 x **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + … + 8 x **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**)/(**N.ChMeas.MIMO.UL.MuPairing.1Layer.RB** + **N.ChMeas.MIMO.UL.MuPairing.2Layers.RB** + … + **N.ChMeas.MIMO.UL.MuPairing.8Layers.RB**) Number of uplink scheduling times = **N.User.Schedule.Ul.Sum** Number of downlink scheduling times = **N.User.Schedule.Dl.Sum**			

Impact relationships with UCN-related functions

Function Name	Function Switch	Reference	Description
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**, and **NRDUCellMultiTrp**.***DataTransMode*** set to **BASIC_MODE** or **DPS_MODE**	*Cell Combination*	With power saving BWP enabled, UEs are more likely to use BWP2 for power saving after the value of the **NRDUCellMultiTrp**.***DataTransMode*** parameter is changed from **BASIC_MODE** to **DPS_MODE** for Cell Combination. Related performance indicators on the base station side deteriorate. For details, see **5.1.3.2 Impacts**.

Function Name	Function Switch	Reference	Description
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	If the **COOR_UE_PWR_SAVE_PRI_ADJ_SW** option of the **NRDUCellCollabServ**.***CommAlgoSwitch*** parameter is selected, the downlink throughput of Co-MM UEs increases. However, the proportion of time in which UEs stay in the power saving state (that is, UEs work on BWP2) may decrease, the value of the **N.Bwp2.Dur.Total** counter decreases, and the UE power saving effect may deteriorate. The SRS SINR measurement value may increase (indicated by a decrease in the proportion of SINR measurement values that fall into small-index ranges).
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station UL CoMP is not performed for UEs working on BWP2 for power saving.
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station DL CoMP is not performed for UEs working on BWP2 for power saving.

Impact relationships with functions related to interference suppression

Function Name	Function Switch	Reference	Description
Interference avoidance for indoor cells	**INDOOR_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***IndoorOutdoorCoo-perateSw*** parameter	*3D Networking Experience Improvement*	After power saving BWP is enabled, the gains brought by interference avoidance for indoor cells decrease. This is because the start position for PDSCH scheduling is not fixed and some of the positions for PDSCH scheduling in indoor and outdoor cells are different. If the positions for PDSCH scheduling between indoor and outdoor cells are completely different after power saving BWP is enabled, enabling the function of interference avoidance for indoor cells leads to an occasional decrease in the data rates of these cells. However, positive network gains are achieved after interference avoidance for indoor cells is enabled.

Function Name	Function Switch	Reference	Description
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	If the **COOR_UE_PWR_SAVE_PRI_ADJ_SW** option of the **NRDUCellCollabServ**.***CommAlgoSwitch*** parameter is selected, the downlink throughput of fast CBF UEs increases. However, the proportion of time in which UEs stay in the power saving state (that is, UEs work on BWP2) may decrease, the value of the **N.Bwp2.Dur.Total** counter decreases, and the UE power saving effect may deteriorate.

Function Name	Function Switch	Reference	Description
External interference avoidance (low-frequency TDD)	**UL_UE_LEVEL_INTRF_AVOID** option of the **NRDUCellIntrfIdent**.***IntrfOptimizationMethod*** parameter and the **UL_MANUAL** or **UL_AUTO** setting of the **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** parameter	*Interference Avoidance*	If **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** is set to **UL_MANUAL** and the BWP2 frequency domain range and interference range overlap, there may be no available resources allocated to BWP2 UEs. In this case, the attempt for reducing the rate threshold for BWP2 UEs can be initiated. If there are still no available resources after several attempts, the UEs are switched to BWP1, affecting UE power saving.

Function Name	Function Switch	Reference	Description
Enhanced interference avoidance for indoor cells	**INDOOR_INTRF_AVOID_ENH_SW** option of the **NRDUCellAlgoSwitch**.***IndoorOutdoorCooperateSw*** parameter	*3D Networking Experience Improvement*	After power saving BWP is enabled, the gains brought by the function of enhanced interference avoidance for indoor cells decrease. This is because the start position for PDSCH scheduling is not fixed and some of the positions for PDSCH scheduling in indoor and outdoor cells are different. If the positions for PDSCH scheduling between indoor and outdoor cells are completely different after power saving BWP is enabled, enabling the function of enhanced interference avoidance for indoor cells leads to an occasional decrease in the average downlink throughput of these cells. However, the downlink spectral efficiency of the entire network improves after enhanced interference avoidance for indoor cells is enabled.

Impact relationships with functions related to radio services

Function Name	Function Switch	Reference	Description
Downlink large-packet alignment pairing for RedCap UEs	**DL_BIG_PKT_ALIGN_PAIR_SW** option of the **NRDUCellRedCap**.***RedCapSchAlgoSwitch*** parameter	*RedCap*	Downlink large-packet alignment pairing for RedCap UEs increases the spectral efficiency of a cell. If power saving BWP is enabled, more UEs may use BWP2. In this case, if scheduling in common PDCCH symbols for RedCap UEs (controlled by the **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter) is also enabled, the uplink CCE allocation success rate of BWP2 UEs decreases and the cell-level uplink CCE allocation success rate also decreases.

Function Name	Function Switch	Reference	Description
Downlink adaptive MU pairing for RedCap UEs	**REDCAP_DL_MU_ADAPT_PAIR_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter	*RedCap*	Downlink adaptive MU pairing for RedCap UEs increases the spectral efficiency of a cell. If power saving BWP is enabled, more UEs may use BWP2. In this case, if scheduling in common PDCCH symbols for RedCap UEs (controlled by the **REDCAP_COMM_PDCCH_SYM_SCH_SW** option of the **NRDUCellRedCap**.***RedCapAlgoSwitch*** parameter) is also enabled, the uplink CCE allocation success rate of BWP2 UEs decreases and the cell-level uplink CCE allocation success rate also decreases.

Function Name	Function Switch	Reference	Description
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	When both the RedCap Introduction and the power saving BWP functions are enabled: ● Power saving BWP does not take effect on RedCap UEs. ● As RedCap UEs and UEs in the power saving BWP state share the 20 MHz bandwidth of the cell, the actual resources available to UEs in the power saving BWP state are affected and the value of the **N.Bwp2.Dur.Total** counter may decrease. ● When the RedCap load is heavy, RedCap UEs occupy SRS resources of BWP2 UEs. As a result, the number of BWP2 UEs decreases. In this case, it is recommended that the **BWP2_SRS_RSV_SW** option of the **NRDUCellRedCap**.***RedCapAl***

Function Name	Function Switch	Reference	Description
			goSwitch parameter be selected to increase SRS resources allocated to BWP2 UEs and improve the energy saving effect.
Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	After the downlink resource isolation function is enabled, if the power saving BWP function takes effect on FWA UEs, the downlink experience of FWA UEs deteriorates more severely. In this case, it is good practice to select the **FWA_BWP2_DISABLE_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter so that power saving BWP will not be triggered for FWA UEs.

Function Name	Function Switch	Reference	Description
Uplink RB reservation	**UL_RB_RSV_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	It is not recommended that uplink RB reservation and power saving BWP be enabled at the same time. If both functions are enabled and the reserved RB resources are not in BWP2, the reserved RB resources cannot be used. When the reserved RB resources are in BWP2, the number of available RBs in BWP2 for UEs performing only data services decreases.
Optimized cooperation with BWP2 and DRX	**VONR_UE_PWR_SAVING_COOP_OPT_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter or **VONR_BWP2_CONGESTION_EXIT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*VoNR*	When optimized cooperation with BWP2 and DRX is enabled, the duration in which BWP2 is used by UEs reduces.
Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	When the low latency and high reliability function and power saving BWP are both enabled, the UEs using the low latency and high reliability function do not work on BWP2.

Function Name	Function Switch	Reference	Description
U-TDOA- and AoA-based high-precision positioning	**AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	Power saving BWP does not take effect on UEs with U-TDOA- and AoA-based high-precision positioning in effect.
SRS field strength-based positioning	**SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	Power saving BWP does not take effect on UEs with SRS field strength-based positioning in effect.
U-TDOA-based positioning TOA fingerprint-based positioning	**UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	Power saving BWP does not take effect on UEs with U-TDOA-based positioning or TOA fingerprint-based positioning in effect.

Impact relationships with functions related to power saving

Function Name	Function Switch	Reference	Description
SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSav-ing**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	If the **QCI_CA_BWP_SWITCH_THLD_CTRL_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected, the QCI-specific switching thresholds are configured for BWP2 UEs, and some cell-level parameters in the threshold calculation methods (**Table 4-5** and **Table 4-6**) for BWP2 UEs in SSSG switching are replaced with QCI-specific parameters. The parameter replacement principles are as follows: ● **NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld*** is replaced with **NRDUCellQciBearer**.***QciBwp1ToBwp2DlThptThld***. ● **NRDUCellCarrMgmt**.***DlStateTransitBufVolThld*** is replaced with **NRDUCellQciBearer**.***QciDlStTransitBufVolThld***.

Function Name	Function Switch	Reference	Description
			● **NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld*** is replaced with **NRDUCellQciBearer**.***QciBwp1ToBwp2UlThptThld***.

Function Name	Function Switch	Reference	Description
DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving.*NrDuCellDrxAlgoSwitch*** parameter	*DRX*	● All UEs working in BWP2 aperiodically report CSI-RS for CM measurement results if all of the following conditions are met: (1) both power saving BWP and DRX are enabled; (2) **NRDUCellPucch**.***CsiDedicatedRbNum*** is set to **RB2**; (3) the **PUCCH_RBRES_ADAPTIVE_SWITCH** option of the **NRDUCellPucch.*PucchAlgoSwitch*** parameter is deselected. Because such UEs aperiodically report CSI-RS for CM measurement results on the PUSCH, uplink data of these UEs may be scheduled continuously. As a result, these UEs cannot enter the DRX sleep time. ● If both power saving BWP and DRX are enabled and

Function Name	Function Switch	Reference	Description
			the **gNBDUDrxParamGroup**.***DrxInactivityTimer*** parameter is set to a value less than 20 ms, UEs may stay in BWP2 and cannot switch to BWP1. ● If both power saving BWP and DRX are enabled and the **BWP2_APER_CSI_UE_DRX_ENH_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter is selected, aperiodic CSI-RS for CM does not take effect on UEs in BWP2 when there is no uplink data transmission in non-On Duration periods. This way, these UEs can enter the DRX sleep time and the number of CSI-RS for CM measurement reporting times decreases, affecting network

Function Name	Function Switch	Reference	Description
			performance. For details about the impact analysis, see *DRX*. ● If both power saving BWP and DRX are enabled and the base station triggers a BWP2-to-BWP1 switching during the DRX sleep time, the DCI for BWP2-to-BWP1 switching may fail to be delivered. As a result, the base station stops downlink scheduling for the UE during the BWP switching, leading to downlink scheduling failures. After the **DRX_BWP2_OPT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, the base station can trigger UE switching from BWP2 to BWP1 only when the DRX inactivity

Function Name	Function Switch	Reference	Description
			timer is running. ● When the **BWP2_PERIOD_CSI_SW** option of the **NRDuCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, periodic PUCCH CSI reporting resources can be allocated to BWP2 UEs. This helps increase the proportion of UEs in the DRX sleep time and improve power saving gains. However, the average uplink and downlink UE throughput decreases, the delay increases, and the measured CQI value may increase or decrease. ● After downlink enhancement for power saving is enabled, the duration in which UEs stay in the narrow-bandwidth BWP2 decreases, accelerating data

Function Name	Function Switch	Reference	Description
			transmission and possibly increasing the proportion of UEs in the DRX sleep time. As a result, the number of false SR detections, BLER, number of retransmissions, and number of used CCEs and PRBs increase. For details about the impact of DRX sleep time on network performance, see *DRX*.
RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	If RF channel intelligent shutdown is enabled for certain NR TDD low-frequency AAUs/RRUs in automatic coverage compensation mode, as described in *Energy Conservation and Emission Reduction*, UEs switch back to BWP1 from BWP2.

Function Name	Function Switch	Reference	Description
Data convergence in slots in power saving scheduling	**POWER_SAVING_SCHEDULE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	When data convergence in slots is enabled for power saving scheduling, a larger value of the **NRDUCellPowerSaving**.***RlcFirstPktDelayThld*** parameter results in a higher degree of data convergence with a longer time. As a result, the number of scheduling failures of BWP2 UEs in a cell increases. When the scheduling failure rate of BWP2 UEs in a cell exceeds the value of the **NRDUCellUePwrSaving**.***Bwp2SchFailRateThld*** parameter, UEs are more likely to switch from BWP2 back to BWP1.

Other impacts:

Function Name	Function Switch	Reference	Description
SUL CoMP	**INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter	*Super Uplink*	UEs working on BWP2 for power saving do not support SUL CoMP.
SUL CoMP	**INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter	*UL and DL Decoupling*	UEs working on BWP2 for power saving do not support SUL CoMP.

Function Name	Function Switch	Reference	Description
UE bandwidth adaptation	**UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter	*Scalable Bandwidth*	When UE bandwidth adaptation is enabled (by selecting the **UE_BW_ADAPTIVE_SW** option of **NRDUCellBwp**.***BwpConfigSwitch***), the frequency-domain position of the narrow-bandwidth BWP2 is restricted to be within a range. In addition, the bandwidth of BWP2 may be adjusted.

Function Name	Function Switch	Reference	Description
NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	When both the function of NR DU cell resource management between network slices and the power saving BWP function are enabled: ● If the TTI-based control policy is configured but **NRDUCellRes.*DlMinRbStartPos*** and **NRDUCellRes.*UlMinRbStartPos*** are not specified for the former function, power saving BWP does not take effect for all UEs in the network slice group in an NR DU cell. ● If the TTI-based control policy is configured and **NRDUCellRes.*DlMinRbStartPos*** or **NRDUCellRes.*UlMinRbStartPos*** is specified for the former function, power saving BWP does not take effect for all UEs in an NR DU cell, regardless of whether these

Function Name	Function Switch	Reference	Description
			UEs are in a network slice group.
Non-gap-assisted measurements for active BWP	**SSB_MEAS_WITHOUT_GAP_SW** option of the **NRDUCellMeas**.***SsbMeasPolicySwitch*** parameter	*SA Mobility Management in Connected Mode*	When the function of non-gap-assisted measurements for active BWP and the power saving BWP function are both enabled, the frequency-domain range of BWP2 is used to calculate the frequency-domain range of the active BWP enabled with non-gap-assisted measurements. In this case, the scenarios where the function of non-gap-assisted measurements for active BWP takes effect are reduced.

5.1.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

Downlink enhancement for power saving: supported only by the MetaAAU

Other functions:

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.4.4 Cells

When **NRDUCell**.***SlotAssignment*** is set to **2_3_DSUUU** or **1_1_DSDSDSDSDS**, the power saving BWP function is not supported.

5.1.4.5 Others

Requirements on UEs

UEs must support multiple (two or more) dedicated BWPs. As defined in 3GPP TS 38.331, a UE supports multiple dedicated BWPs if its capability field contains **bwp-SameNumerology**.

5.1.5 Operation and Maintenance

5.1.5.1 Data Configuration

5.1.5.1.1 Data Preparation

The following tables describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-5 Parameters used for activation (power saving BWP)

Parameter Name	Parameter ID	Option	Setting Notes
Enhanced UE Power Saving Switch	**NRDUCellUePwrSaving**.***UePowerSavingSwitch***	UE_PWR_SAVING_ENHANCE_SW	Select this option.
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	BWP2_SWITCH	Select this option.
BWP1 to BWP2 DL Throughput Threshold	**NRDUCellUePwrSaving**.***Bwp1ToBwp2DlThptThld***	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
BWP1 to BWP2 UL Throughput Threshold	**NRDUCellUePwrSaving**.***Bwp1ToBwp2UlThptThld***	None	Use the recommended value.
CA Deactivate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellDeactThldSlotNum***	None	Use the recommended value.
CA Deactivate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.***CaUlScellDeactThldSlotNum***	None	Use the recommended value.
CA Activate and BWP Switch DL Thld Slot Num	**NRDUCellCarrMgmt**.***CaDlScellActThldSlotNum***	None	Use the recommended value.
CA Activate and BWP Switch UL Thld Slot Num	**NRDUCellCarrMgmt**.***CaUlScellActThldSlotNum***	None	Use the recommended value.
DL State Transition Buffer Volume Threshold	**NRDUCellCarrMgmt**.***DlStateTransitBufVolThld***	None	The value **20** is recommended.
BWP2 Maximum RB Number	**NRDUCellUePwrSaving**.***Bwp2MaxRbNum***	None	Set this parameter based on the network plan. This parameter can only be set to **RB51** or **RB64** in low-frequency TDD scenarios. To set the BWP2 bandwidth to 20 MHz, set this parameter to **RB51**.
BWP1 to BWP2 Evaluation Period	**NRDUCellUePwrSaving**.***Bwp1ToBwp2EvaluatePeriod***	None	Use the recommended value.

Table 5-6 Parameters used for activation (BWP switching reliability optimization)

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	BWP_SWITCHING_OPT_SW	This option is selected by default.

Table 5-7 Parameters used for activation (BWP2 scheduling failure rate threshold)

Parameter Name	Parameter ID	Option	Setting Notes
BWP2 Scheduling Failure Rate Threshold	**NRDUCellUePwrSaving**.***Bwp2SchFailRateThld***	None	Use the recommended value.

Table 5-8 Parameters used for activation (power saving BWP CCE adjustment policy)

Parameter Name	Parameter ID	Option	Setting Notes
Power Saving BWP CCE Adjustment Policy	**NRDuCellPdcch**.***PwrSaveBwpCceAdjustPolicy***	None	If power saving is preferred, the value **PWR_SAVING_BWP_NOT_CONSIDERED** is recommended. If performance is preferred, the value **PWR_SAVING_BWP_CONSIDERED** is recommended.

Table 5-9 Parameters used for activation (exit of weak-coverage UEs from BWP2)

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	WEAK_COV_BWP2_EXIT_SW	Select this option if you want to enable exit of weak-coverage UEs from BWP2.

Parameter Name	Parameter ID	Option	Setting Notes
Weak Coverage Threshold Offset	**NRDUCellUePwrSaving**.***WeakCoverageThldOffset***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Initial DMRS RSRP Weak Coverage Threshold	**NRDUCellDmrs**.***InitDmrsRsrpWeakCovThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Initial DMRS SINR Weak Coverage Threshold	**NRDUCellDmrs**.***InitDmrsSinrWeakCovThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Weak Coverage CQI Threshold	**NRDUCellUePwrSaving**.***WeakCoverageCqiThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.
Weak Coverage UL SINR Threshold	**NRDUCellUePwrSaving**.***WeakCoverageUlSinrThld***	None	Configure this parameter if enabling exit of weak-coverage UEs from BWP2 is required. Use the recommended value.

Table 5-10 Parameters used for activation (DCI scheduling optimization for switching from BWP2 to BWP1)

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	BWP2_SWITCHING_DCI_SCH_OPT_SW	It is recommended that this option be selected to enable DCI scheduling optimization for switching from BWP2 to BWP1.

Table 5-11 Parameters used for activation (QCI-level configuration for power saving BWP)

Parameter Name	Parameter ID	Option	Setting Notes
QCI BWP2 Indicator	**NRDUCellQciBearer**.***QciBwp2Ind***	None	If power saving BWP needs to be disabled separately for ToB services, set this parameter to **NOT_SUPPORT** for corresponding QCIs.

Table 5-12 Parameters used for activation (QCI-level threshold control for BWP switching)

Parameter Name	Parameter ID	Option	Setting Notes
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	QCI_CA_BWP_SWITCH_THLD_CTRL_SW	Select this option if the thresholds for BWP switching needs to be changed for QCI-specific bearers.
QCI BWP1 to BWP2 DL Throughput Threshold	**NRDUCellQciBearer**.***QciBwp1ToBwp2DlThptThld***	None	Use the recommended value.
QCI BWP1 to BWP2 UL Throughput Threshold	**NRDUCellQciBearer**.***QciBwp1ToBwp2UlThptThld***	None	Use the recommended value.
QCI DL State Transit Buffer Volume Threshold	**NRDUCellQciBearer**.***QciDlStTransitBufVolThld***	None	Use the recommended value. This parameter is invalid when it is set to **65535**.

Table 5-13 Parameters used for activation (downlink enhancement for power saving)

Parameter Name	Parameter ID	Option	Setting Notes
DL Link Adaptation Enhancement Switch	**NRDUCellPdsch**.***DlLinkAdaptEnhancementSw***	PWR_SAVING_DOWNLINK_ENH_SW	This option is selected by default.

Table 5-14 Parameters used for activation (periodic CSI resource allocation for BWP2 UEs)

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	BWP2_PERIOD_CSI_SW	Select this option if UE power saving is required.

Table 5-15 Parameters used for activation (Type of UEs [in terms of the networking mode] on which power saving BWP can take effect)

Parameter Name	Parameter ID	Option	Setting Notes
BWP2 Networking Mode	**NRDUCellUePwrSaving**.***Bwp2NetworkingMode***	None	Use the recommended value.

Table 5-16 Parameters used for activation (SRS resource reservation for BWP2 UEs in RedCap scenarios)

Parameter Name	Parameter ID	Option	Setting Notes
RedCap Algorithm Switch	**NRDUCellRedCap**.***RedCapAlgoSwitch***	BWP2_SRS_RSV_SW	It is recommended that this option be selected when the RedCap load is heavy.

Table 5-17 Parameters used for activation (BWP2 PUCCH frequency-domain position adjustment)

Parameter Name	Parameter ID	Option	Setting Notes
PUCCH Algorithm Extension Switch	**NRDUCellPucch**.***PucchAlgoExtSwitch***	BWP2_PUCCH_FREQ_POS_ADJ	Select this option if BWP2 PUCCH frequency-domain position adjustment is required.

5.1.5.1.2 Using MML Commands

A cell automatically resets when the status of the **BWP2_SWITCH** or **BWP2_PERIOD_CSI_SW** option is changed. Before turning on the **BWP2_SWITCH** switch, ensure that the license for the UE Power Saving feature has been installed on the base station. Otherwise, the cell cannot be activated after an automatic reset.

Before using MML commands, refer to **5.1.4 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Power saving BWP

```
//Turning on the UE power saving enhancement switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-1;
//Turning on the power saving BWP switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP2_SWITCH-1;
//Configuring the downlink and uplink throughput thresholds for BWP1-to-BWP2 switching
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp1ToBwp2DlThptThld=1000,
Bwp1ToBwp2UlThptThld=250;
//(Optional) Configuring the maximum number of RBs for BWP2 when the BWP2 bandwidth needs to be
set to 20 MHz
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp2MaxRbNum=RB51;
//Setting the number of slots used for calculating the downlink and uplink traffic volume thresholds for CA
SCell deactivation and BWP switching
MOD NRDUCELLCARRMGMT: NrDuCellId=1, CaDlScellDeactThldSlotNum=4, CaUlScellDeactThldSlotNum=1;
//Setting the number of slots used for calculating the downlink and uplink traffic volume thresholds for CA
SCell activation and BWP switching
MOD NRDUCELLCARRMGMT: NrDuCellId=1, CaDlScellActThldSlotNum=16, CaUlScellActThldSlotNum=4;
//Configuring the threshold of buffered traffic volume for downlink state transitions
MOD NRDUCELLCARRMGMT: NrDuCellId=1, DlStateTransitBufVolThld=20;
//Configuring the period for determining whether to switch from BWP1 to BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp1ToBwp2EvaluatePeriod=10;
```

- BWP switching reliability optimization

```
//Turning on the BWP switching optimization switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP_SWITCHING_OPT_SW-1;
```

- BWP2 scheduling failure rate threshold

```
//Configuring the threshold expressed as a probability of failure to schedule UEs in BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp2SchFailRateThld=4;
```

- Power saving BWP CCE adjustment policy

```
//Configuring the power saving BWP CCE adjustment policy (using the power saving preferred scenario as
an example)
MOD NRDUCELLPDCCH: NrDuCellId=1,
PwrSaveBwpCceAdjustPolicy=PWR_SAVING_BWP_NOT_CONSIDERED;
```

- Exit of weak-coverage UEs from BWP2

```
//Turning on the switch for exit of weak-coverage UEs from BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=WEAK_COV_BWP2_EXIT_SW-1;
//Configuring the initial DMRS RSRP weak coverage threshold and initial DMRS SINR weak coverage
threshold
MOD NRDUCELLDMRS: NrDuCellId=1, InitDmrsRsrpWeakCovThld=-130, InitDmrsSinrWeakCovThld=0;
//Configuring the weak coverage threshold offset, weak coverage CQI threshold, and weak coverage uplink
```



```
SINR threshold
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, WeakCoverageThldOffset=5, WeakCoverageCqiThld=5, WeakCoverageUlSinrThld=5;
```

- DCI scheduling optimization for switching from BWP2 to BWP1

```
//Turning on the switch for DCI scheduling optimization for switching from BWP2 to BWP1
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP2_SWITCHING_DCI_SCH_OPT_SW-1;
```

- QCI-level configuration for power saving BWP

```
//Setting the NRDUCellQciBearer.QciBwp2Ind parameter to NOT_SUPPORT for a QCI (QCI 82 is used as an example) so that power saving BWP can be disabled separately for ToB services
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=82, QciBwp2Ind=NOT_SUPPORT;
```

- QCI-level threshold control for BWP switching

```
//Enabling QCI-level threshold control for BWP switching (using a single bearer of QCI 80 as an example)
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=80,ServiceDiffAlgoSwitch=QCI_CA_BWP_SWITCH_THLD_CTRL_SW-1;
//Configuring the QCI-specific downlink and uplink throughput thresholds for BWP1-to-BWP2 switching
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=80, QciBwp1ToBwp2DlThptThld=1000, QciBwp1ToBwp2UlThptThld=250;
//Configuring the QCI-specific threshold of buffered traffic volume for downlink state transitions
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=80, QciDlStTransitBufVolThld=20;
```

- Downlink enhancement for power saving

```
//Enabling downlink enhancement for power saving
MOD NRDUCELLPDSCH: NrDuCellId=1, DlLinkAdaptEnhancementSw=PWR_SAVING_DOWNLINK_ENH_SW-1;
```

- Periodic CSI resource allocation for BWP2 UEs

```
//Enabling periodic CSI resource allocation for BWP2 UEs
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP2_PERIOD_CSI_SW-1;
```

- Type of UEs (in terms of the networking mode) on which power saving BWP can take effect

```
//Configuring the type of UEs (in terms of the networking mode) on which power saving BWP can take effect
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, Bwp2NetworkingMode=SA_AND_NSA;
```

- SRS resource reservation for BWP2 UEs in RedCap scenarios

```
//Enabling SRS resource reservation for BWP2 UEs in RedCap scenarios
MOD NRDUCELLREDCAP: NrDuCellId=1, RedCapAlgoSwitch=BWP2_SRS_RSV_SW-1;
```

- BWP2 PUCCH frequency-domain position adjustment

```
//Enabling BWP2 PUCCH frequency-domain position adjustment
MOD NRDUCELLPUCCH: NrDuCellId=1, PucchAlgoExtSwitch=BWP2_PUCCH_FREQ_POS_ADJ-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- BWP switching reliability optimization

```
//Turning off the BWP switching optimization switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP_SWITCHING_OPT_SW-0;
```

- Exit of weak-coverage UEs from BWP2

```
//Turning off the switch for exit of weak-coverage UEs from BWP2
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=WEAK_COV_BWP2_EXIT_SW-0;
```

- DCI scheduling optimization for switching from BWP2 to BWP1

```
//Turning off the switch for DCI scheduling optimization for switching from BWP2 to BWP1
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP2_SWITCHING_DCI_SCH_OPT_SW-0;
```

- QCI-level threshold control for BWP switching

```
//Disabling QCI-level threshold control for BWP switching (using a single bearer of QCI 80 as an example)
MOD NRDUCELLQCIBEARER: NrDuCellId=1,
Qci=80,ServiceDiffAlgoSwitch=QCI_CA_BWP_SWITCH_THLD_CTRL_SW-0;
```

- Downlink enhancement for power saving

```
//Disabling downlink enhancement for power saving
MOD NRDUCELLPDSCH: NrDuCellId=1, DlLinkAdaptEnhancementSw=PWR_SAVING_DOWNLINK_ENH_SW-0;
```

- Periodic CSI resource allocation for BWP2 UEs

```
//Disabling periodic CSI resource allocation for BWP2 UEs
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP2_PERIOD_CSI_SW-0;
```

- SRS resource reservation for BWP2 UEs in RedCap scenarios

```
//Disabling SRS resource reservation for BWP2 UEs in RedCap scenarios
MOD NRDUCELLREDCAP: NrDuCellId=1, RedCapAlgoSwitch=BWP2_SRS_RSV_SW-0;
```

- BWP2 PUCCH frequency-domain position adjustment

```
//Disabling BWP2 PUCCH frequency-domain position adjustment
MOD NRDUCELLPUCCH: NrDuCellId=1, PucchAlgoExtSwitch=BWP2_PUCCH_FREQ_POS_ADJ-0;
```

- Power saving BWP

```
//Turning off the power saving BWP switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP2_SWITCH-0;
//Turning off the UE power saving enhancement switch. This switch also controls other functions. Before
performing this operation, check whether this switch needs to be turned off.
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-0;
```

5.1.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.5.2 Activation Verification

Tracing Signaling

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 In the navigation tree on the left, choose **Uu Interface Trace**.

Step 3 On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If both **BWP-Downlink** and **BWP-Uplink** contain two BWP IDs, the UE is configured with two BWPs.

----End

The BWP2 PUCCH frequency-domain position adjustment function can be verified by observing the PUCCH frequency-domain position. The PUCCH frequency-domain position can be observed through the fields in the **pucch-Config** IE in the RRC_RECFG message.

Counter Observation

Observe the following counters to check whether the power saving BWP function has taken effect. If the value of the **N.Bwp2.Dur.Total** counter is not 0, power saving BWP has taken effect.

5.1.5.3 Network Monitoring

After the power saving BWP function is enabled, observe the following counters for network monitoring:

<table>
<tr><th>Counter ID</th><th>Counter Name</th><th>Monitoring Purpose</th></tr>
<tr><td>1911825117</td><td>N.Bwp2.Dur.Total</td><td>After the power saving BWP function is enabled, this counter can be used to measure the total duration of all UEs working in BWP2 in a cell.</td></tr>
<tr><td>1911825121</td><td>N.BWP.ThpVol.DL</td><td rowspan="6">After the power saving BWP function is enabled, these counters can be used to measure the average throughput of UEs working in BWP0 and BWP1 in a cell.
UEs working in BWP0 are those working in the initial BWP.</td></tr>
<tr><td>1911825125</td><td>N.BWP.ThpVol.DL.LastSlot</td></tr>
<tr><td>1911825128</td><td>N.BWP.ThpTime.DL.RmvLastSlot</td></tr>
<tr><td>1911825129</td><td>N.BWP.ThpVol.UL</td></tr>
<tr><td>1911825123</td><td>N.BWP.ThpVol.UE.UL.SmallPkt</td></tr>
<tr><td>1911825127</td><td>N.BWP.ThpTime.UE.UL.RmvSmallPkt</td></tr>
</table>

Average downlink throughput of UEs working in BWP0 and BWP1 = (**N.BWP.ThpVol.DL – N.BWP.ThpVol.DL.LastSlot**) x 1000/ **N.BWP.ThpTime.DL.RmvLastSlot**

Average uplink throughput of UEs working in BWP0 and BWP1 = (**N.BWP.ThpVol.UL – N.BWP.ThpVol.UE.UL.SmallPkt**) x 1000/ **N.BWP.ThpTime.UE.UL.RmvSmallPkt**

NOTE

- The gain of UE power saving is closely related to the BWP capability of UEs in a cell and the traffic model. Therefore, the gain of UE power saving needs to be observed and evaluated on the UE side, not by using the **N.Bwp2.Dur.Total** counter.
- When the switch for power saving BWP is off, the average UE throughputs (indicated by **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**), targeted for measurement of all UEs in a cell, differ from the average UE throughputs in BWP0 and BWP1 in terms of the measurement dimensions. Therefore, the final measurement results differ slightly.
- The average throughputs of UEs working in BWP0 and BWP1 is related to the traffic model and the measured values may be 0.

The indicators described in **5.1.3.2 Impacts** can be used for network monitoring of other functions.

5.2 Reducing the Number of SCCs

5.2.1 Principles

5.2.1.1 Overview

This function uses standard protocol-compliant procedures to reduce the number of SCCs for UEs to achieve overheating protection and UE power saving, as shown in **Figure 5-5**. The process of reducing the number of SCCs for overheating protection and that for UE power saving are different. **Table 5-18** describes the conditions for individual processes to take effect. The function of reducing the number of SCCs for overheating protection described in this section complies with 3GPP Release 15 and does not support NSA networking. For details about the functions applicable to NSA networking, see the 3GPP Release 14-compliant function SCG release and addition based on UE overheating status reporting described in *EPC-based NSA Basics*.

Figure 5-5 Schematic diagram of reducing the number of SCCs

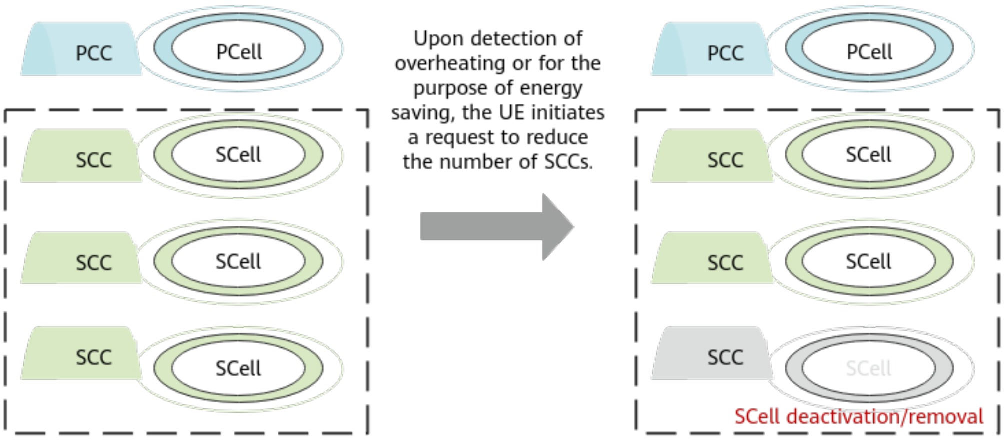


Table 5-18 Conditions for the function of reducing the number of SCCs for overheating protection and that for UE power saving to take effect

Process	Parameter	Scenario	UE Capability
3GPP Release 15-compliant reducing the number of SCCs for overheating protection (**5.2.1.2 Process of Reducing the Number of SCCs for UE Overheating Protection**)	Both the **OVERHEATING_ASSISTANCE_INFO_SW** and **UE_SCC_DEACTIVATE_SW** options of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter are selected.	• Low-frequency SA networking • High-frequency SA networking • Inter-FR CA in SA networking	The **overheatingInd** capability defined in 3GPP Release 15
3GPP Release 16-compliant reducing the number of SCCs for UE power saving (**5.2.1.3 Process of Reducing the Number of SCCs for UE Power Saving**)	• The **UE_SCC_DEACTIVATE_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected. • The **UAI_NSA_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected in low-frequency NSA networking.	• Low-frequency SA/NSA networking • High-frequency SA/NSA networking • Inter-FR CA in SA networking	The **maxCC-Preference-r16** capability defined in 3GPP Release 16

The settings of the preceding options take effect only on UEs that newly access or are newly handed over to the cell.

When both the conditions for the function of reducing the number of SCCs for overheating protection and that for UE power saving are met, both functions can take effect. If the UE uses both the **OverheatingAssistance** and **maxCC-Preference-r16** fields in the UEAssistanceInformation message to indicate the number of SCCs to be reduced, the base station uses the smaller one of the reported values. That is, the expected number of SCCs that can be activated in the downlink is min{reducedCCsDL, reducedCCsDL-r16}, and the expected number of SCCs that can be activated in the uplink is min{reducedCCsUL, reducedCCsUL-r16}.

The function of reducing the number of SCCs is not used for trial in low-frequency band scenarios, and the function of reducing the number of SCCs for overheating protection is not used for trial in FWA-specific high-frequency SA networking scenarios. The function of reducing the number of SCCs is used for trial in other high-frequency band scenarios and inter-FR CA scenarios.

NOTE

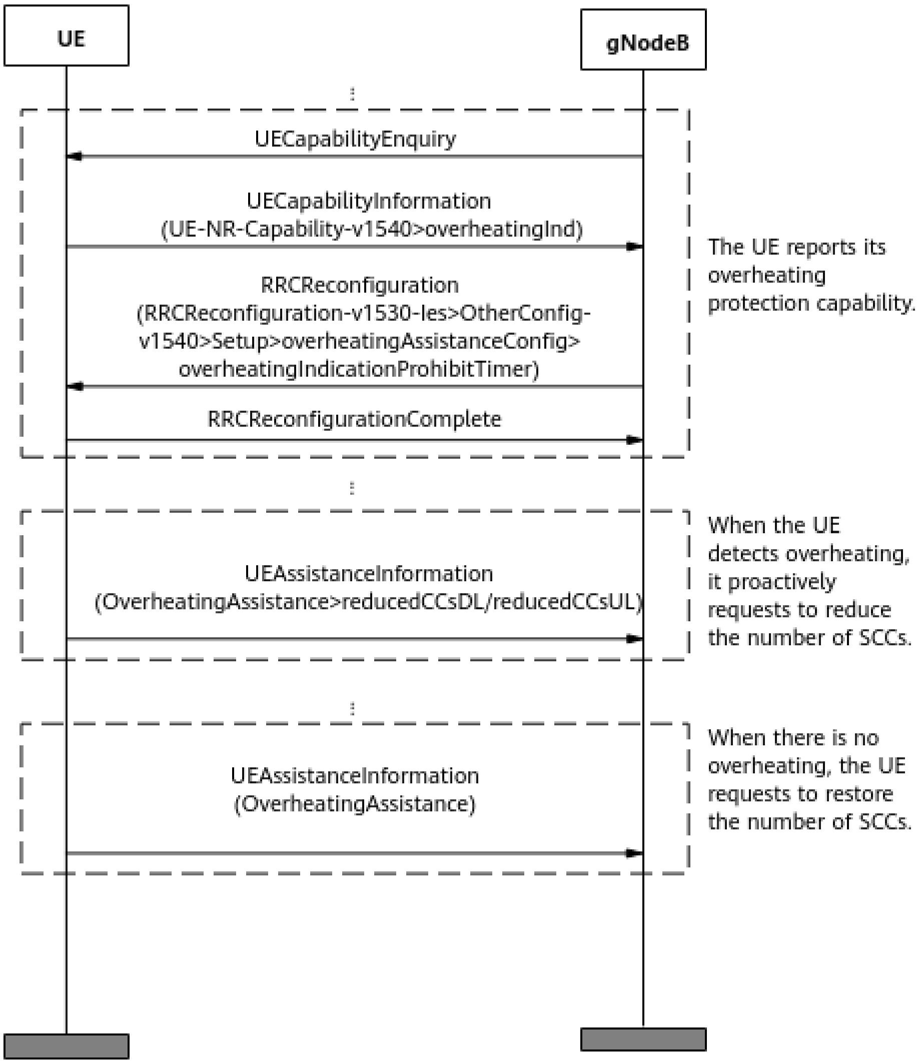
Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

5.2.1.2 Process of Reducing the Number of SCCs for UE Overheating Protection

Figure 5-6 shows the process of UE overheating protection.

Figure 5-6 UE overheating protection process



1. The UE reports its overheating protection capability.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.

   b. The UE returns a UECapabilityInformation message to the base station. In this message, the **overheatingInd** field indicates that the UE supports overheating protection.

   c. The base station sends an RRCReconfiguration message containing the **overheatingIndicationProhibitTimer** field to the UE to configure the overheating indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report overheating messages, which can be set using the **NRCellUeTimer**.***OverheatingIndProhibitTmr*** parameter. This prevents the UE from frequently reporting the overheating indication.

   d. The UE returns an RRCReconfigurationComplete message to the base station.

2. When overheating occurs, the UE makes a request to reduce the number of SCCs.

   When the UE detects overheating, it sends a UEAssistanceInformation message to the base station. If all of the following conditions are met, the UE overheats and expects the base station to reduce the number of SCCs configured for the UE:

   – The **OverheatingAssistance** field contains **reducedCCsDL** and **reducedCCsUL**.
   – The value of **reducedCCsDL** is less than the maximum number of downlink SCCs supported by the UE, and the value of **reducedCCsUL** is less than the maximum number of uplink SCCs supported by the UE.

   When the UE proactively initiates a request to reduce the number of SCCs, the base station deactivates a certain number of SCCs as requested by the UE. In addition, in inter-FR CA, if the value of **reducedCCsDL** or **reducedCCsUL** reported by the UE is **0**, the base station proactively removes high-frequency SCells for the UE due to UE overheating.

3. When there is no overheating, the UE can request to restore the number of SCCs.

   When the UE in overheating protection detects that it no longer overheats, it sends a UEAssistanceInformation message to the base station. If the **OverheatingAssistance** field does not contain any information, the UE does not overheat and expects the base station to restore the number of SCCs allowed to be activated for the UE.

   After the UE initiates an SCC restoration procedure, the restoration is not performed immediately. Instead, it acts according to the SCell activation, SCell configuration, mechanism. For details about SCell activation and SCell configuration, see the section "SCell Management" in *Carrier Aggregation*.

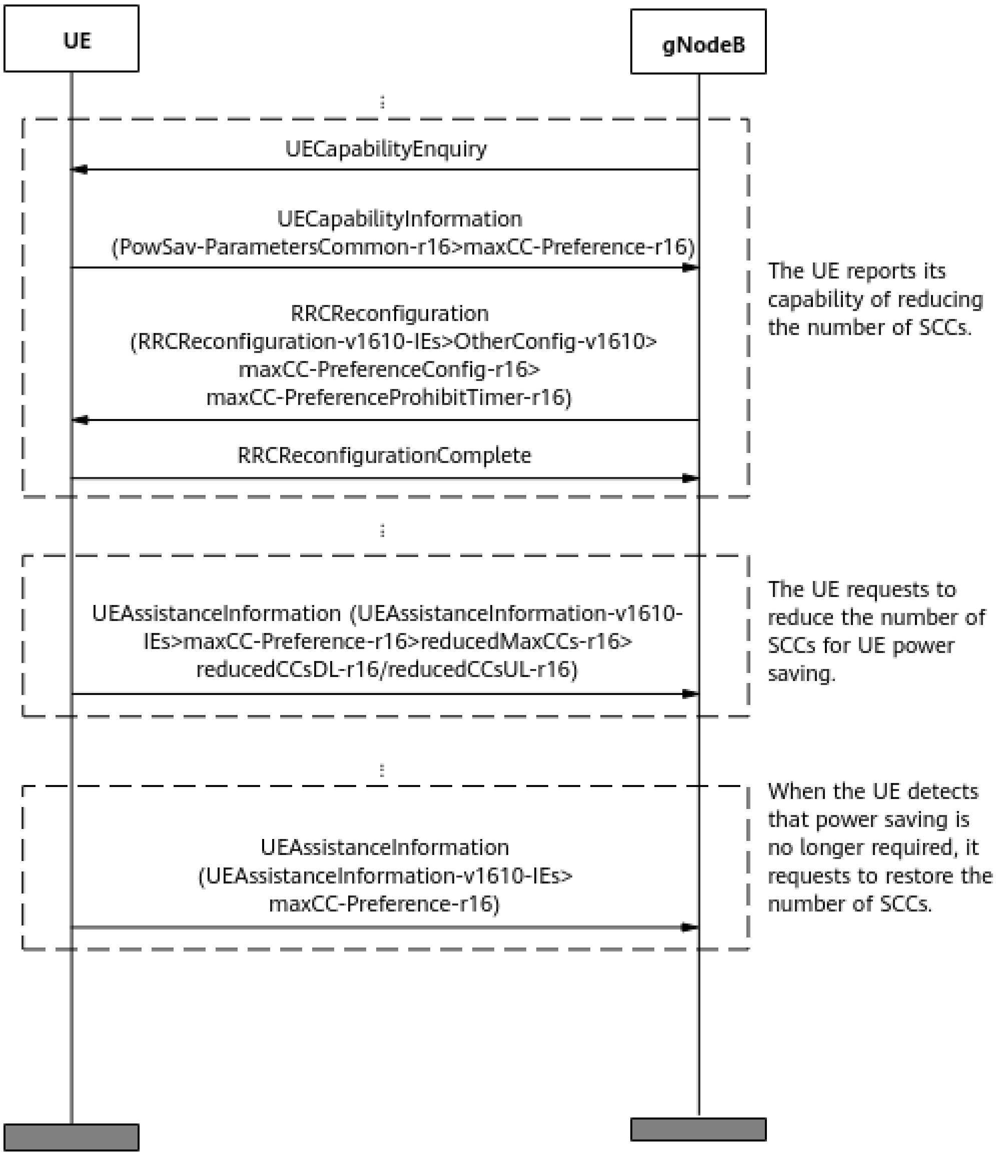
NOTE

The number of activated SCCs allowed in the uplink does not exceed the number of activated SCCs allowed in the downlink.

5.2.1.3 Process of Reducing the Number of SCCs for UE Power Saving

Figure 5-7 shows the process of reducing the number of SCCs for UE power saving.

Figure 5-7 Process of reducing the number of SCCs for UE power saving



1. The UE reports its 3GPP Release 16-compliant capability for reducing the number of SCCs.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.
   b. The UE returns a UECapabilityInformation message to the base station. In this message, the **maxCC-Preference-r16** field indicates that the UE supports the 3GPP Release 16-compliant capability for reducing the number of SCCs.
   c. The base station sends an RRCReconfiguration message containing the **maxCC-PreferenceProhibitTimer-r16** field to the UE to configure the SCC reduction indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report SCC reduction messages. This prevents the UE from frequently reporting the SCC reduction indication.

d. The UE returns an RRCReconfigurationComplete message to the base station.

2. The UE proactively makes a request to reduce the number of SCCs for UE power saving.

   The UE sends a UEAssistanceInformation message to the base station for UE power saving. If all of the following conditions are met, the base station is expected to reduce the number of SCCs for UE power saving:

   – The **maxCC-Preference-r16** field carries **reducedMaxCCs-r16>reducedCCsDL-r16** and **reducedMaxCCs-r16>reducedCCsUL-r16**.
   – The value of **reducedCCsDL-r16** is less than the maximum number of downlink SCCs supported by the UE, and the value of **reducedCCsUL-r16** is less than the maximum number of uplink SCCs supported by the UE.

   When the UE proactively initiates a request to reduce the number of SCCs, the base station deactivates a certain number of SCCs as requested by the UE. In addition, in inter-FR CA, if the value of **reducedCCsDL** or **reducedCCsUL** reported by the UE is 0, the base station proactively removes high-frequency SCells for the UE due to UE overheating.

   In NSA networking, if the values of **reducedCCsDL-r16** and **reducedCCsUL-r16** reported by a UE are both 0, the gNodeB proactively triggers the SCG removal procedure for power saving. For details, see *NSA Mobility Management*.

3. When power saving is no longer required, the UE can initiate SCC restoration.

   When power saving is no longer required, the UE sends a UEAssistanceInformation message to the base station. If the **maxCC-Preference-r16** field does not contain any information, the UE does not require power saving and expects the base station to restore the number of SCCs allowed to be activated for the UE.

   After the UE initiates an SCC restoration procedure, the restoration is not performed immediately. Instead, it acts according to the SCell activation, SCell configuration, or SCG addition mechanism. For details about SCell activation and SCell configuration, see the section "SCell Management" in *Carrier Aggregation*. For details about the SCG addition procedure, see *NSA Mobility Management*.

   NOTE

   The number of activated SCCs allowed in the uplink does not exceed the number of activated SCCs allowed in the downlink.

5.2.1.4 Blacklist and Whitelist Control over Reducing the Number of SCCs for UE Power Saving

The function of reducing the number of SCCs for UE power saving defined in 3GPP Release 16 supports the blacklist and whitelist mechanism.

- If the **UE_SCC_DEACTIVATE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter is selected, the function of reducing the number of SCCs is disabled for specified UEs. That is, this function does not take effect on the specified UEs and can take effect on other UEs.
- If the **UE_SCC_DEACTIVATE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the function

of reducing the number of SCCs is enabled for specified UEs. That is, this function can take effect on the specified UEs and does not take effect on other UEs.

The **UE_SCC_DEACTIVATE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UE_SCC_DEACTIVATE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

5.2.2 Network Analysis

5.2.2.1 Benefits

When the function of reducing the number of SCCs is enabled, the power consumption of communication modules of UEs can be reduced. The specific reduction in power consumption is related to the number of reduced SCCs. The power reduction is also related to the chips of UEs. For example, the power consumption of Huawei UEs can be reduced by up to 20%.

Assume that the function of reducing the number of SCCs is enabled. When the **UE_SCC_DEACTIVATE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter or the **UE_SCC_DEACTIVATE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the number of UEs with reducing the number of SCCs in effect decreases, the average cell and UE throughputs increase, the value of the **N.CA.SCell.Act.Att**, **N.CA.SCell.Act.Succ**, **N.CA.SCell.Deact.Att**, or **N.CA.SCell.Deact.Succ** counter may decrease, and the power consumption of UEs without reducing the number of SCCs in effect increases.

5.2.2.2 Impacts

The impacts between this function and other functions are described in **5.2.3.2.3 Function Impacts**.

- On the UE side:

  The UE proactively makes a request to reduce the number of SCCs and any impact on the UE side does not involve the base station. After this function is enabled, if a UE initiates SCC reduction on the network, the base station reduces the number of SCCs allowed to be activated for the UE. As a result, the user-perceived service rate decreases along with UE power saving.
- On the base station side, the possible impacts are as follows:
  - The average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) decreases.
  - The average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases.
  - The average uplink UE throughput (**User Uplink Average Throughput (DU)**) decreases.
  - The average downlink UE throughput (**User Downlink Average Throughput (DU)**) decreases.

- The value of **N.CA.SCell.Act.Att** or **N.CA.SCell.Act.Succ** increases.
- The value of **N.CA.SCell.Deact.Att** or **N.CA.SCell.Deact.Succ** increases.
- If the function of reducing the number of SCCs is enabled in NSA networking, values of the following counters increase when SCG removal is triggered for UE power saving:
  - **N.NsaDc.SgNB.Add.Att** or **N.NsaDc.SgNB.Add.Succ**
  - **N.NsaDc.SgNB.Rel**

5.2.3 Requirements

5.2.3.1 Licenses

There are no license requirements.

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter **INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The function of reducing the number of SCCs takes effect only when carrier aggregation is enabled. At least one of the following functions must be enabled: ● Intra-band CA ● Intra-FR inter-band CA ● Inter-gNodeB CA ● Inter-FR CA
Intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter **INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	

Function Name	Function Switch	Reference	Description
Inter-FR CA (Trial)	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The function of reducing the number of SCCs takes effect only when carrier aggregation is enabled. At least one of the following functions must be enabled: ● Intra-band CA ● Intra-FR inter-band CA ● Inter-gNodeB CA ● Inter-FR CA

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	Inter-FR CA (Trial)	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the function of reducing the number of SCCs is enabled in SA-networking inter-FR CA scenarios, values of the following counters increase when high-frequency SCell removal is triggered for overheating protection or UE power saving: ● **N.CA.SCell.Add.Att** or **N.CA.SCell.Add.Succ** ● **N.CA.SCell.Rmv.Att** or **N.CA.SCell.Rmv.Succ**
High-frequency TDD	Flexible carrier management (high-frequency TDD)	**NRDUCellCarrMgmt**.***Fr2CaCarrMgmtPolicy*** set to **USER_TRAFFIC_BASED**	*Carrier Aggregation*	● If an SCell is deactivated for overheating protection or UE energy saving for more than 5s, dynamic SCell removal will not be triggered. ● The gNodeB does not trigger dynamic SCell configuration for UEs in the overheated or energy saving state.

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.2.3.4 Others

UEs must support the **overheatingInd** capability defined in 3GPP Release 15 or the **maxCC-Preference-r16** capability defined in 3GPP Release 16.

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Table 5-19 describes the parameters used for function activation.

Table 5-19 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	UE_SCC_DEACTIVATE_SW	Select this option.
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	OVERHEATING_ASSISTANCE_INFO_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	UAI_NSA_SW	Select this option in low-frequency NSA networking.
Overheating Indication Prohibit Timer	**NRCellUeTimer**.***OverheatingIndProhibitTmr***	None	Use the recommended value.
Blacklist Control Switch	**gNBUeCompatSw**.***BlacklistCtrlSwitch***	UE_SCC_DEACTIVATE_SW_OFF	Select this option if a blacklist mechanism needs to be used to disable the function of reducing the number of SCCs for specified UEs.
Whitelist Control Switch	**gNBUeCompatSw**.***WhitelistCtrlExtSwitch***	UE_SCC_DEACTIVATE_SW_ON	Select this option if a whitelist mechanism needs to be used to enable the function of reducing the number of SCCs for specified UEs.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to **5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- The **MOD GNBUECOMPATSW** command is a high-risk command. Running this command changes the feature activation states for specified UEs.
- The **UE_SCC_DEACTIVATE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UE_SCC_DEACTIVATE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

Activation Command Examples

```
//Turning on the switch for UE SCC deactivation
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UE_SCC_DEACTIVATE_SW-1;
//Turning on the overheating assistance information processing switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=OVERHEATING_ASSISTANCE_INFO_SW-1;
//Configuring the overheating indication prohibition timer
MOD NRCELLUETIMER: NrCellId=1, OverheatingIndProhibitTmr=S30;
//Disabling the function of reducing the number of SCCs for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=UE_SCC_DEACTIVATE_SW_OFF-1;
//Enabling the function of reducing the number of SCCs for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UE_SCC_DEACTIVATE_SW_ON-1;
```

Turning on the NSA UAI switch in low-frequency NSA networking

```
//Turning on the NSA UAI switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UAI_NSA_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for UE SCC deactivation
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UE_SCC_DEACTIVATE_SW-0;
//Turning off the overheating assistance information processing switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=OVERHEATING_ASSISTANCE_INFO_SW-0;
//Disabling the blacklist mechanism of reducing the number of SCCs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=UE_SCC_DEACTIVATE_SW_OFF-0;
//Disabling the whitelist mechanism of reducing the number of SCCs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UE_SCC_DEACTIVATE_SW_ON-0;
```

Turning off the NSA UAI switch in low-frequency NSA networking

```
//Turning off the NSA UAI switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UAI_NSA_SW-0;
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Tracing Signaling

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 In the navigation tree on the left of the displayed **Signaling Trace Management** page, choose **Uu Interface Trace**.

Step 3 On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. If **overheatingAssistanceConfig>overheatingIndicationProhibitTimer** or **maxCC-**

PreferenceConfig-r16>maxCC-PreferenceProhibitTimer-r16 exists in the message, this function has taken effect.

----End

5.2.4.3 Network Monitoring

Subscribe to CHR logs. The value of the **MethodCount** field corresponding to **MethodId** whose value is **1** in the PRIVATE_UE_RELEASE event block indicates the number of times a specific UE reports SCC reduction for overheating protection or UE power saving.

The indicators described in **5.2.2.2 Impacts** can be used for network monitoring of this function.

5.3 Maximum-Bandwidth Reduction

5.3.1 Principles

5.3.1.1 Overview

This function reduces the maximum bandwidth of UEs to save power and provide overheating protection for UEs. This function applies only to low-frequency UEs. The process of maximum-bandwidth reduction for overheating protection and that for UE power saving are different. **Table 5-20** describes the conditions for individual processes to take effect.

Table 5-20 Conditions for the maximum-bandwidth reduction function for overheating protection and that for UE power saving to take effect

Process	Parameter	UE Capability
3GPP Release 15-compliant maximum-bandwidth reduction for overheating protection (described in **5.3.1.2 Process of Maximum-Bandwidth Reduction for UE Overheating Protection**)	• Both the **OVERHEATING_ASSISTANCE_INFO_SW** and **UE_BW_REDUCE_SW** options of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter are selected. • In NSA networking, the **UAI_NSA_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter also needs to be selected.	The **overheatingInd** capability defined in 3GPP Release 15

Process	Parameter	UE Capability
3GPP Release 16-compliant maximum-bandwidth reduction for UE power saving (described in **5.3.1.3 Process of Maximum-Bandwidth Reduction for UE Power Saving**)	• The **UE_BW_REDUCE_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected. • In NSA networking, the **UAI_NSA_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter also needs to be selected.	The **maxBW-Preference-r16** capability defined in 3GPP Release 16

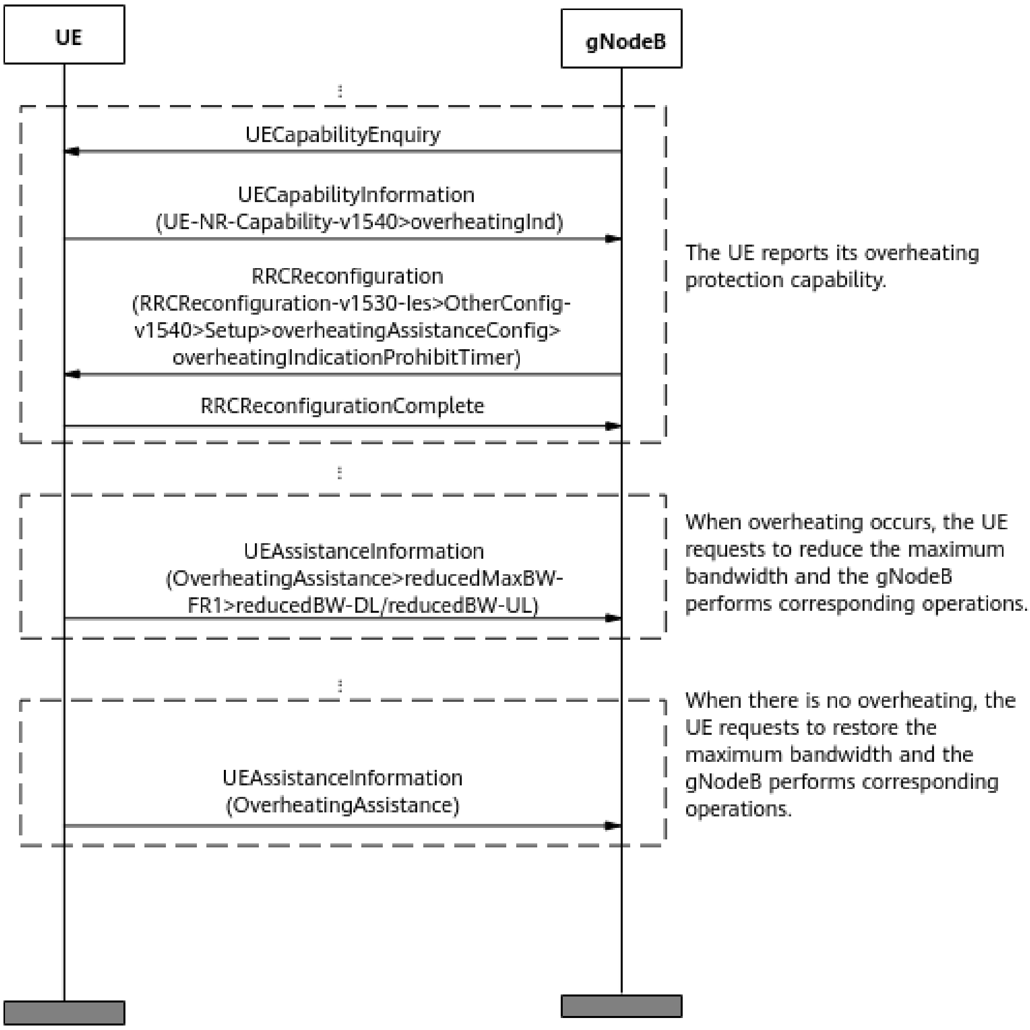
The settings of the preceding options take effect only on UEs that newly access or are newly handed over to the cell.

When both the conditions for the maximum-bandwidth reduction function for overheating protection and that for UE power saving are met, both functions can take effect. If the UE uses both the **OverheatingAssistance** and **maxBW-Preference-r16** fields in the UEAssistanceInformation message to indicate that the maximum bandwidth needs to be reduced, the base station uses the smaller one of the reported values. That is, the expected maximum downlink bandwidth is min{reducedBW-DL, reducedBW-DL-r16}, and the expected maximum uplink bandwidth is min{reducedBW-UL, reducedBW-UL-r16}.

5.3.1.2 Process of Maximum-Bandwidth Reduction for UE Overheating Protection

Figure 5-8 shows the process of UE overheating protection.

Figure 5-8 Process of UE overheating protection



1. The UE reports its overheating protection capability.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

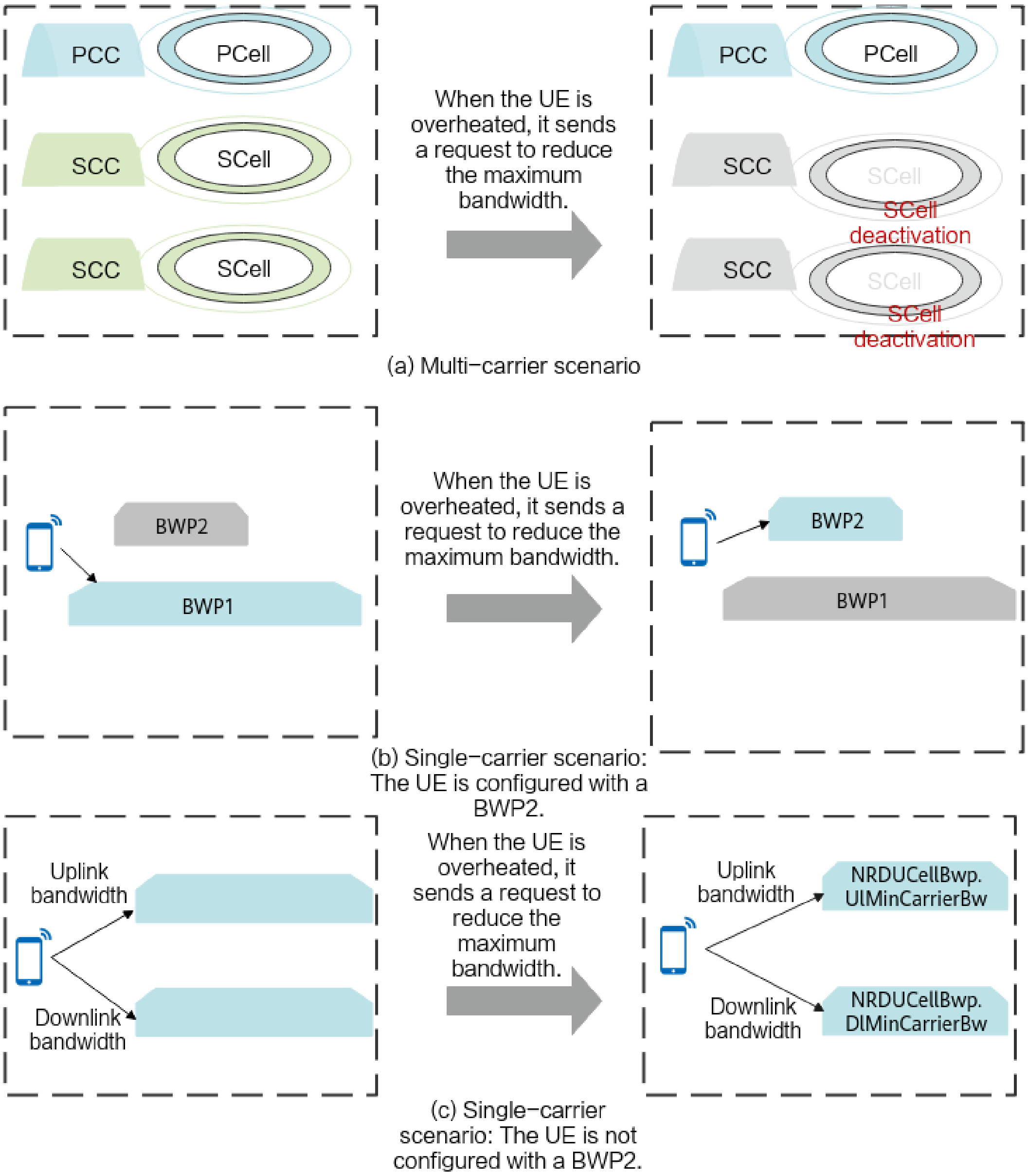
   a. The base station sends a UECapabilityEnquiry message to the UE.
   b. The UE sends a UECapabilityInformation message to the base station. In this message, the **overheatingInd** field indicates that the UE supports overheating protection.
   c. The base station sends an RRCReconfiguration message containing the **overheatingIndicationProhibitTimer** field to the UE to configure the overheating indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report overheating messages, which can be set using the **NRCellUeTimer**.***OverheatingIndProhibitTmr*** parameter. This prevents the UE from frequently reporting the overheating indication.
   d. The UE sends an RRCReconfigurationComplete message to the base station.
2. When the UE is overheated, it sends a request to reduce the maximum bandwidth.

When the UE detects overheating, it sends a UEAssistanceInformation message to the base station. If the **OverheatingAssistance** field in the message contains **reducedBW-DL** and **reducedBW-UL**, the UE is overheated and the base station is expected to reduce the maximum bandwidth configured for the UE.

After receiving the request, the base station performs the following operations (as illustrated in **Figure 5-9**):

- In multi-carrier scenarios, the base station deactivates all SCCs of the UE.
- In single-carrier scenarios:
  - If a narrow bandwidth BWP2 is configured for the UE, the base station switches the UE to BWP2 and does not allow the UE to switch to the full bandwidth BWP1 unless the UE initiates a request to restore the maximum bandwidth.
  - If BWP2 is not configured for the UE and the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is selected, the base station configures the uplink and downlink BWPs with the bandwidths specified by **NRDUCellBwp**.***UlMinCarrierBw*** and **NRDUCellBwp**.***DlMinCarrierBw***, respectively. If the UE does not support these bandwidths, the base station does not reconfigure the bandwidths for the UE. For details about the setting notes of the **NRDUCellBwp**.***UlMinCarrierBw*** and **NRDUCellBwp**.***DlMinCarrierBw*** parameters, see "UE Bandwidth Adaptation" in *Scalable Bandwidth*.
  - In other cases, the base station does not respond to the bandwidth reduction request.

Figure 5-9 Schematic diagram of maximum-bandwidth reduction for overheating protection



3. When the UE is no longer overheated, it sends a request to restore the maximum bandwidth.

   When the UE in the overheating protection state detects that it is no longer overheated, the UE sends a UEAssistanceInformation message to the base station. (The restriction imposed by the overheating indication prohibition timer must be met.) If the **OverheatingAssistance** field does not contain any information, the UE does not overheat and expects the base station to restore the maximum bandwidth.

   After the UE sends a request to restore the maximum bandwidth, the base station performs the following operations to restore the maximum bandwidth:

   – In multi-carrier scenarios, the base station allows SCell activation for the UE. Whether to activate the SCCs depends on the traffic volume. For details, see SCell activation in *Carrier Aggregation*.

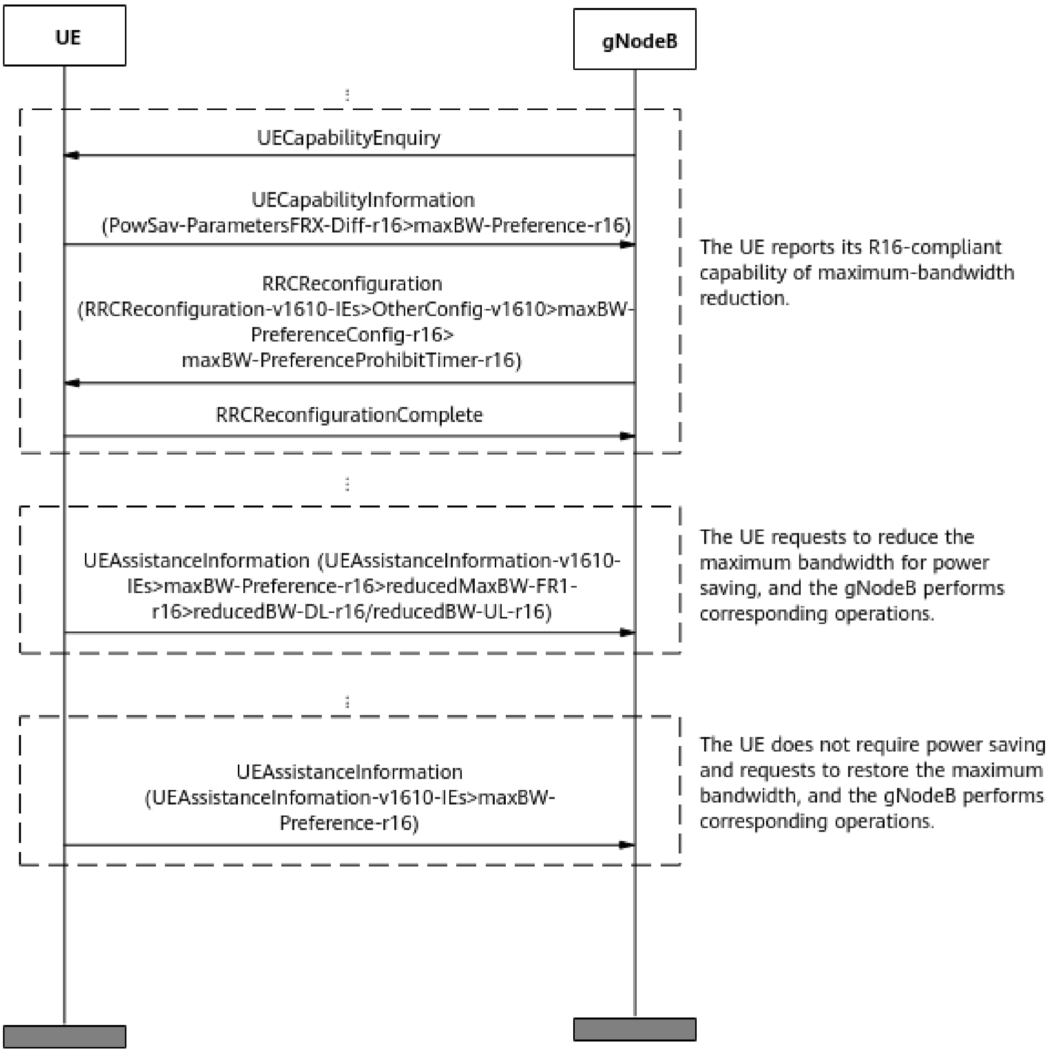
– In single-carrier scenarios:

  - If a narrow-bandwidth BWP2 is configured for a UE, the base station allows the UE to switch to the full-bandwidth BWP1. Whether to switch to BWP1 depends on the traffic volume. For details, see **5.1 Power Saving BWP (Low-Frequency TDD)**.
  - If BWP2 is not configured for the UE and the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is selected, the base station reconfigures the BWP bandwidth for the UE through an RRC reconfiguration message. In this case, the BWP bandwidth is the highest bandwidth supported by the UE and does not exceed the cell bandwidth.
  - In other cases, the base station does not respond to the bandwidth restoration request.

5.3.1.3 Process of Maximum-Bandwidth Reduction for UE Power Saving

Figure 5-10 shows the process of maximum-bandwidth reduction for UE power saving.

Figure 5-10 Process of maximum-bandwidth reduction for UE power saving



1. The UE reports its 3GPP Release 16-compliant capability for maximum-bandwidth reduction.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.
   b. The UE sends a UECapabilityInformation message to the base station. In this message, the **maxBW-Preference-r16** field indicates that the UE supports the 3GPP Release 16-compliant capability for maximum-bandwidth reduction.
   c. The base station sends an RRCReconfiguration message containing the **maxBW-PreferenceProhibitTimer-r16** field to the UE to configure the maximum-bandwidth reduction indication prohibition timer for the UE. This timer specifies the minimum interval at which the base station allows the UE to report maximum-bandwidth reduction messages. This prevents the UE from frequently reporting the maximum-bandwidth reduction indication.
   d. The UE sends an RRCReconfigurationComplete message to the base station.

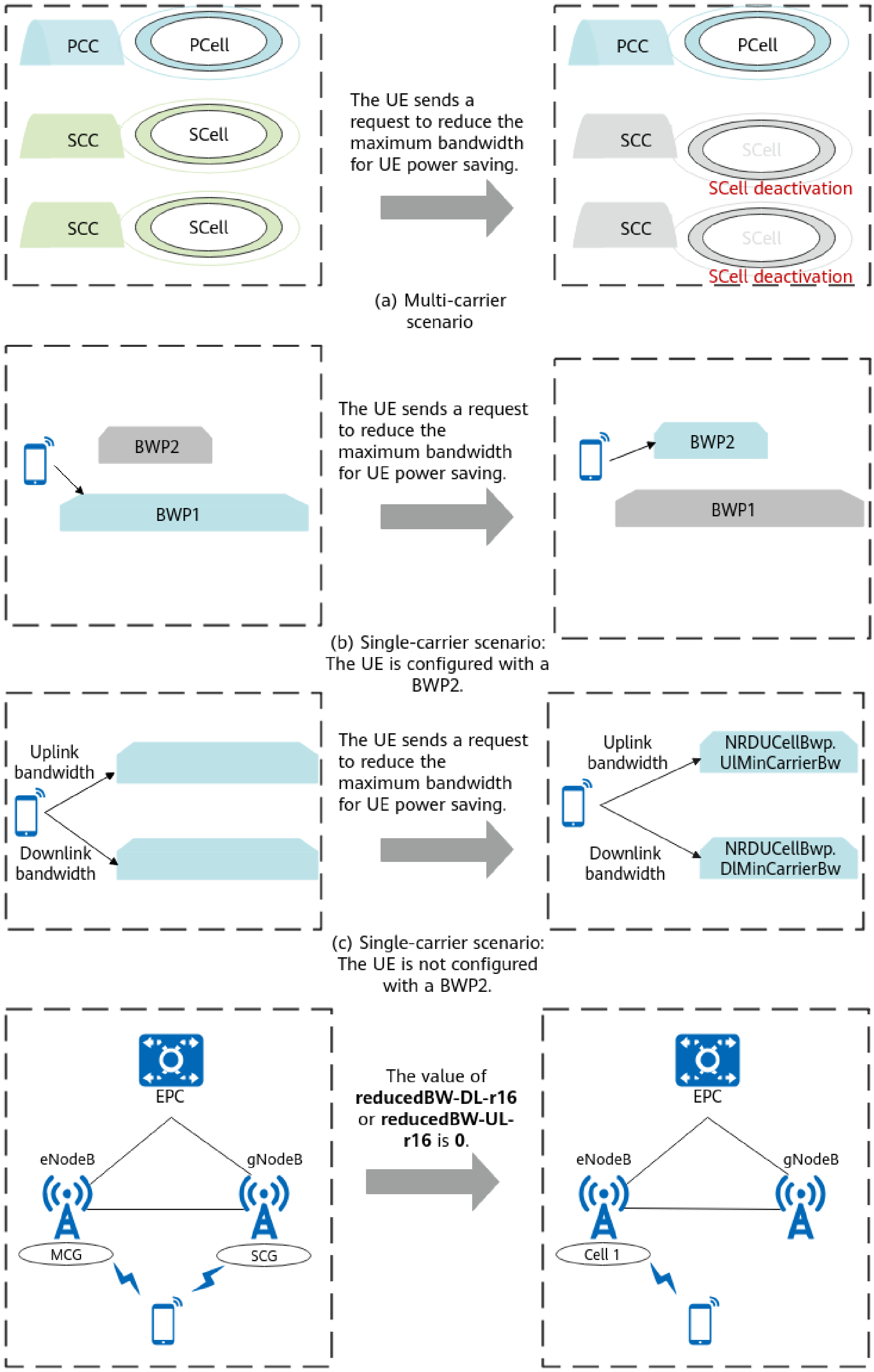
2. When the UE requires power saving, it sends a request to reduce the maximum bandwidth.

   The UE sends a UEAssistanceInformation message to the base station for power saving. If the **maxBW-Preference-r16** field in the message contains **reducedBW-DL-r16** and **reducedBW-UL-r16**, the UE requires power saving and the base station is expected to reduce the maximum bandwidth configured for the UE.

   The UE initiates a request to reduce the maximum bandwidth, and the base station decreases the maximum bandwidth (as illustrated in **Figure 5-11**).

   – In multi-carrier scenarios, the base station deactivates all SCCs of the UE.
   – In single-carrier scenarios:
     ■ If a narrow bandwidth BWP2 is configured for the UE, the base station switches the UE to BWP2 and does not allow the UE to switch to the full bandwidth BWP1 unless the UE initiates a request to restore the maximum bandwidth.
     ■ If BWP2 is not configured for the UE and the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is selected, the base station configures the uplink and downlink BWPs with the bandwidths specified by **NRDUCellBwp**.***UlMinCarrierBw*** and **NRDUCellBwp**.***DlMinCarrierBw***, respectively. If the UE does not support these bandwidths, the base station does not reconfigure the bandwidths for the UE. For details about the setting notes of the **NRDUCellBwp**.***UlMinCarrierBw*** and **NRDUCellBwp**.***DlMinCarrierBw*** parameters, see "UE Bandwidth Adaptation" in *Scalable Bandwidth*.
     ■ In other cases, the base station does not respond to the bandwidth reduction request.
   – In NSA networking, when the value of **reducedBW-DL-r16** or **reducedBW-UL-r16** reported by a UE is 0, the gNodeB proactively triggers the SCG removal procedure for power saving. For details, see *NSA Mobility Management*.

Figure 5-11 Schematic diagram of maximum-bandwidth reduction for UE power saving



(d) NSA scenario: The value of **reducedBW-DL-r16** or **reducedBW-UL-r16** is **0**.

3. When power saving is no longer required, the UE can send a request to restore the maximum bandwidth.

   When power saving is no longer required, the UE sends a UEAssistanceInformation message to the base station. If the **maxBW-Preference-r16** field does not contain any information, the UE does not require power saving and expects the base station to restore the maximum bandwidth.

   After the UE sends a request to restore the maximum bandwidth, the base station performs the following operations to restore the maximum bandwidth:

   – In multi-carrier scenarios, the base station allows SCell activation for the UE. Whether to activate the SCCs depends on the traffic volume. For details, see SCell activation in *Carrier Aggregation*.
   – In single-carrier scenarios:
     - If a narrow-bandwidth BWP2 is configured for a UE, the base station allows the UE to switch to the full-bandwidth BWP1. Whether to switch to BWP1 depends on the traffic volume. For details, see **5.1 Power Saving BWP (Low-Frequency TDD)**.
     - If BWP2 is not configured for the UE and the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is selected, the base station reconfigures the BWP bandwidth for the UE through an RRC reconfiguration message. In this case, the BWP bandwidth is the highest bandwidth supported by the UE and does not exceed the cell bandwidth.
     - In other cases, the base station does not respond to the bandwidth restoration request.

5.3.2 Network Analysis

5.3.2.1 Benefits

When the function of maximum-bandwidth reduction is enabled, the power consumption of UEs' communication modules can be reduced. The specific reduction in power consumption is related to the bandwidth in effect.

5.3.2.2 Impacts

The impacts between this function and other functions are described in **5.3.3.2.3 Function Impacts**.

This function may have the following impacts:

- The average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) decreases.
- The average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases.
- The average uplink UE throughput (**User Uplink Average Throughput (DU)**) decreases.
- The average downlink UE throughput (**User Downlink Average Throughput (DU)**) decreases.

- The number of RRC reconfiguration messages increases, and therefore the service drop rate (**Service Call Drop Rate (CU)**) increases.

5.3.3 Requirements

5.3.3.1 Licenses

There are no license requirements for basic functions.

5.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

None

5.3.3.2.2 Mutually Exclusive Functions

None

5.3.3.2.3 Function Impacts

Impact relationships with CA-related functions

Function Name	Function Switch	Reference	Description
Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When a UE requests maximum-bandwidth reduction, the base station deactivates all SCCs of the UE in multi-carrier scenarios.
Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	

Function Name	Function Switch	Reference	Description
Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*CA Experience Improvement*	
3CC aggregation	**NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL3CC**	*CA Experience Improvement*	
Downlink fast intra-gNodeB CA	**CA_ULTRA_LOW_LATENCY_SPLIT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*CA Experience Improvement*	When a UE requests maximum-bandwidth reduction, the base station deactivates all SCCs of the UE in multi-carrier scenarios.
Downlink fast inter-gNodeB CA	**INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoExtSwitch*** parameter	*CA Experience Improvement*	

Function Name	Function Switch	Reference	Description
Inter-FR CA (Trial)	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	● If the **OVERHEATING_ASSISTANCE_INFO_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected, the base station removes high-frequency SCells for a UE when the UE makes a request to reduce the bandwidth due to overheating. ● If the **UE_BW_REDUCE_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected, the function of maximum-bandwidth reduction does not take effect on an

Function Name	Function Switch	Reference	Description
			inter-FR CA UE when the UE makes a request to reduce the bandwidth only due to power saving.

Impact relationships with other functions

Function Name	Function Switch	Reference	Description
Power saving BWP (low-frequency TDD)	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	When a UE requests maximum-bandwidth reduction, if BWP2 is configured for the UE in single-carrier scenarios, the base station switches the UE to BWP2.

Function Name	Function Switch	Reference	Description
UE bandwidth adaptation	**UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter	*Scalable Bandwidth*	When a UE requests maximum-bandwidth reduction in single-carrier scenarios, BWP2 is not configured for the UE, and the **UE_BW_ADAPTIVE_SW** option of the **NRDUCellBwp**.***BwpConfigSwitch*** parameter is selected, the base station configures uplink and downlink BWPs with the bandwidths specified by **NRDUCellBwp**.***UlMinCarrierBw*** and **NRDUCellBwp**.***DlMinCarrierBw***, respectively, for the UE through an RRC reconfiguration message.

5.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.3.4 Others

UEs must support the **overheatingInd** capability defined in 3GPP Release 15 or the **maxBW-Preference-r16** capability defined in 3GPP Release 16.

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

Table 5-21 describes the parameters used for function activation.

Table 5-21 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	UE_BW_REDUCE_SW	Select this option.
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	OVERHEATING_ASSISTANCE_INFO_SW	Select this option.
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	UAI_NSA_SW	Select this option in NSA networking.

Parameter Name	Parameter ID	Option	Setting Notes
Overheating Indication Prohibit Timer	**NRCellUeTimer**.***OverheatingIndProhibitTmr***	None	Set this parameter to its recommended value.

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the UE bandwidth reduction switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UE_BW_REDUCE_SW-1;
//Turning on the overheating assistance information processing switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=OVERHEATING_ASSISTANCE_INFO_SW-1;
//Configuring the overheating indication prohibition timer
MOD NRCELLUETIMER: NrCellId=1, OverheatingIndProhibitTmr=S30;
```

Turning on the NSA UAI switch in NSA networking

```
//Turning on the NSA UAI switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UAI_NSA_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the UE bandwidth reduction switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UE_BW_REDUCE_SW-0;
//Turning off the overheating assistance information processing switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=OVERHEATING_ASSISTANCE_INFO_SW-0;
```

Turning off the NSA UAI switch in NSA networking

```
//Turning off the NSA UAI switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UAI_NSA_SW-0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 Observe the RRC reconfiguration message. If the RRC reconfiguration message contains "overheatingAssistanceConfig>overheatingIndicationProhibitTimer" or "maxBW-PreferenceConfig-r16>maxBW-PreferenceProhibitTimer-r16", the UE has reported its capabilities to the base station.

Step 3 Observe the UEAssistanceInformation message. If the UEAssistanceInformation message reported by the UE contains the **reducedBW-DL**/**reducedBW-UL** or **reducedBW-DL-r16**/**reducedBW-UL-r16** field, the UE has initiated a bandwidth reduction request.

----End

5.3.4.3 Network Monitoring

The indicators described in **5.3.2.2 Impacts** can be used for network monitoring of this function.

6 Space Domain

Generally, higher power consumption occurs for UEs that use a larger quantity of antennas (the quantity of radio frequency chains) to receive data. As defined in 3GPP specifications, the number of receive antennas used by a UE cannot be less than the maximum number of downlink MIMO layers. For example, a UE supporting a maximum of four downlink MIMO layers must enable at least four receive antennas to receive downlink data. Therefore, when the actual traffic volume is not large, the following solutions can be used to reduce the maximum number of downlink MIMO layers for the UE and the number of receive antennas used by the UE, thereby reducing the UE power consumption. MIMO in this section refers to SU-MIMO.

- **6.1 Reducing the Number of MIMO Layers**: The maximum number of MIMO layers is configured at the cell level. That is, all BWPs of the serving cell correspond to one maximum number of MIMO layers. The base station reduces the number of MIMO layers for UEs to achieve power saving and overheating protection.
- **6.2 BWP Spatial Adaptation (Low-Frequency TDD)**: 3GPP Release 16 introduces the configuration of the maximum number of MIMO layers based on the downlink BWP. UEs can adaptively select the maximum number of MIMO layers through BWP switching. BWP switching is triggered based on the traffic volume. When a UE switches from BWP1 to BWP2, the base station reduces the maximum number of downlink MIMO layers of the UE for power saving and overheating protection.

6.1 Reducing the Number of MIMO Layers

6.1.1 Principles

6.1.1.1 Overview

This function uses standard protocol-compliant procedures for reducing the number of MIMO layers in the uplink and downlink to achieve overheating protection and UE power saving, as shown in **Figure 6-1**. This function applies only to low-frequency UEs. The process of reducing the number of MIMO layers for overheating protection and that for UE power saving are different. **Table 6-1** describes the conditions for individual processes to take effect.

Figure 6-1 Schematic diagram of reducing the number of MIMO layers

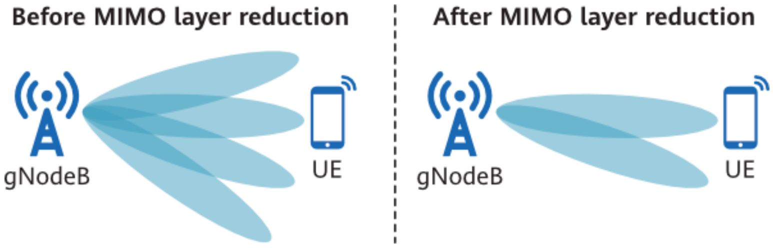


Table 6-1 Conditions for the function of reducing the number of MIMO layers for overheating protection and that for UE power saving to take effect

Process	Parameter	UE Capability
3GPP Release 15-compliant reducing the number of MIMO layers for overheating protection (described in **5.2.1.2 Process of Reducing the Number of SCCs for UE Overheating Protection**)	• Both the **OVERHEATING_ASSISTANCE_INFO_SW** and **UE_MIMO_LAYER_REDUCE_SW** options of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter are selected. • In NSA networking, the **UAI_NSA_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter also needs to be selected.	The **overheatingInd** capability defined in 3GPP Release 15
3GPP Release 16-compliant reducing the number of MIMO layers for UE power saving (described in **5.2.1.3 Process of Reducing the Number of SCCs for UE Power Saving**)	• The **UE_MIMO_LAYER_REDUCE_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected. • In NSA networking, the **UAI_NSA_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter also needs to be selected.	The **maxMIMO-LayerPreference-r16** capability defined in 3GPP Release 16

The settings of the preceding options take effect only on UEs that newly access or are newly handed over to the cell.

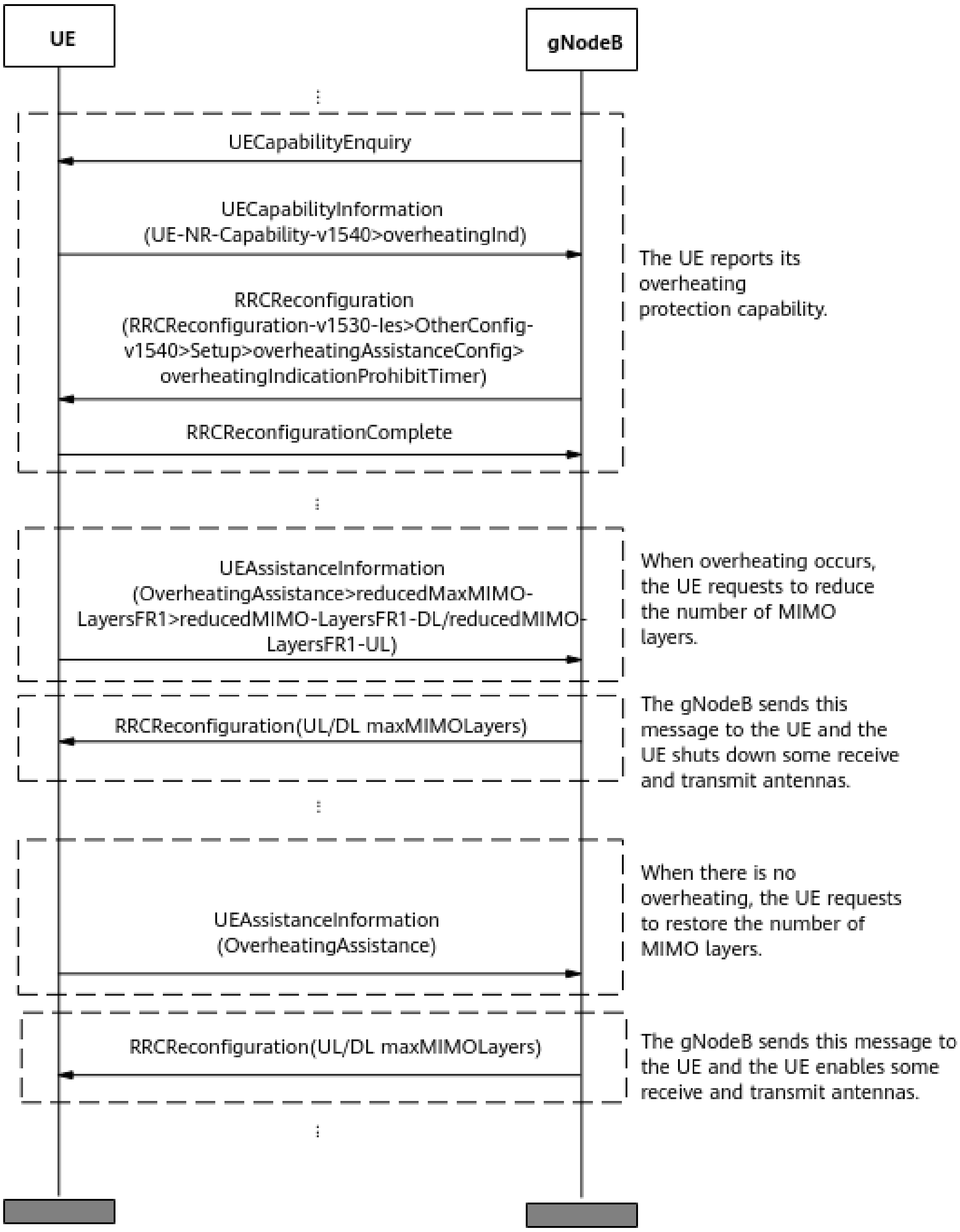
When the conditions for the function of reducing the number of MIMO layers for overheating protection and those for the function of reducing the number of

MIMO layers for UE power saving are met, both functions can take effect. If the UE uses both the **OverheatingAssistance** and **maxMIMO-LayerPreference-r16** fields in the UEAssistanceInformation message to indicate the number of MIMO layers to be reduced, the base station uses the smaller one of the reported values. That is, the expected number of MIMO layers in the downlink is min{reducedMIMO-LayersFR1-DL, reducedMIMO-LayersFR1-DL-r16}, and the expected number of MIMO layers in the uplink is min{reducedMIMO-LayersFR1-UL, reducedMIMO-LayersFR1-UL-r16}.

6.1.1.2 Process of Reducing the Number of MIMO Layers for UE Overheating Protection

Figure 6-2 shows the process of UE overheating protection.

Figure 6-2 UE overheating protection process



1. The UE reports its overheating protection capability.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.
   b. The UE returns a UECapabilityInformation message to the base station. In this message, the **overheatingInd** field indicates that the UE supports overheating protection.
   c. The base station sends an RRCReconfiguration message containing the **overheatingIndicationProhibitTimer** field to the UE to configure the

overheating indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report overheating messages for MIMO layer reduction, which can be set using the **NRCellUeTimer**.***OverheatingIndProhibitTmr*** parameter. This prevents the UE from frequently reporting the overheating indication.

d. The UE returns an RRCReconfigurationComplete message to the base station.

2. When the UE overheats, it proactively initiates a request for reducing the number of MIMO layers.

   When the UE detects overheating, it sends a UEAssistanceInformation message to the base station. If all of the following conditions are met, the UE overheats and expects the base station to reduce the number of uplink and downlink MIMO layers:

   – The **OverheatingAssistance** field carries **reducedMIMO-LayersFR1-DL** and **reducedMIMO-LayersFR1-UL**.
   – The value of **reducedMIMO-LayersFR1-DL** is less than the maximum number of downlink MIMO layers supported by the UE, or the value of **reducedMIMO-LayersFR1-DL/reducedMIMO-LayersFR1-UL** is less than the maximum number of uplink MIMO layers supported by the UE.

3. The UE automatically shuts down some transmit and receive antennas.

   After a UE proactively initiates MIMO layer reduction and the function of reducing the number of MIMO layers is enabled, the base station sends the UE an RRCReconfiguration message that contains fields indicating the maximum number of MIMO layers in the uplink and downlink (or either uplink or downlink). Upon receiving this message, the UE shuts down some transmit and receive antennas to save power.

4. When there is no overheating, the UE makes a request to restore the number of MIMO layers.

   When the UE in overheating protection detects that it no longer overheats, it sends a UEAssistanceInformation message to the base station. If the **OverheatingAssistance** field does not contain any information, the UE does not overheat and expects the base station to restore the number of uplink and downlink MIMO layers.

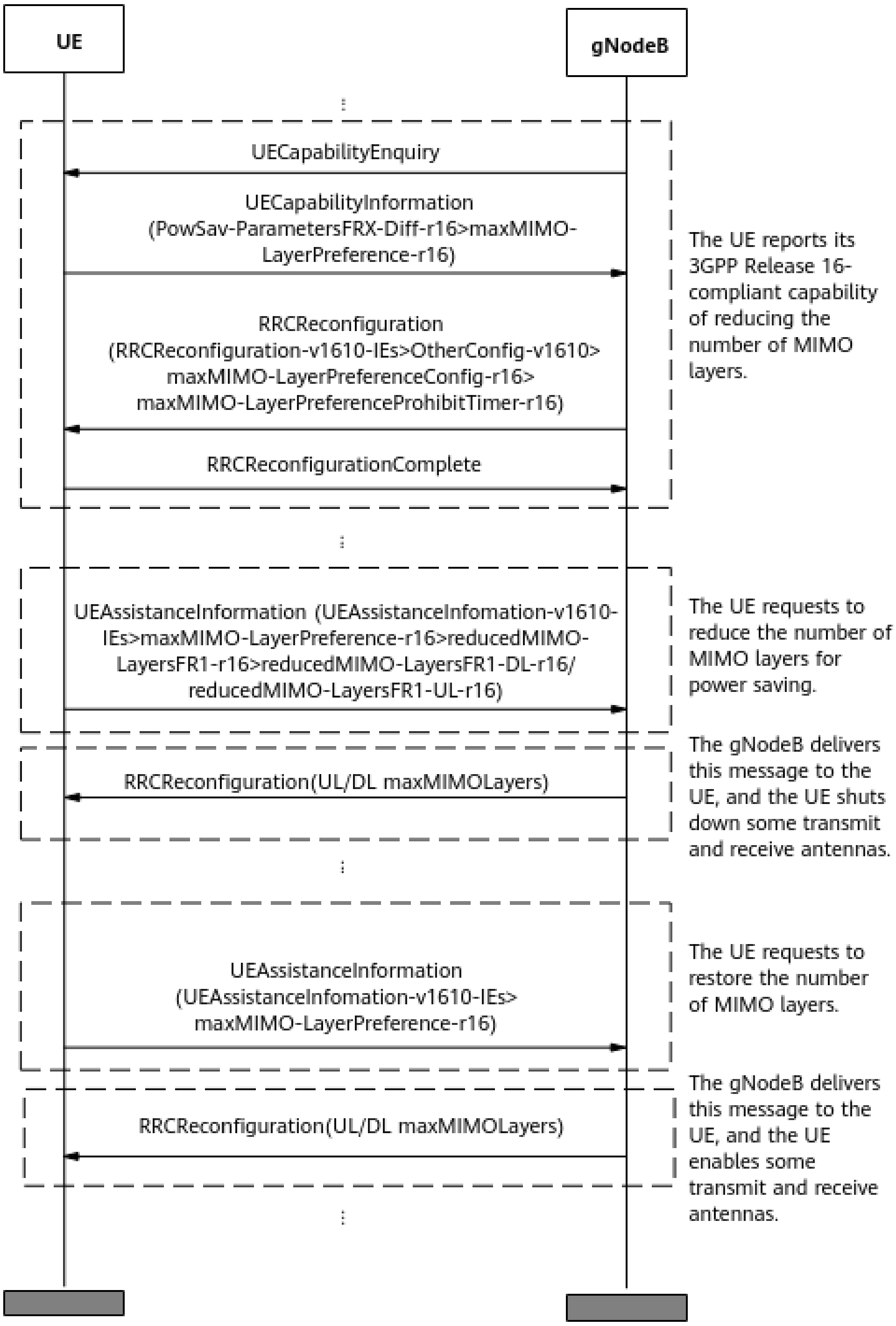
5. The UE turns on the transmit and receive antennas that have been shut down.

   After a UE proactively initiates MIMO layer restoration and the function of reducing the number of MIMO layers is enabled, the base station sends the UE an RRCReconfiguration message that contains fields indicating the maximum number of MIMO layers in the uplink and downlink (or either uplink or downlink). After receiving the message, the UE turns on the transmit and receive antennas that have been turned off.

6.1.1.3 Process of Reducing the Number of MIMO Layers for UE Power Saving

Figure 6-3 shows the process of reducing the number of MIMO layers for UE power saving.

Figure 6-3 Process of reducing the number of MIMO layers for UE power saving



1. The UE reports its 3GPP Release 16-compliant capability for reducing the number of MIMO layers.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.

   b. The UE returns a UECapabilityInformation message to the base station. In this message, the **maxMIMO-LayerPreference-r16** field indicates that the UE supports the 3GPP Release 16-compliant capability for reducing the number of MIMO layers.

c. The base station sends an RRCReconfiguration message containing the **maxMIMO-LayerPreferenceProhibitTimer-r16** field to the UE to configure the MIMO layer reduction indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report MIMO layer reduction messages for power saving. This prevents the UE from frequently reporting the MIMO layer reduction indication.

d. The UE returns an RRCReconfigurationComplete message to the base station.

2. The UE proactively requests for MIMO layer reduction for power saving.

   The UE sends a UEAssistanceInformation message to the base station for power saving. If all the following conditions are met, the base station is expected to reduce the number of MIMO layers for UE power saving:

   – The **maxMIMO-LayerPreference-r16** field carries **reducedMIMO-LayersFR1-DL-r16** and **reducedMIMO-LayersFR1-UL-r16**.

   – The value of **reducedMIMO-LayersFR1-DL-r16** is less than the maximum number of downlink MIMO layers supported by the UE, or the value of **reducedMIMO-LayersFR1-UL-r16** is less than the maximum number of uplink MIMO layers supported by the UE.

3. The UE automatically shuts down some transmit and receive antennas.

   After a UE proactively initiates MIMO layer reduction and the function of reducing the number of MIMO layers is enabled, the base station sends the UE an RRCReconfiguration message that contains fields indicating the maximum number of MIMO layers in the uplink and downlink (or either uplink or downlink). Upon receiving this message, the UE shuts down some transmit and receive antennas to save power.

4. When power saving is no longer required, the UE can initiate MIMO layer restoration.

   When a UE in the power saving state detects that power saving is no longer required, the UE sends a UEAssistanceInformation message to the base station. If the **maxMIMO-LayerPreference-r16** field does not contain any information, the UE does not require power saving and expects the base station to restore the number of uplink and downlink MIMO layers.

5. The UE turns on the transmit and receive antennas that have been shut down.

   After a UE proactively initiates MIMO layer restoration and the function of reducing the number of MIMO layers is enabled, the base station sends the UE an RRCReconfiguration message that contains fields indicating the maximum number of MIMO layers in the uplink and downlink (or either uplink or downlink). After receiving the message, the UE turns on the transmit and receive antennas that have been turned off.

6.1.1.4 Blacklist and Whitelist Control over Reducing the Number of MIMO Layers for UE Power Saving

The function of reducing the number of MIMO layers for UE power saving defined in 3GPP Release 16 supports the blacklist and whitelist mechanism.

- If the **UE_MIMO_LAYER_REDUCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter is selected, the function of reducing the number of MIMO layers for UE power saving is disabled for specified UEs. That is, this function does not take effect on the specified UEs and can take effect on other UEs.

- If the **UE_MIMO_LAYER_REDUCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the function of reducing the number of MIMO layers for UE power saving is enabled for specified UEs. That is, this function can take effect on the specified UEs and does not take effect on other UEs.

The **UE_MIMO_LAYER_REDUCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UE_MIMO_LAYER_REDUCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be both selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter.

6.1.2 Network Analysis

6.1.2.1 Benefits

When SU-MIMO is enabled and the maximum number of layers for a single UE is greater than 2, enabling the function of reducing the number of MIMO layers can reduce power consumption of communication modules of UEs by a certain amount, depending on the number of reduced MIMO layers. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains. For details about SU-MIMO, see *MIMO (TDD)*.

Assume that the function of reducing the number of MIMO layers is enabled. When the **UE_MIMO_LAYER_REDUCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter or the **UE_MIMO_LAYER_REDUCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the number of UEs with reducing the number of MIMO layers in effect decreases, the average cell and UE throughputs increase, the service drop rate decreases, the proportions of rank 1 and rank 2 may decrease, the proportions of rank 3 and rank 4 may increase, and the number of UEs without reducing the number of MIMO layers in effect increases.

6.1.2.2 Impacts

The impacts between this function and other functions are described in **6.1.3.2.3 Function Impacts**.

After this function is enabled, the values of the following indicators change:

- The average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) decreases.
- The average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) decreases.
- The average uplink UE throughput (**User Uplink Average Throughput (DU)**) decreases.
- The average downlink UE throughput (**User Downlink Average Throughput (DU)**) decreases.

- The proportion of rank 1 (Rank1 Rate of DL TB) and the proportion of rank 2 (Rank2 Rate of DL TB) increase.
- The proportion of rank 3 (Rank3 Rate of DL TB) and the proportion of rank 4 (Rank4 Rate of DL TB) decrease.
- The number of RRC connection reconfiguration messages may increase, which may increase the service drop rate (**Service Call Drop Rate (CU)**).

Indicator Description	Formula
Proportion of rank 1 (Rank1 Rate of DL TB)	**N.PDSCH.NbrTbDl.Rank1**/ (**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) x 100%
Proportion of rank 2 (Rank2 Rate of DL TB)	**N.PDSCH.NbrTbDl.Rank2**/ (**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) x 100%
Proportion of rank 3 (Rank3 Rate of DL TB)	**N.PDSCH.NbrTbDl.Rank3**/ (**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) x 100%
Proportion of rank 4 (Rank4 Rate of DL TB)	**N.PDSCH.NbrTbDl.Rank4**/ (**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) x 100%

6.1.3 Requirements

6.1.3.1 Licenses

There are no license requirements.

6.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.1.3.2.1 Prerequisite Functions

None

6.1.3.2.2 Mutually Exclusive Functions

None

6.1.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
SU-MIMO	**DL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch**.***SuMimoMultipleLayerSw*** parameter	*MIMO (TDD)*	When both SU-MIMO and MIMO layer reduction are enabled, the value of the **N.ChMeas.MIMO.DL.Transmission.Layer** counter may decrease.

Function Name	Function Switch	Reference	Description
Inter-FR CA (Trial)	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	● If the **OVERHEATING_ASSISTANCE_INFO_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected, the base station removes high-frequency SCells for a UE when the UE makes a request to reduce the number of MIMO layers due to overheating. ● If the **UE_MIMO_LAYER_REDUCE_SW** option of the **NRCellAlgoSwitch**.***UeAssistanceInfoSwitch*** parameter is selected, the function of reducing the number of

Function Name	Function Switch	Reference	Description
			MIMO layers does not take effect for an inter-FR CA UE when the UE makes a request to reduce the number of MIMO layers only due to power saving.

6.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.1.3.4 Others

UEs must support the **overheatingInd** capability defined in 3GPP Release 15 or the **maxMIMO-LayerPreference-r16** capability defined in 3GPP Release 16.

6.1.4 Operation and Maintenance

6.1.4.1 Data Configuration

6.1.4.1.1 Data Preparation

Table 6-2 describes the parameters used for function activation.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	OVERHEATING_ASSISTANCE_INFO_SW	Select this option.
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	UE_MIMO_LAYER_REDUCE_SW	Select this option.
UE Assistance Information Switch	**NRCellAlgoSwitch**.***UeAssistanceInfoSwitch***	UAI_NSA_SW	Select this option in NSA networking.
Overheating Indication Prohibit Timer	**NRCellUeTimer**.***OverheatingIndProhibitTmr***	None	Use the recommended value.
Blacklist Control Switch	**gNBUeCompatSw**.***BlacklistCtrlSwitch***	UE_MIMO_LAYER_REDUCE_SW_OFF	Select this option if a blacklist mechanism needs to be used to disable the function of reducing the number of MIMO layers for specified UEs.
Whitelist Control Switch	**gNBUeCompatSw**.***WhitelistCtrlExtSwitch***	UE_MIMO_LAYER_REDUCE_SW_ON	Select this option if a whitelist mechanism needs to be used to enable the function of reducing the number of MIMO layers for specified UEs.

6.1.4.1.2 Using MML Commands

Before using MML commands, refer to 6.1.3 Requirements and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- The **MOD GNBUECOMPATSW** command is a high-risk command. Running this command changes the feature activation states for specified UEs.
- The **UE_MIMO_LAYER_REDUCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UE_MIMO_LAYER_REDUCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

Activation Command Examples

```
//Turning on the overheating assistance information processing switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=OVERHEATING_ASSISTANCE_INFO_SW-1;
//Turning on the UE MIMO layer reduction switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UE_MIMO_LAYER_REDUCE_SW-1;
//Configuring the overheating indication prohibition timer
MOD NRCELLUETIMER: NrCellId=1, OverheatingIndProhibitTmr=S30;
//Enabling blacklist control over "the function of reducing the number of MIMO layers" for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=UE_MIMO_LAYER_REDUCE_SW_OFF-1;
//Enabling whitelist control over "the function of reducing the number of MIMO layers" for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UE_MIMO_LAYER_REDUCE_SW_ON-1;
```

Turning on the NSA UAI switch in NSA networking

```
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UAI_NSA_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the overheating assistance information processing switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=OVERHEATING_ASSISTANCE_INFO_SW-0;
//Turning off the UE MIMO layer reduction switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UE_MIMO_LAYER_REDUCE_SW-0;
//Disabling the blacklist control over "the function of reducing the number of MIMO layers" for specified
UEs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=UE_MIMO_LAYER_REDUCE_SW_OFF-0;
//Disabling the whitelist control over "the function of reducing the number of MIMO layers" for specified
UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UE_MIMO_LAYER_REDUCE_SW_ON-0;
```

Turning off the NSA UAI switch in NSA networking

```
//Turning off the NSA UAI switch
MOD NRCELLALGOSWITCH: NrCellId=1, UeAssistanceInfoSwitch=UAI_NSA_SW-0;
```


6.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.1.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 Observe the RRC reconfiguration message. If the RRC reconfiguration message contains **overheatingAssistanceConfig > overheatingIndicationProhibitTimer** or **maxMIMO-LayerPreferenceConfig-r16 > maxMIMO-LayerPreferenceProhibitTimer-r16**, the UE has reported its capabilities to the base station.

Step 3 Observe the UEAssistanceInformation message. If the UEAssistanceInformation message reported by the UE contains the **reducedMIMO-LayersFR1-DL**, **reducedMIMO-LayersFR1-UL**, **reducedMIMO-LayersFR1-DL-r16**, or **reducedMIMO-LayersFR1-UL-r16** field, the UE has initiated a request for reducing the number of MIMO layers.

----End

6.1.4.3 Network Monitoring

Subscribe to CHR logs. The value of the **MethodCount** field corresponding to **MethodId** with the value of **0** in the PRIVATE_UE_RELEASE event block indicates the number of times MIMO layer reduction is reported by UEs for overheating protection or power saving.

The indicators described in **6.1.2 Network Analysis** can be used for network monitoring of this function.

6.2 BWP Spatial Adaptation (Low-Frequency TDD)

6.2.1 Principles

The BWP spatial adaptation function complies with 3GPP Release 16 and is controlled by the **BWP_MIMO_ADPT_SW** option of **NRDUCellUePwrSaving**.***BwpPwrSavingSw***. This function takes effect only on low-frequency UEs in SA networking.

The mechanism of BWP spatial adaptation is as follows:

A UE that requires power saving reports the maximum number of downlink MIMO layers to be used to the base station. The base station selects an appropriate BWP for the UE based on this reported number and the maximum numbers of downlink MIMO layers configured for individual BWPs. After switching to this BWP, the number of downlink MIMO layers may decrease, thereby reducing UE power consumption, as illustrated in **Figure 6-4**. In addition, for a power-saving BWP, the base station configures SRS resources based on the 1T2R UE type. After the UE is switched to the power saving BWP, some uplink transmit links can be shut down, as illustrated in **Figure 6-5**.

Figure 6-4 Schematic diagram of BWP spatial adaptation in the downlink

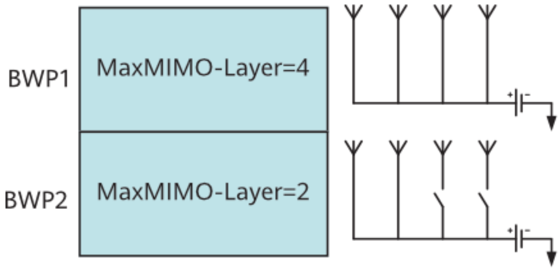
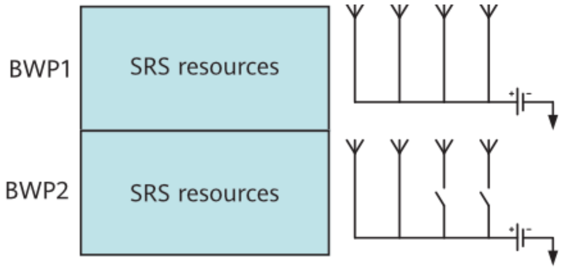


Figure 6-5 Schematic diagram of BWP spatial adaptation in the uplink



For details about BWP spatial adaptation for UEs that support the **maxLayersMIMO-Adaptation-r15** capability, see *Device-Pipe Synergy*.

6.2.2 Network Analysis

6.2.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

6.2.2.2 Impacts

The impacts between this function and other functions are described in **6.2.3.2.3 Function Impacts**.

BWP spatial adaptation depends on the power saving BWP function. When BWP spatial adaptation takes effect, some transmit and receive antennas of UEs in the power saving BWPs will be shut down. Therefore, when BWP spatial adaptation takes effect, the impact of power saving BWP on the network will be further aggravated. For details about the impacts of power saving BWP on the network, see **5.1.3.2 Impacts**.

BWP spatial adaptation may have the following impacts:

- The average uplink and downlink cell throughputs decrease, and the average uplink and downlink UE throughputs decrease significantly.

- The delay increases.
- The uplink and downlink HARQ retransmission rates and uplink and downlink IBLERs and RBLERs increase or decrease.
- The proportions of rank 1 and rank 2 increase.
- The proportions of rank 3 and rank 4 may decrease.
- The service drop rate increases.
- The average MCS index for downlink scheduling decreases, and the average MCS index for uplink scheduling may increase or decrease.

The following table lists the preceding indicators and their calculation formulas.

Indicator Description	Formula
Average uplink cell throughput	**Cell Uplink Average Throughput (DU)**
Average downlink cell throughput	**Cell Downlink Average Throughput (DU)**
Average uplink UE throughput	**User Uplink Average Throughput (DU)**
Average downlink UE throughput	**User Downlink Average Throughput (DU)**
Delay	**N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
Uplink HARQ retransmission rate	(**N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**)/ (**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB** + **N.UL.SCH.HalfPiBPSK.TB.Retrans** + **N.UL.SCH.QPSK.TB.Retrans** + **N.UL.SCH.16QAM.TB.Retrans** + **N.UL.SCH.64QAM.TB.Retrans** + **N.UL.SCH.256QAM.TB.Retrans**) × 100%

Indicator Description	Formula
Downlink HARQ retransmission rate	(**N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB** + **N.DL.SCH.QPSK.TB.Retrans** + **N.DL.SCH.16QAM.TB.Retrans** + **N.DL.SCH.64QAM.TB.Retrans** + **N.DL.SCH.256QAM.TB.Retrans**) × 100%
Uplink IBLER	(**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink IBLER	(**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%
Uplink RBLER at the MAC layer	(**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) × 100%
Downlink RBLER at the MAC layer	(**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB**) × 100%

Indicator Description	Formula
Proportions of TBs transmitted with different ranks	• Rank1 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank1**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank2 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank2**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank3 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank3**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank4 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank4**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank5 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank5**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank6 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank6**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank7 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank7**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100% • Rank8 Rate of DL TB = **N.PDSCH.NbrTbDl.Rank8**/(**N.PDSCH.NbrTbDl.Rank1** + ... + **N.PDSCH.NbrTbDl.Rank8**) × 100%
Service drop rate	**N.PDUSession.AbnormRel**/(**N.PDUSession.AbnormRel** + **N.PDUSession.NormRel**) × 100%
Average MCS index for uplink scheduling	(0 × **N.ChMeas.PUSCH.MCS.0** + 1 × **N.ChMeas.PUSCH.MCS.1** + 2 × **N.ChMeas.PUSCH.MCS.2** + 3 × **N.ChMeas.PUSCH.MCS.3** + ... + 26 × **N.ChMeas.PUSCH.MCS.26** + 27 × **N.ChMeas.PUSCH.MCS.27** + 28 × **N.ChMeas.PUSCH.MCS.28**)/(**N.ChMeas.PUSCH.MCS.0** + **N.ChMeas.PUSCH.MCS.1** + **N.ChMeas.PUSCH.MCS.2** + **N.ChMeas.PUSCH.MCS.3** + ... + **N.ChMeas.PUSCH.MCS.26** + **N.ChMeas.PUSCH.MCS.27** + **N.ChMeas.PUSCH.MCS.28**)

Indicator Description	Formula
Average MCS index for downlink scheduling	(0 × **N.ChMeas.PDSCH.MCS.0** + 1 × **N.ChMeas.PDSCH.MCS.1** + 2 × **N.ChMeas.PDSCH.MCS.2** + 3 × **N.ChMeas.PDSCH.MCS.3** + ... + 26 × **N.ChMeas.PDSCH.MCS.26** + 27 × **N.ChMeas.PDSCH.MCS.27** + 28 × **N.ChMeas.PDSCH.MCS.28**)/(**N.ChMeas.PDSCH.MCS.0** + **N.ChMeas.PDSCH.MCS.1** + **N.ChMeas.PDSCH.MCS.2** + **N.ChMeas.PDSCH.MCS.3** + ... + **N.ChMeas.PDSCH.MCS.26** + **N.ChMeas.PDSCH.MCS.27** + **N.ChMeas.PDSCH.MCS.28**)

6.2.3 Requirements

6.2.3.1 Licenses

This function requires the power saving BWP (low-frequency TDD) function. For details about the license requirements of the power saving BWP (low-frequency TDD) function, see **5.1.4.1 Licenses**. This function is not under other license control.

6.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference
Power saving BWP (low-frequency TDD)	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*

6.2.3.2.2 Mutually Exclusive Functions

None

6.2.3.2.3 Function Impacts

None

6.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.2.3.4 Others

UEs must support the **maxLayersMIMO-Adaptation-r16** capability defined in 3GPP Release 16.

6.2.4 Operation and Maintenance

6.2.4.1 Data Configuration

6.2.4.1.1 Data Preparation

Table 6-3 describes the parameters used for function activation.

Table 6-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	BWP_MIMO_ADPT_SW	To enable BWP spatial adaptation, select this option.

6.2.4.1.2 Using MML Commands

Before using MML commands, refer to **6.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling BWP spatial adaptation
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP_MIMO_ADPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling BWP spatial adaptation
MOD NRDUCELLUEPWRSAVING: NrDuCellId=1, BwpPwrSavingSw=BWP_MIMO_ADPT_SW-0;
```

6.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.2.4.2 Activation Verification

Tracing Signaling

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. The **Signaling Trace Management** page is displayed.

Step 2 In the navigation tree on the left of the displayed **Signaling Trace Management** page, choose **Uu Interface Trace**.

Step 3 On the displayed **Uu Interface Trace** page, check the RRC_RECFG message. Observe the BWP2 configuration. If the **BWP-DownlinkDedicated > PDSCH-Config > maxMIMO-Layers-r16** field exists, this function has taken effect.

----End

6.2.4.3 Network Monitoring

After this function is enabled, monitor the changes in related counters by referring to **6.2.2.2 Impacts**.

7 Other Power Saving Solutions

7.1 Fast RRC Connection Release

7.1.1 Principles

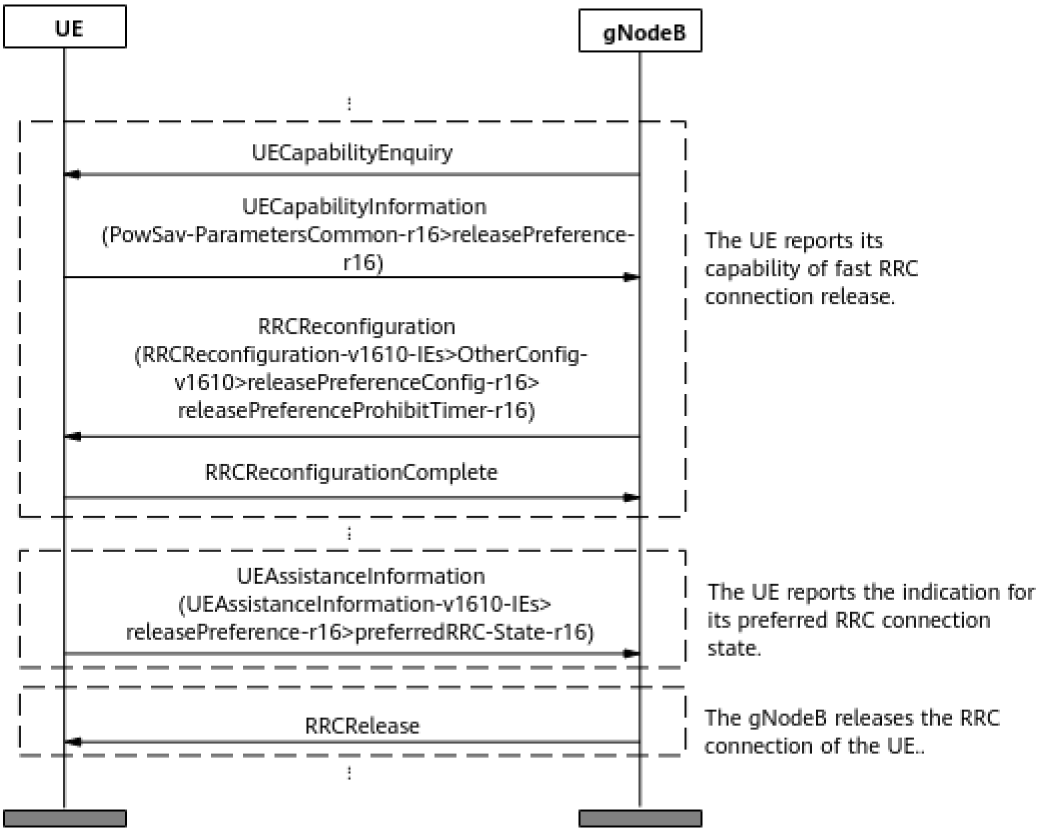
To reduce UE power consumption when a UE stays in the RRC_CONNECTED state but does not transmit data, 3GPP Release 16 introduces the fast RRC connection release function. This function requires that UEs support the **releasePreference-r16** capability defined in 3GPP Release 16. If fast RRC connection release is required for UEs that do not support this capability, device-pipe synergy can be used for these UEs. For details, see "Fast RRC Connection Release" in *Device-Pipe Synergy*. Fast RRC connection release described in this document does not apply to RedCap UEs. For details about the similar function supported by RedCap UEs, see descriptions of RedCap UE release preference in *RedCap*.

NOTE

If the **REDCAP_RELEASE_PREFER_CTRL_SW** option of the **NRCellRedCap**.***RedCapAlgoSwitch*** parameter is selected, the fast RRC connection release function described in this document can also take effect on RedCap UEs.

Fast RRC connection release in compliance with 3GPP Release 16 is controlled by the **UE_RELEASE_PREFERENCE_SW** option of the **NRCellAlgoSwitch**.***UePwrSavingSwitch*** parameter. **7.1.1 Principles** shows the procedure for this function.

Figure 7-1 Fast RRC connection release procedure



The process of this function is as follows:

1. The UE and base station exchange capability information.

   After an RRC connection is set up between the UE and the base station, the following operations are performed:

   a. The base station sends a UECapabilityEnquiry message to the UE.
   b. The UE returns a UECapabilityInformation message to the base station. The **releasePreference-r16** field in the message indicates that the UE supports fast RRC connection release.
   c. The base station sends an RRCReconfiguration message containing the **releasePreferenceProhibitTimer-r16** field to the UE to configure the RRC connection release indication prohibition timer for the UE. This timer specifies the minimum interval at which the UE can report RRC connection release indication messages. This prevents the UE from frequently reporting the indication.
   d. The UE returns an RRCReconfigurationComplete message to the base station.

2. The UE proactively reports the indication for its preferred RRC connection state.

   After the fast RRC connection release function is enabled, when detecting that the RRC connection can be released, the UE does not need to wait for the inactivity timer on the base station side to expire. Instead, the UE sends a UEAssistanceInformation message to the base station, requesting for an RRC connection release and a switch to the RRC_IDLE or RRC_INACTIVE state. For details about the RRC_INACTIVE state, see *SA Mobility Management in Inactive Mode*.

The preferred RRC connection state indications proactively reported by the UE include **idle**, **inactive**, **connected**, and **outOfConnected**. **outOfConnected** indicates that the UE expects to exit the RRC_CONNECTED state and the base station determines whether to release the UE to the RRC_IDLE or RRC_INACTIVE state.

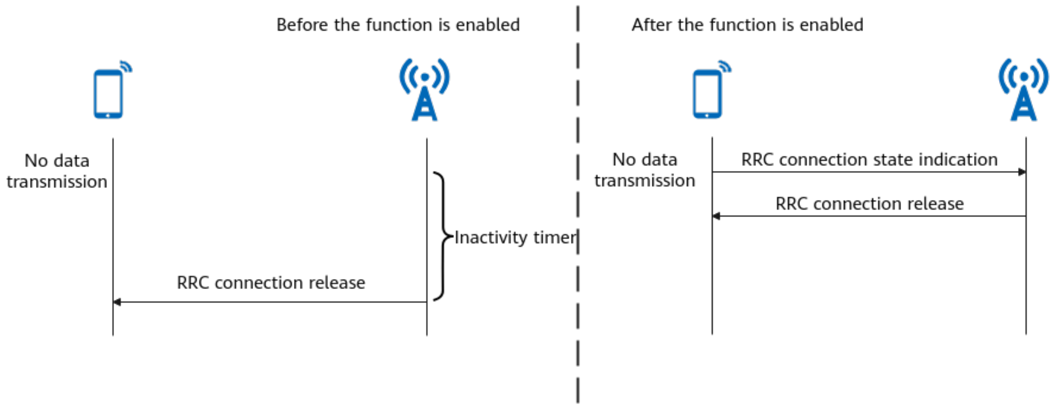
3. Upon receiving this message, the base station releases the RRC connection for the UE, thereby reducing UE power consumption.

   If a UE sends a UEAssistanceInformation message to the base station, requesting to change from the RRC_CONNECTED state to the RRC_INACTIVE state, but the conditions for the UE to enter the RRC_INACTIVE state are not met, then the base station switches the UE to the RRC_IDLE state. If the conditions are met, the base station switches the UE to the RRC_INACTIVE state.

   If the UE is performing positioning or VoNR services, the base station ignores the RRC connection state indication of the UE and does not deliver an RRC connection release message.

Figure 7-2 shows the differences before and after the fast RRC connection release function is enabled.

Figure 7-2 Comparison before and after the fast RRC connection release function is enabled



Blacklist and Whitelist Control over Fast RRC Connection Release

- If the **UE_RELEASE_PREFERENCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter is selected, the function of fast RRC connection release is disabled for specified UEs. That is, this function does not take effect on the specified UEs and can take effect on other UEs. This option applies only to non-RedCap UEs and 1RX RedCap UEs.
- If the **UE_RELEASE_PREFERENCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the function of fast RRC connection release is enabled for specified UEs. That is, this function can take effect on the specified UEs and does not take effect on other UEs. This option applies only to non-RedCap UEs. For details about whitelist control over non-1RX RedCap UEs, see *RedCap*.

The **UE_RELEASE_PREFERENCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UE_RELEASE_PREFERENCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be both selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter.

7.1.2 Network Analysis

7.1.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

Assume that the fast RRC connection release function is enabled. When the **UE_RELEASE_PREFERENCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter or the **UE_RELEASE_PREFERENCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter is selected, the number of UEs with fast RRC connection release in effect decreases, the delay decreases, the numbers of times UEs enter RRC_IDLE mode and RRC_INACTIVE mode in the cell decrease, the number of times RRC connections are resumed for UEs in the cell decreases, the number of UEs in RRC_CONNECTED mode in the cell increases, the number of UE context setup attempts and the number of successful UE context setups in the cell decrease, and the power consumption of UEs without fast RRC connection release in effect increases.

7.1.2.2 Impacts

The impacts between this function and other functions are described in **7.1.3.2.3 Function Impacts**.

This function has the following impacts on the network:

- The numbers of times that the RRC_IDLE and RRC_INACTIVE states are entered, and the RRC connection is resumed may increase. The number of RRC_CONNECTED UEs may decrease. The number of UE context setup attempts and the number of successful UE context setups may increase.
- The number of RRC connection setup times may increase because the number of times UEs enter the RRC_IDLE state increases.
- In scenarios with ongoing services, the delay may increase if a UE initiates a fast RRC connection release. The specific impact is closely related to the RRC connection release policy of the UEs. For details about the impact of the RRC_INACTIVE state on the network, see *SA Mobility Management in Inactive Mode*.
- If UEs frequently report messages for fast RRC connection release, the CPU load of the main control board may increase.

The involved counters are as follows:

- Counters related to RRC connection setups: **N.RRC.SetupReq.Att**, **N.RRC.Setup**, and **N.RRC.SetupReq.Succ**.
- Counters related to the number of RRC connection resume times: **N.RRC.ResumeReq.Att**, **N.RRC.Resume**, and **N.RRC.ResumeReq.Succ**

- Counters related to the number of UEs in the RRC_CONNECTED state: **N.User.RRCConn.Max**, **N.User.RRCConn.Avg**, and **N.User.NsaDc.PSCell.Avg**
- Number of UE context setup attempts in a cell: **N.UECntx.Est.Att**
- Number of successful UE context setup times in a cell: **N.UECntx.Est.Succ**
- Packet Delay in DU = **N.QoS.DL.PktDelayAirInterface.Time**/ **N.QoS.DL.PktDelayAirInterface.Num** + **N.QoS.DL.PktDelaygNBDU.Time**/ **N.QoS.DL.PktDelaygNBDU.Num**
- Board CPU usage: **VS.BBUBoard.CPULoad.Mean** and **VS.BBUBoard.CPULoad.Max**

7.1.3 Requirements

7.1.3.1 Licenses

There are no license requirements.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

None

7.1.3.2.2 Mutually Exclusive Functions

None

7.1.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
SA mobility management in inactive mode	**RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.*InactiveStrategySwitch*** parameter	*SA Mobility Management in Inactive Mode*	When the RRC_INACTIVE state function is enabled, fast RRC connection release allows UEs to quickly switch from the RRC_CONNECTED state to the RRC_INACTIVE state.

7.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.1.3.4 Others

UEs must support the **releasePreference-r16** capability defined in 3GPP Release 16.

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

Table 7-1 describes the parameters used for function activation.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
UE Power Saving Switch	**NRCellAlgoSwitch**.***UePwrSavingSwitch***	UE_RELEASE_PREFERENCE_SW	To enable fast RRC connection release, select this option.
Blacklist Control Switch	**gNBUeCompatSw**.***BlacklistCtrlSwitch***	UE_RELEASE_PREFERENCE_SW_OFF	Select this option if a blacklist mechanism needs to be used to disable the fast RRC connection release function for specified UEs.
Whitelist Control Switch	**gNBUeCompatSw**.***WhitelistCtrlExtSwitch***	UE_RELEASE_PREFERENCE_SW_ON	Select this option if a whitelist mechanism needs to be used to enable the fast RRC connection release function for specified UEs.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to **7.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- The **MOD GNBUECOMPATSW** command is a high-risk command. Running this command changes the feature activation states for specified UEs.
- The **UE_RELEASE_PREFERENCE_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter and the **UE_RELEASE_PREFERENCE_SW_ON** option of the **gNBUeCompatSw**.***WhitelistCtrlExtSwitch*** parameter cannot be selected for UEs of the same type specified by the **gNBUeCompatSw**.***UeInfoIndex*** parameter at the same time.

Activation Command Examples

```
//Enabling fast RRC connection release
MOD NRCELLALGOSWITCH: NrCellId=1, UePwrSavingSwitch=UE_RELEASE_PREFERENCE_SW-1;
//Enabling blacklist control over fast RRC connection release for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=UE_RELEASE_PREFERENCE_SW_OFF-1;
//Enabling whitelist control over fast RRC connection release for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UE_RELEASE_PREFERENCE_SW_ON-1;
```

Deactivation Command Examples

```
//Disabling fast RRC connection release
MOD NRCELLALGOSWITCH: NrCellId=1, UePwrSavingSwitch=UE_RELEASE_PREFERENCE_SW-0;
//Disabling blacklist control over fast RRC connection release for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1,
ParamCtrlType=ALL,BlacklistCtrlSwitch=UE_RELEASE_PREFERENCE_SW_OFF-0;
//Disabling whitelist control over fast RRC connection release for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=2,
ParamCtrlType=ALL,WhitelistCtrlExtSwitch=UE_RELEASE_PREFERENCE_SW_ON-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

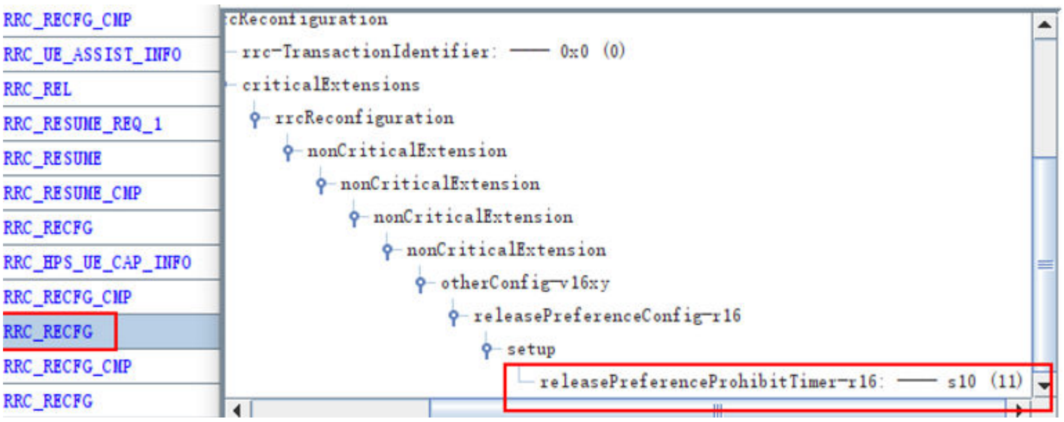
NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

Create a Uu interface tracing task on the MAE-Access. Enable a UE supporting the **releasePreference-r16** capability defined in 3GPP Release 16 to initially access the network. Check the Uu interface tracing result. If the RRC_RECFG message carries the **releasePreferenceConfig** field in which **releasePreferenceProhibitTimer-r16** is set to a value other than 0, as shown in **Figure 7-3**, fast RRC connection release has taken effect. In other cases, fast RRC connection release does not take effect.

Figure 7-3 releasePreferenceConfig field



7.1.4.3 Network Monitoring

Subscribe to CHR logs. If the value of the **RelTrigger** field in the **PRIVATE_UE_RELEASE** event block is **UE**, the UE proactively requests an RRC connection release to switch to the RRC_IDLE state. After the fast RRC connection release function takes effect, the number of times that this event block is triggered increases.

Subscribe to CHR logs. If the value of the **RelTrigger** field in the **PRIVATE_RRC_RELEASE_SUSPEND** event block is **UE**, the UE proactively requests an RRC connection release to switch to the RRC_INACTIVE state. After the fast RRC connection release function takes effect, the number of times this event block is triggered increases.

This function reduces power consumption of the UE communication module. The gains must be observed and evaluated on the UE side.

After this function is enabled, monitor the changes in related counters by referring to **7.1.2.2 Impacts**.

7.2 Inactive RA-SDT

7.2.1 Principles

When a UE is running services with a small data volume, excessive power is consumed if it stays in the RRC_CONNECTED state for a long time. To address this issue, 3GPP Release 17 introduces the inactive small data transmission (SDT) function. **This function allows RRC_INACTIVE UEs to perform services with a small amount of data. This function increases the duration in which UEs stay in states other than RRC_CONNECTED, to achieve UE power saving.**

3GPP Release 17 support two SDT modes:

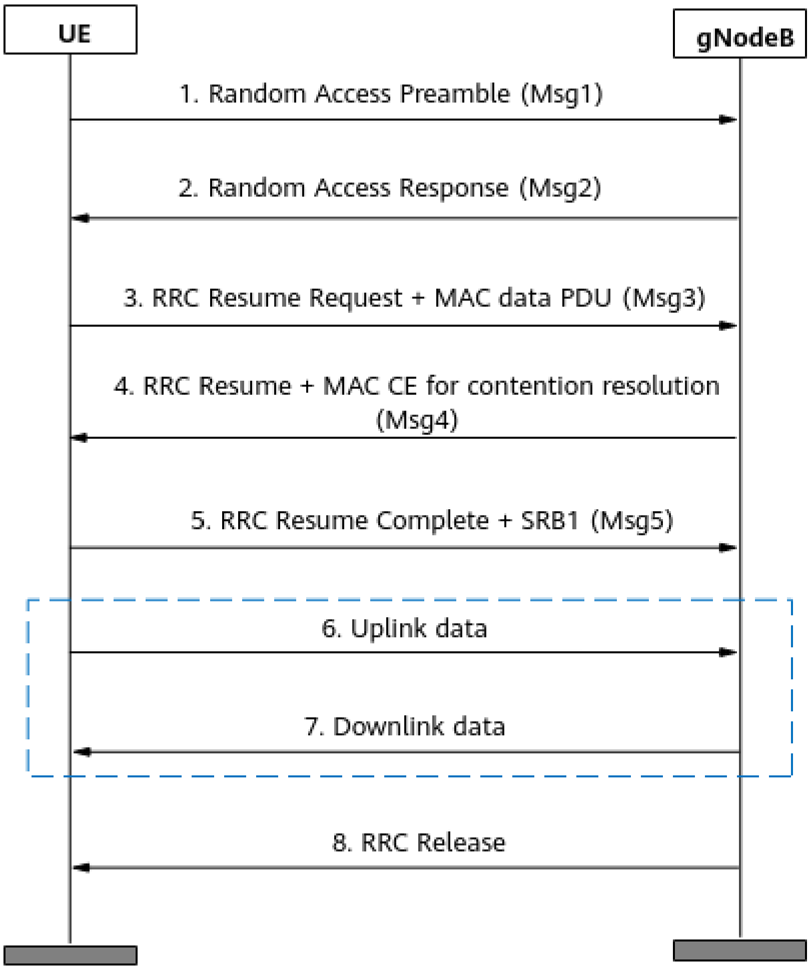
- Random access-based SDT (RA-SDT)
- Configured grant-based SDT (CG-SDT)

Currently, Huawei base stations implement the inactive SDT function in RA-SDT mode. This function takes effect only on non-RedCap UEs.

Inactive RA-SDT is controlled by the **UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter. Assume that this function is selected. When releasing a UE supporting the inactive RA-SDT function to the RRC_INACTIVE state, the gNodeB includes **SuspendConfig->sdt-Config-r17** in the RRC Release message sent to the UE. In addition, **sdt-DRB-List** in the *sdt-Config-r17* IE contains the DRBs that are allowed to transmit uplink data through the SDT procedure.

For an RRC_INACTIVE UE, if the RRC Release message used to release the UE and enable the UE to enter the RRC_INACTIVE state contains the configuration of the SDT DRBs set by the gNodeB, inactive RA-SDT takes effect during the next RA procedure, as illustrated in **Figure 7-4**. In this way, RRC_INACTIVE UEs can perform services with a small amount of data, reducing power consumption.

Figure 7-4 Procedure for inactive RA-SDT



1. When there is uplink data transmission and all the following conditions are met, a UE can initiate RA using the eMBB SDT preamble and starts the T319a timer.
   – The RSRP of the UE is greater than the value of **NRDUCellSelConfig**.***RaSdtRsrpThld***.

– The uplink data volume of the UE is less than or equal to the value of **NRDUCellSelConfig**.***RaSdtDataVolumeThld***.

NOTE

The T319a timer is described as follows:

- If the UE receives an RRC Resume, RRC Setup, or RRC Release message from the gNodeB, the UE stops the T319a timer.
- If the T319a timer expires, the UE switches to the RRC_IDLE state.

2. The gNodeB performs the RA response.
3. After the UE receives the random access response (RAR) from the gNodeB, it sends the RRC resume request signaling and uplink data to the gNodeB through the Msg3 message. The MCS index for Msg3 is specified by the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter.

   If a UE has no uplink or downlink data to transmit before the inactivity timer expires, it is released to the RRC_INACTIVE state.
4. The gNodeB delivers an RRC Resume message and a MAC CE for contention resolution.
5. The UE responds with an RRC Resume Complete message and SRB1.
6. Data is transmitted in the uplink and downlink.
7. After data transmission is complete, the gNodeB sends an RRC Release message to release the UE and enable the UE to enter the RRC_INACTIVE state.

In the procedure, note the following:

In step 3, the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter has the following impacts:

- A smaller value of the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter results in a smaller volume of uplink data that can be contained in Msg3, a higher probability of successful Msg3 demodulation during RA-SDT, and a lower requirement for the signal quality (RSRP value of the UE) of the serving cell of the UE.
- A larger value of the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter results in a larger volume of uplink data that can be contained in Msg3, a lower probability of successful Msg3 demodulation during RA-SDT, and a higher requirement for the signal quality of the serving cell of the UE.

The UE RSRP and the **NRDUCellSelConfig**.***RaSdtRsrpThld*** parameter value affect the function effectiveness of SDT on UEs. Therefore, when setting the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter, you must take the **NRDUCellSelConfig**.***RaSdtRsrpThld*** parameter value into consideration. **Table 7-2** describes the configuration requirements for **NRDUCellSelConfig**.***RaSdtRsrpThld*** when the **NRDUCellPusch**.***RaSdtMsg3Mcs*** parameter is set to different values in different interference scenarios.

Table 7-2 Configuration requirements for **NRDUCellSelConfig**.***RaSdtRsrpThld***

Scenario	Value of NRDUCellPusch.*RaSdtMsg3Mcs*	Value of NRDUCellSelConfig.*RaSdtRsrpThld*
Medium- and low-interference scenario (–120 dBm ≤ **N.UL.NI.Avg** < –100 dBm)	MCS1 to MCS3	≥ -110 dBm
	MCS4 to MCS6	≥ -107 dBm
	MCS7 to MCS9	≥ -102 dBm
	MCS10 to MCS11	≥ -98 dBm
High-interference scenario (**N.UL.NI.Avg** ≥ –100 dBm)	MCS1 to MCS3	≥ -100 dBm
	MCS4 to MCS6	≥ -95 dBm
	MCS7 to MCS9	≥ -90 dBm
	MCS10 to MCS11	≥ -85 dBm

7.2.2 Network Analysis

7.2.2.1 Benefits

This function reduces the power consumption of communication modules of UEs. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

7.2.2.2 Impacts

The impacts between this function and other functions are described in **7.2.3.2.3 Function Impacts**.

- After inactive RA-SDT is enabled, the contention-based random access success rate fluctuates because the access-related counters related to inactive RA-SDT are measured in the original contention-based random access counters.
- In different interference scenarios, if the **NRDUCellSelConfig**.***RaSdtRsrpThld*** parameter is set to a value less than the required value listed in **Table 7-2**, Msg3 demodulation may fail and UEs may fail to access the network.
- When the random access load is heavy and the inactive RA-SDT function is enabled, as the number of available PRACH preambles decreases for UEs on which inactive RA-SDT does not take effect, the probability of contention conflicts increases, the access delay may increase for UEs on which inactive RA-SDT does not take effect, and the contention-based random access success rate (indicated by **N.RA.Contention.Resolution.Succ**/**N.RA.Contention.Att** x 100%) may decrease.

For details about counters related to inactive RA-SDT, see **Counter Observation**.

7.2.3 Requirements

7.2.3.1 Licenses

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>Sales Unit</th></tr>
<tr><td>FOFD-030210</td><td>UE Power Saving</td><td>NR0S00UEPS00</td><td>per Cell</td></tr>
<tr><td colspan="4">For the license control item NR0S00UEPS00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
– When the license sales number pre-check function is enabled, the UE_PWR_SAVING_ENHANCE_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures.
– When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.
The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Mobility management in inactive mode	**RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*SA Mobility Management in Inactive Mode*	The **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter can be selected only when the **RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter is selected.

7.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	IAB deployment	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	If the **NRDUCell**.***DeployPosition*** parameter is set to **DEPLOY_IAB**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter cannot be selected.
Low-frequency TDD	Extended Cell Range	**NRDUCell**.***CellRadius*** set to a value greater than **14500**	*Extended Cell Range*	If the **NRDUCell**.***CellRadius*** parameter is set to a value greater than **14500**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter cannot be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	INACTIVE RA-SDT	**REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*RedCap*	If the **REDCAP_RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter is selected, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter cannot be selected.
Low-frequency TDD	Low-power high-accuracy positioning	**LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	If the **LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter is selected, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter cannot be selected.
Low-frequency TDD	High speed mobility	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	If the **NRDUCell**.***HighSpeedFlag*** parameter is set to **HIGH_SPEED**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategy-Switch*** parameter cannot be selected.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PRACH configuration index	**NrDUCellPrach**.***PrachConfigurationIndex*** set to **0**, **1**, **2**, or **200**	None	If the **NrDUCellPrach**.***PrachConfigurationIndex*** parameter is set to **0**, **1**, **2**, or **200**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
Low-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	If the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
Low-frequency TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	If the **NRDUCell**.***NrDuCellNetworkingMode*** parameter is set to **HYPER_CELL_COMBINE_MODE**, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.

7.2.3.2.3 Function Impacts

None

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.2.3.4 Others

Requirements for UEs: UEs must support the 3GPP-defined **RA-SDT-r17** capability.

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

Table 7-3 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
UE Power Saving Switch	**NRDUCellUePwrSaving**.***UePowerSavingSwitch***	Select the **UE_PWR_SAVING_ENHANCE_SW** option of this parameter if inactive RA-SDT is required.

Parameter Name	Parameter ID	Setting Notes
INACTIVE Strategy Switch	**NRCellAlgoSwitch**.***InactiveStrategySwitch***	Select the **RRC_INACTIVE_RA_SDT_SW** option of this parameter if inactive RA-SDT is required.
RA-SDT Data Volume Threshold	**NRDUCellSelConfig**.***RaSdtDataVolumeThld***	Use the recommended value.
RA-SDT RSRP Threshold	**NRDUCellSelConfig**.***RaSdtRsrpThld***	For details about the configurations in different interference scenarios, see **Table 7-2**.
RA-SDT MSG3 MCS	**NRDUCellPusch**.***RaSdtMsg3Mcs***	

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to **7.2.3.2 Functions** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the UE power saving enhancement switch
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-1;
//Turning on the RRC Inactive RA-SDT Switch
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=RRC_INACTIVE_RA_SDT_SW-1;
//Configuring the RSRP threshold and data volume threshold for RA-SDT
MOD NRDUCELLSELCONFIG: NrDuCellId=0, RaSdtRsrpThld=-100, RaSdtDataVolumeThld=BYTE100;
//Configuring the MCS index for RA-SDT Msg3
MOD NRDUCELLPUSCH: NrDuCellId=0, RaSdtMsg3Mcs=MCS7;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling inactive RA-SDT
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=RRC_INACTIVE_RA_SDT_SW-0;
//Turning off the UE power saving enhancement switch. This switch also controls other functions. Before performing this operation, check whether this switch needs to be turned off.
MOD NRDUCELLUEPWRSAVING: NrDuCellId=0, UePowerSavingSwitch=UE_PWR_SAVING_ENHANCE_SW-0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

Tracing Signaling

Step 1 On the MAE-Access, create and start a Uu interface tracing task.

Step 2 Check that RRC_Release signaling is displayed in the Uu interface tracing result.

If the RRC_Release message contains **SuspendConfig->sdt-Config-r17**, it indicates that the inactive RA-SDT function has taken effect. Otherwise, it indicates that the inactive RA-SDT function has not taken effect.

----End

Counter Observation

Observe the following counters to check the inactive RA-SDT function. If the value of **N.RA.SDT.Contention.Msg3** or **N.RA.SDT.Contention.Resolution.Succ** is not 0, it indicates that inactive RA-SDT has taken effect.

Counter ID	Counter Name
1911840400	**N.RA.SDT.Contention.Resp**
1911840401	**N.RA.SDT.Contention.Msg3**
1911840402	**N.RA.SDT.Contention.Att**
1911840403	**N.RA.SDT.Contention.Resolution.Succ**

7.2.4.3 Network Monitoring

The indicators described in **7.2.2 Network Analysis** can be used for network monitoring of this function.

7.3 UE Power Saving Based on Inactive State

7.3.1 Principles

UE Power Saving Based on Inactive State consists of two functions, RRC_INACTIVE UE power saving based on the inactivity timer and automatic RNA planning based on the cell list, and is controlled by the **SA_UE_POWER_SAVING_SW** option of the **NRCellAlgoSwitch**.***UePwrSavingSwitch*** parameter. The UE Power Saving Based on Inactive State feature takes effect on UEs in the RRC_INACTIVE state. Therefore, mobility management in inactive mode must be enabled before this feature is enabled. For details, see *SA Mobility Management in Inactive Mode*.

RRC_INACTIVE UE Power Saving Based on the Inactivity Timer

If a UE has no data to transmit, the UE is released and enters the RRC_IDLE or RRC_INACTIVE state when the inactivity timer specified by the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter expires. If a UE supporting the RRC_INACTIVE state is released and enters the RRC_INACTIVE state in advance when there is no data transmission, the UE can quickly transit to the RRC_CONNECTED state when a service is initiated. This has little impact on services. In this context, RRC_INACTIVE UE power saving based on the inactivity timer is introduced.

If mobility management in inactive mode has been enabled for a cell and the cell is in an RNA, **the NRDUCellQciBearer.*InactiveUeInactivityTimer* parameter specifies the QCI-specific length of the inactivity timer for a UE that meets all of the following conditions. If the inactivity timer expires, the UE is released and enters the RRC_IDLE or RRC_INACTIVE state.**

- A DRB has been set up for the UE.
- The UE has received the *Core Network Assistance Information for RRC INACTIVE* IE from the core network.
- The UE supports the RRC_INACTIVE state.
- The UE works in non-MICO mode. MICO stands for mobile initiated connection only.

The timer can be set to a proper value so that the UE can enter the RRC_IDLE or RRC_INACTIVE state in advance. This reduces the duration in which the UE stays in the RRC_CONNECTED state and achieves UE power saving. If this parameter is set to **0**, this parameter does not take effect, and the UE inactivity timer specified by the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter takes effect. If the UE is running services with multiple QCIs, the maximum value of the UE inactivity timers for different QCIs takes effect.

NOTE

The AMF uses the *MICO Mode Indication* IE to indicate whether the UE is in MICO mode. For details about the MICO mode, see section "Mobile Initiated Connection Only (MICO) mode" in 3GPP TS 23.501.

Automatic RNA Planning Based on the Cell List

To enable UEs that support the RRC_INACTIVE state to enter the RRC_INACTIVE state, an RNA needs to be planned. **RNA planning based on RNA IDs described in *SA Mobility Management in Inactive Mode* is labor-intensive and complex. Therefore, Huawei introduces automatic RNA planning based on the cell list to simplify RNA planning and activation of features related to the RRC_INACTIVE state.**

This function is controlled by the **RNA_POLICY_SELECT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter. Assume that this function is enabled and a UE is about to be released and transit from the RRC_CONNECTED state to the RRC_INACTIVE state. The gNodeB selects target cells based on the handover history of the UE and handover history of the serving cell. If the number of target cells is less than the value of the **gNBMobilityCommParam**.***ConToInacCellThdInCellList*** parameter, the gNodeB releases the UE to make it transit from the RRC_CONNECTED state to the

RRC_IDLE state. Otherwise, the gNodeB continues to select target cells based on the configured neighboring NR cells, fills the selected target cells in the cellList field in the RRC Release message, and sends the message to the UE. The UE uses these cells as an RNA for transition to the RRC_INACTIVE state. Specifically, a maximum of 32 target cells can be selected.

7.3.2 Network Analysis

7.3.2.1 Benefits

When the **NRDUCellQciBearer**.***InactiveUeInactivityTimer*** parameter is set to a value other than **0** and less than the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter value, the power consumption of communication modules of certain UEs can be reduced. The reduction in power consumption can be observed only on the UE side with the help of professional instruments. Because different UEs have varying chips and service types, it is recommended that UE vendors observe and evaluate the power consumption gains.

7.3.2.2 Impacts

The impacts between this function and other functions are described in **7.3.3.2.3 Function Impacts**.

- When the **NRDUCellQciBearer**.***InactiveUeInactivityTimer*** parameter is set to a value other than **0** and less than the **NRDUCellQciBearer**.***UeInactivityTimer*** parameter value, the following is true:

  Assume that a UE that supports the RRC_INACTIVE state does not perform any services. The earlier time it is released or transits to the RRC_INACTIVE state, the more frequently it initiates RRC connection setup or resume requests. In addition, when the number of normal RRC releases or the number of transitions from the RRC_CONNECTED state to the RRC_INACTIVE state increases, values of counters related to RRC resume (for example, the **N.RRC.Resume** counter) increase, and the average and maximum numbers of UEs in the RRC_INACTIVE state in the cell (indicated by **N.User.Inactive.Avg** and **N.User.Inactive.Max**, respectively) increase.
- After automatic RNA planning based on the cell list is enabled, if the RNA range increases, the number of RAN-initiated paging messages in a cell (indicated by **N.Paging.RAN.Att**) increases, and the number of RNA update attempts in a cell (indicated by **N.RRC.RnaUpdate.Att**) and the number of successful RNA updates in a cell (indicated by **N.RRC.RnaUpdate.Succ**) decrease. A decrease in the RNA range leads to the opposite effects.
- For details about the impacts of the RRC_INACTIVE state, see *SA Mobility Management in Inactive Mode*.

7.3.3 Requirements

7.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081252	UE Power Saving Based on Inactive State	NR0S00UPSB00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Mobility management in inactive mode	**RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.*InactiveStrategySwitch*** parameter	*SA Mobility Management in Inactive Mode*	Before enabling functions in the UE Power Saving Based on Inactive State feature, you must enable mobility management in inactive mode.

7.3.3.2.2 Mutually Exclusive Functions

None

7.3.3.2.3 Function Impacts

None

7.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.3.4 Others

This feature requires that the RRC_INACTIVE state is supported. For details, see section "Others" in *SA Mobility Management in Inactive Mode*.

This feature has the following requirements for the core network:

- When the core network receives a UE CONTEXT RELEASE REQUEST message with the cause value "UE in RRC_INACTIVE state not reachable" from the last serving gNodeB, the core network must be able to release the UE to the RRC_IDLE state and perform core network paging.
- When the core network receives an NG interface message with the cause value "UE Context Transfer" from the last serving gNodeB, the core network must be able to initiate a bearer reestablishment procedure on the new gNodeB.

7.3.4 Operation and Maintenance

7.3.4.1 Data Configuration

7.3.4.1.1 Data Preparation

Table 7-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 7-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
UE Power Saving Switch	**NRCellAlgoSwitch**.***UePwrSavingSwitch***	Select the **SA_UE_POWER_SAVING_SW** option.
Inactive UE Inactivity Timer	**NRDUCellQciBearer**.***InactiveUeInactivityTimer***	Use the recommended value.
INACTIVE Strategy Switch	**NRCellAlgoSwitch**.***InactiveStrategySwitch***	Select the **RNA_POLICY_SELECT_SW** option.
Connect to Inactive Cell Thld in Cell List	**gNBMobilityCommParam**.***ConToInacCellThdInCellList***	Set this parameter based on live network conditions. The smaller the parameter value, the higher the probability of UEs transiting from the RRC_CONNECTED state to the RRC_INACTIVE state. The larger the parameter value, the lower the probability of UEs transiting from the RRC_CONNECTED state to the RRC_INACTIVE state.

7.3.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the SA_UE_POWER_SAVING_SW
MOD NRCELLALGOSWITCH: NrCellId=0,UePwrSavingSwitch=SA_UE_POWER_SAVING_SW-1;
//Setting the InactiveUeInactivityTimer
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=1, InactiveUeInactivityTimer=5;
//Turning on the RNA_POLICY_SELECT_SW
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=RNA_POLICY_SELECT_SW-1;
MOD GNBMOBILITYCOMMPARAM: ConToInacCellThdInCellList=0;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling the InactiveUeInactivityTimer
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=1, InactiveUeInactivityTimer=0;
//Turning off the RNA_POLICY_SELECT_SW
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=RNA_POLICY_SELECT_SW-0;
//Turning off the SA_UE_POWER_SAVING_SW
MOD NRCELLALGOSWITCH: NrCellId=0,UePwrSavingSwitch=SA_UE_POWER_SAVING_SW-0;
```

7.3.4.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.3.4.3 Activation Verification

RRC_INACTIVE UE Power Saving Based on the Inactivity Timer

If the following is true after the **NRDUCellQciBearer**.***InactiveUeInactivityTimer*** parameter is modified, RRC_INACTIVE UE power saving based on the inactivity timer has taken effect: The values of counters related to RRC resume (for example, the **N.RRC.Resume** counter) increase, the RRC connection setup success rate increases, and the average and maximum numbers of UEs in the RRC_INACTIVE state in the cell (indicated by **N.User.Inactive.Avg** and **N.User.Inactive.Max**, respectively) increase.

Automatic RNA Planning Based on the Cell List

If the IE *RAN-NotificationAreaInfo* in the RRC Release message contains the cellList field, automatic RNA planning based on the cell list has taken effect.

7.3.4.4 Network Monitoring

The indicators described in **7.2.2 Network Analysis** can be used for network monitoring of this function.

8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *5G Networking and Signaling*
- *Co-MM (Low-Frequency TDD)*
- *CoMP*
- *DRX*
- *NSA Mobility Management*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Inactive Mode*
- *URLLC*
- *VoNR*
- *Super Uplink*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *Device-Pipe Synergy*
- *Positioning*
- *Interference Avoidance*
- *High Speed Mobility*
- *Energy Conservation and Emission Reduction*
- *Scalable Bandwidth*
- *3D Networking Experience Improvement*
- *UL and DL Decoupling*
- *Network Slicing*
- *Channel Management*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Carrier Aggregation*

- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Multi-Frequency Smart Aggregation*
- *3900 & 5900 Series Base Station Product Documentation*
- **(Video) Frequency-Domain UE Power Saving**
- **(Video) Time-Domain UE Power Saving**
- **(Video) Time-Domain UE Power Saving - Smart Preallocation**
- *CA Experience Improvement*
- *Power Control*
- *FWA*
- *Random Access Control*
- *RedCap*
- *Cell Combination*
- *MIMO (TDD)*
- *Extended Cell Range*
- *EPC-based NSA Basics*
- *Turbo Uplink (Low-Frequency TDD)*
- *Hyper Cell*
- *Fusion Cell (Low-Frequency TDD)*
- *License Control Item Lists*
- *Multi-Operator Sharing*
- *Ultimate Video Experience*
- *Ultimate Cloud X Experience*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*

